Group No: 33

DESIGN AND DEVELOPMENT OF A FRAMEWORK FOR FLIGHT DYNAMICS AND CONTROL FOR SMALL CONVENTIONAL SURVEILLANCE UAVS

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Declaration of the candidate and supervisor

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Abstract

Unmanned Aerial Vehicles (UAVs) are widely used among vast range of applications in both civil and military aerospace industry. Yet existing literature remain limited on systematic design methodologies that integrate geometric configuration with flight dynamic behaviour. This report presents an inclusive research effort builds toward addressing this gap. An integrated Python-based modular framework for the systematic generation of geometrical configuration, and design, analysis and evaluation of flight dynamics and control behaviour of small conventional surveillance UAVs, An experimental setup for precise measurement of airframe geometry and inertia properties with quantification of error to ensure the reliability of measurements, a UAV prototype with modified fuselage and empennage sections and instrumented to conduct tests to extract flight data and validate the framework, a parametric sensitivity analysis to systematically assess the influence of geometric variations on inherent stability and dynamic behaviour of the UAV have been carried out. The impact of geometric variations on modal behaviour of an UAV laid out the basis to propose a methodological approach to fine-tune the dynamic behaviour of an UAV. The sensitivity analysis showed that main wing sweep angle, tail volume ratio and tail lift coefficient vs angle of attack curve gradient are significant drivers of short period dynamic mode while main wing sweep and wing aerofoil mainly effects the phugoid dynamic mode characteristics. These works collectively enables the UAV design practice and study on flight dynamic and control more efficient and advanced.

Keywords - Flight dynamics and control, Flying qualities, Sensitivity analysis, Inertia measurement, PID tuning.

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1. Introduction

Unmanned aerial vehicles (UAVs) are increasingly emerging across both civil and military domains for agriculture, surveying, disaster response, surveillance, reconnaissance and more remote applications. Their rapid expansion demonstrates as one of fastest growing platforms in modern aerospace engineering. Small fixed wing UAVs display favourable endurance and mobility of payloads for respective missions. Advancement in UAV design have been developed in propulsion systems, avionics, lightweight composite materials and autonomous control systems and etc that has broadened the scope of UAVs. The effectiveness of a UAV largely depends on performance, dynamic and control behaviour. Small surveillance UAVs differ significantly from larger manned aircraft in terms of scale, operating conditions, certification standards and reliance on automated control systems, making direct application of existing flight dynamics models and flying quality standards inadequate. Many UAVs are developed through iterative flight testing and numerous prototypes in expense of time, resources and cost rather than attaining to the underlying airframe geometry. Both manual and autonomous operations of an UAV depend on the flight dynamic characteristics and flying qualities where these are fundamentally governed by the geometric configuration, mass distribution and inertia properties. The vital relationship between dynamic behaviour and UAV design has received limited systematic treatment in literature particularly in small, fixed wing UAVs.

This breach in the systematic interconnection between UAV geometric configuration and dynamic behaviour and flying qualities leads to the deficiencies in flying qualities to be identified only after fabrication and testing increasing the time and cost of the development. Addressing this problem demands a integrated programmable framework that merges aerodynamic design, stability analysis, flying quality evaluation, and stability augmentation within an automated pipeline, experimental capability for measuring airframe geometry and inertia properties, validated flight dynamic model from real flight data, and sensitivity analysis that quantifies the deviation of dynamic characteristics with geometric parameters, none of which are comprehensively established in the small UAV context.

In response, this research presents,

- a UAV design framework developed with modular workflow of UAV design, performance analysis, static and dynamic stability analyses, and stability augmentation.

- a dedicated experimental setup for the accurate measurement of airframe geometry and mass inertia properties, with full uncertainty quantification.
- a UAV modification followed by instrumentation and flight data retrieval to provide the empirical basis for framework validation.
- a systematic sensitivity analysis to examine the influence of geometric parameter variations on the dynamic mode characteristics.

This study is limited to conventional UAVs that are flown in low Reynold number conditions and the dynamic stability analysis done within the linearized small perturbation about cruising trimmed state. The flying quality assessment references MIL-F-8785C and MIL-HDBK-1797, adapted as appropriate to the UAV context to focus on relative trends in damping ratios, natural frequencies, and time constants. The flights tests are done in calm-air conditions to acknowledge the linearized model and the generalization of the proposed geometric design methodology. Within these bounds, the work establishes a geometry-driven pathway from configuration definition to dynamic behavior of the UAV, with implications for both manually operated and autonomous UAV platforms.

1.1 Aim and Objectives

1.1.1 Aim

To design and develop a framework for small conventional surveillance UAVs that enables reliable analysis of flight dynamics, stability, and control behaviour.

1.1.2 Objective

- Develop the Flight Dynamic Framework using governing principles.
- Fabrication of the experimental setup to measure geometric and inertia properties of a small UAV.
- Experimentally validating the Fight Dynamic model for selected modes.
- Propose a methodological approach to achieve the expected flying qualities for surveillance UAVs and develop a stability augmentation system.

2. Literature Review

The preliminary design stage of fixed wing aircraft has been studied comprehensively using empirical methods and computational simulations since 1940s with the interest of military advancements. Studies explain a widely adopted analytical workflow for aircraft design which includes the tail sizing, drag estimation, stability and control, and control-surface sizing criteria as foundational references in this work [1] [2] [3]. Formulation of stability and control workflow of an aircraft is descriptively explained in various studies that enables the adaptation it for conventional UAVs [4] [5]. Vortex Lattice Method (VLM) tool offers stability derivative and eigenvalue computation capabilities and previous studies have used them as standalone tool requiring manual configuration and single point analysis of a design workflow. VLM are also used in general low Reynolds number aircraft design process for aerodynamic analysis due to their low computational requirements and reliable accuracy in subsonic and low angle of attack scenarios [6] [7]. Coupling VLM tools in a programmable environment enables more design alternatives for more geometric variations to sensitivity analysis of the stability and control behavior of UAV. Flying quality of UAVs lacks a well-documented standards but at occasions on MIL-SPEC standards (MIL-F-8785C, MIL-HDBK-1797) has been applied for study relative trends in dynamic mode characteristics [8] [9]. Geometric variations done on the aerodynamic design of an aircraft to achieve required aerodynamic and dynamic mode characteristics has been reported [10] [11] [12]. Geometric variations in UAVs, such as changes in wing aspect ratio, sweep angle, dihedral, and airfoil camber, alter dynamic mode characteristics including phugoid, short-period, Dutch roll, spiral, and roll subsidence modes. Studies demonstrates, increased aspect ratio enhances phugoid damping but may reduce short-period frequency, while sweep angles improve roll stability by modifying dihedral effects. Wing morphing and bi-wing configurations further influence aeroelastic flutter boundaries and lift-dependent modes, boosting lift coefficients by up to 27% at high angles of attack. These findings, derived from wind tunnel tests and Computational Fluid Dynamics (CFD) analyses, underscore the need for geometry-optimized designs to ensure flight stability [13] [14] [15]. These findings emphasize the importance of incorporating sensitivity analysis in UAV design to understand the influence of geometric variations on stability and dynamic modes.

Experimental determination of UAV geometric dimensions, centre of gravity (CG), and inertia properties play an important role in improving the accuracy of flight dynamics and stability analysis. Previous studies show that although CAD-based estimation methods provide initial values for mass properties and dimensions, experimental validation is necessary to account for manufacturing deviations, component integration effects, and real-world mass distribution variations [16] [17] [18]. Various experimental techniques such as bifilar pendulum, compound pendulum, torsional pendulum, and torque-twist rigs have been successfully applied for small UAV inertia measurements, demonstrating acceptable accuracy and repeatability when damping and fixture effects are carefully considered [16] [19] [20]. CG determination methods including weight-and-balance approaches and suspension-based plumb-line techniques are commonly used for fixed-wing UAVs due to their simplicity and reliability [21] [22]. Literature also highlights the importance of validating Computer-Aided Design (CAD) geometry using physical measurements or experimental setups to improve the fidelity of aerodynamic and stability models [18] [22]. These findings emphasize the need for a carefully designed experimental setup capable of accurately measuring UAV geometry, inertia properties, and CG location for reliable dynamic and stability analysis.

The reviewed literature shows that the flying qualities of a UAV can be significantly improved through the implementation of properly designed control systems. [4] [5] [40] [41] Studies show that parameters such as damping ratio, natural frequency and settling time in transfer function approach are directly related to the location of system poles in the s-plane [4] [5] [27]. Therefore, introducing feedback control systems allows engineers to shift these poles into stable regions and improve overall flight performance. Among the available control approaches, PID (Proportional-Integral-Derivative) controllers remain one of the most practical and widely used solutions for UAV stabilization and flying quality enhancement [5]. Modern UAV platforms such as ArduPilot, PX4 Autopilot and LibrePilot already implement PID-based stability augmentation systems [39][36], demonstrating the practical success of this approach. Furthermore, optimization techniques such as genetic algorithms, Model Predictive Control based approaches [41], fuzzy logic-based [40] algorithms and simulation environments like MATLAB/Simulink and Python Control library are increasingly used to obtain optimal PID parameters for enhanced flying qualities and reliable UAV operation. However, from these, PID approach is chosen for the project to reduce the complexity yet makes it highly useful with existing Flight-controllers. Designing a flight controller is out of scope of this project.

3. Methodology

3.1 Flight Dynamic and Control Framework

3.1.1 Framework Architecture and Objectives

The stability of an aircraft can be divided into two sections.

- Static stability
- Dynamic stability

Stability is the tendency of the aircraft to maintain or return to its original condition after a disturbance. Static stability defines the initial reaction of an aircraft while the dynamic stability is about behaviour over time after a disturbance.

The design logic of the horizontal and vertical tail sizing is used as the opening module of the flight dynamics and control framework. Starting with tail sizing, the static stability of the UAV design is evaluated. Then the performance of the design is evaluated and checked for its reliability. The control surface sizing and the dynamic stability evaluation is done afterward. Following the dynamic stability evaluation, the flying qualities of the UAV are evaluated. Based on the results of flying qualities, the geometric finetuning or the stability augmentation is performed.

The framework is implemented in Python and ordered as a series of modular units of design stages of an airframe. Each module reads prerequisite parameters and mission parameters as inputs, then performs relevant calculations and analysis, finally exports the outputs that are consumed by the downstream modules. The overall workflow is presented in APPENDIX A: Flowcharts of the Framework.

The framework is built to ensure each module is designed in a consistent manner where a module reads outputs from upstream modules, perform its design calculations and then write its outputs required for the downstream modules. This enables the independent execution of a module as a standalone when necessary. This was architecture as a sequential workflow of design modules and decision logic drives the iteration when the defined criteria are not satisfied. As the terminal criterion, flying qualities of the design will be evaluated and iterate the design process with geometric refinements or if the geometric refinement is limited dynamic characteristics will be addressed through control system design. The framework consists of modules of matching plot and design of wing, fuselage, propulsion and Landing gears. The original contribution lies with the integration of these standalone modules into a unified workflow with development of tail design and static stability evaluation, aerodynamic drag and performance analysis, control surface design and dynamic stability evaluation followed by flying quality evaluation and stability augmentation to a complete UAV design framework.

3.1.2 Modules and Their Theoretical Background

3.1.2.1 Mission Requirements and Basic Configuration

Mission Requirements and basic configuration of the UAV are defined here. Range, endurance, cruise speed, ceiling altitude are defined as mission requirements. The basic UAV configuration is defined as conventional tractor fixed wing with conventional inverted T tail section.

3.1.2.2 Weight Estimation and Matching Plot

Then the mission requirements were further translated into initial weight estimation reviewing existing designs. The matching plot is constructed using stall speed, max speed, rate of climb, take off run and ceiling constraints to select a feasible design point yielding the wing loading and power to weight ratio. This module initializes the sizing of the UAV geometry.

3.1.2.3 Wing Design

In this module, wing sizing is taken place based on the data retrieved from the selected design point. Aerofoil is selected considering the lift coefficient requirement, from a database developed beforehand using XFOIL software. Wing chord at tip and root, taper ratio, aerodynamic data of finite wing, twist angle, wing incidence and other requisite wing data is calculated here.

3.1.2.4 Fuselage Design

A tadpole-type fuselage is designed using this module. The geometry is selected based on the minimum drag coefficient produced by the body. The fuselage contribution towards the pitching moment is also calculated within this module. As similar to the wing design module, the outputs required for downstream modules are saved at the end.

3.1.2.5 Propulsion System Design

Propulsion system design module has been developed to select appropriate propeller, motor and battery specifications and can calculate the thrust available across the flight envelope.

3.1.2.6 Landing Gear Design

The landing gear is designed considering ground clearance, take off rotation, ground overturn, and load acting on the nose and main landing gears. The tricycle type LGs (Landing Gears) is being sized in this module. Wheelbase, wheel track, overturn angle, height of the landing gears and other vital geometrical and functional details are calculated.

3.1.2.7 Horizontal Tail Design

The horizontal tail design module is scripted to design the tail as relevant as to achieve static longitudinal stability. Optimum tail arm is fed from the fuselage design module to have lesser drag contribution. The tail area is calculated from the tail volume ratio which can be changed upon the requirement of the user.

$$V_H = \frac{S_H \cdot l_H}{S \cdot \bar{c}} \tag{01}$$

(The terms are used as in conventional terminology. Refer APPENDIX B: Nomenclature)

The aerofoils that are applicable for the horizontal tail analysed. Each aerofoil's data is extracted from the pre-built database using XFOIL [23]. The required tail lift for the cruising is calculated. Considering aerofoil lift slope, the required finite wing angle of attack for horizontal tail is calculated.

Cruise lift coefficient:

$$C_{L_C} = \frac{2W_{ac}}{\rho_{\text{ceiling}} V_{\text{cruise}}^2 S_w} \tag{02}$$

Trim moment coefficient of the fuselage:

$$C_{m_{f,\text{trim}}} = C_{m0_f} + C_{m\alpha_f}\, i_w \tag{03}$$

(where $i_w$is in radians)

Total zero-lift pitching moment coefficient of wing–fuselage:

$$C_{m0_{wf}} = C_{m_{ac,w}} + C_{m_{f,\text{trim}}} \tag{04}$$

Horizontal tail lift coefficient:

$$C_{L_h} = \frac{C_{m0_{wf}} + C_{m0_p} + C_{L_C}(h - h_0)}{\eta_h V_H} \tag{05}$$

Horizontal tail lift-curve slope (finite wing correction):

$$C_{L_{\alpha h}} = \frac{C_{l_{\alpha h}}}{1 + \frac{C_{l_{\alpha h}}}{\pi AR_h}} \tag{06}$$

Horizontal tail angle of attack:

$$\alpha_h = \frac{C_{L_h}}{C_{L_{\alpha h}}} \tag{07}$$

in radians.

The angle is then corrected using Prandtl Lifting Line theory. This calculation is done all suggested aerofoils for horizontal tail. To decide the tail setting angle (tail incidence) to the fuselage, the downwash effect of the wing on the tail is considered.

Downwash at zero angle of attack:

$$\varepsilon_0 = \frac{2C_{L0_w}}{\pi AR_w} \tag{08}$$

Downwash gradient:

$$\frac{d\varepsilon}{d\alpha} = \frac{2C_{L_{\alpha w}}}{\pi AR_w} \tag{09}$$

Total downwash angle:

$$\varepsilon = \varepsilon_0 + \left(\frac{d\varepsilon}{d\alpha}\right)\alpha_w \tag{10}$$

in radians.

Tail incidence angle:

$$i_h = \alpha_h - \alpha_w + i_w + \varepsilon \tag{11}$$

all angles in degrees.

Given the tail area, the geometrical properties of the horizontal tail are then calculated.

Horizontal tail span:

$$b_h = \sqrt{AR_h S_h} \tag{12}$$

Horizontal tail mean chord:

$$\bar{c}_h = \frac{S_h}{b_h} \tag{13}$$

Horizontal tail root chord:

$$C_{h,\text{root}} = \frac{3\bar{c}_h(1 + TR_h)}{2(1 + TR_h + TR_h^2)} \tag{14}$$

Horizontal tail tip chord

$$C_{h,\text{tip}} = C_{h,\text{root}}\, TR_h \tag{15}$$

The vertical placement of the horizontal tail with reference to the wing mean aerodynamic chord is calculated considering the wake region of the wing. The wing wake reduces the tail efficiency while the downwash decreases the effective angle of attack of tail.

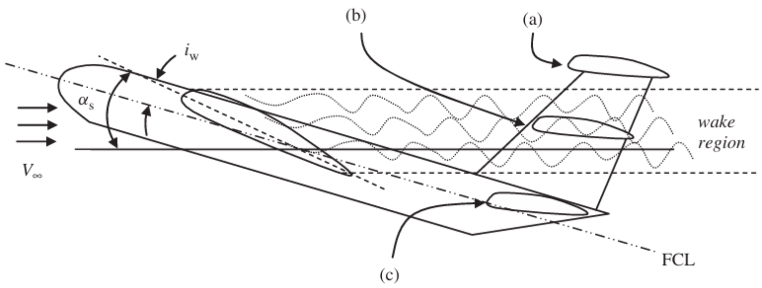

Figure 3.1: Placement of tail in wing wake region

a – Out of wake and downwash

b – Inside wake and downwash

c – Out of wake but affected from downwash

Since the downwash is considered in calculating tail incidence, location c is considered as the design is conventional.

$$h_t < l.\tan(\alpha_s - i_w - 3°) \tag{16}$$

The tail should be positioned vertically less than h_t height from wing mean aerodynamic center.

The wing, and propulsion contribution for the longitudinal moment of the UAV is then calculated. The fuselage contribution to the pitching moment is calculated in the fuselage design module and fed to this module. Multiple horizontal tail design alterations (combinations of different aerofoils, aspect ratios, taper ratios, and tail incidence) are considered, and the longitudinal moment coefficient is calculated for each alteration.

$$C_{m_{0,w}} = C_{m_{ac,w}} + C_{L_{0,w}}(h - h_0) \tag{17}$$

$$C_{m_{\alpha,w}} = C_{L_{\alpha,w}}(h - h_0) \tag{18}$$

$$C_{m_{0,p}} = \frac{2\, T_{cruise}\, z_{prop,cg}}{\rho\, V_{cruise}^2\, C_{w,mean}\, S_w} \tag{19}$$

$$C_{m_{0,h}} = \eta_h\, V_H\, C_{L_{\alpha,h}}\, (\varepsilon_0 + i_w - i_h) \tag{20}$$

$$C_{m_{\alpha,h}} = -\eta_h\, V_H\, C_{L_{\alpha,h}}\, \left(1 - \frac{d\varepsilon}{d\alpha}\right) \tag{21}$$

Total pitching moment:

$$C_{m0} = C_{m0_w} + C_{m0_h} + C_{m0_f} + C_{m0_p} \tag{22}$$

$$C_{m_\alpha} = C_{m_{\alpha w}} + C_{m_{\alpha h}} + C_{m_{\alpha f}} \tag{23}$$

(units: per radian)

Trim angle and the static margin is calculated afterwards.

Trim angle:

$$\theta_{\text{trim}} = -\frac{C_{m0}}{C_{m_\alpha}} \tag{24}$$

in radians.

Neutral point:

$$NP = h_0 - \frac{C_{m_{\alpha f}}}{C_{L_{\alpha w}}} + \frac{\eta_h V_H C_{L_{\alpha h}} \left(1 - \frac{d\varepsilon}{d\alpha}\right)}{C_{L_{\alpha w}}} \tag{25}$$

Static margin:

$$SM = NP - h \tag{26}$$

For a designs, the requirements that satisfies are,

- $-1.5 \leq C_{m_\alpha} \leq -0.3$
- $C_{m_0} \geq 0$

- Static margin, $0.1 \leq SM \leq 0.3$
- Trim angle should be nearly the wing cruising angle.

Design alterations which satisfy these will be saved in a spreadsheet to user to view and select as per their will. Upon the selection of the user horizontal tail design outputs will be saved for downstream modules. As visualization, the framework will plot the pitching moment coefficient variation with wing angle of attack as seen in Figure 3.2. The design filtering criteria can be changed as per the user requirements.

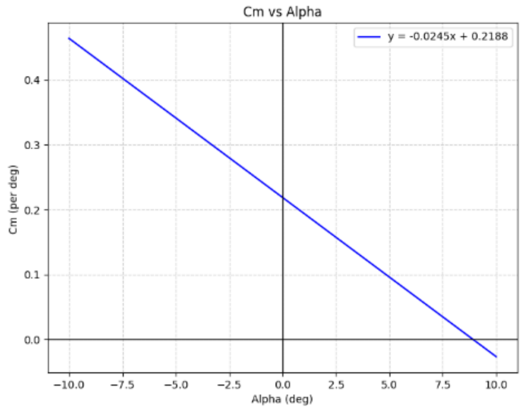

Figure 3.2: Cm vs alpha graph

3.1.2.8 Vertical Tail Sizing

The vertical tail sizing is done considering the tail volume coefficient as previously with target of achieving sufficient directional stability and spin recovery criteria. As in the H-tail design, multiple design alterations are calculated and saved in a spreadsheet here also. Once the geometry of the V-tail is selected, it is validated against the empirical spin recovery guidelines directional stability requirement. More than 50% of the area of the V-tail should be outside the wake region of the H-tail, where the H-tail wake region is considered to lie between two lines as shown in Figure 3.3. The first line is drawn at the horizontal tail trailing edge with an orientation of 30°. The second line is drawn at the horizontal tail leading edge with an orientation of 60° [2] .

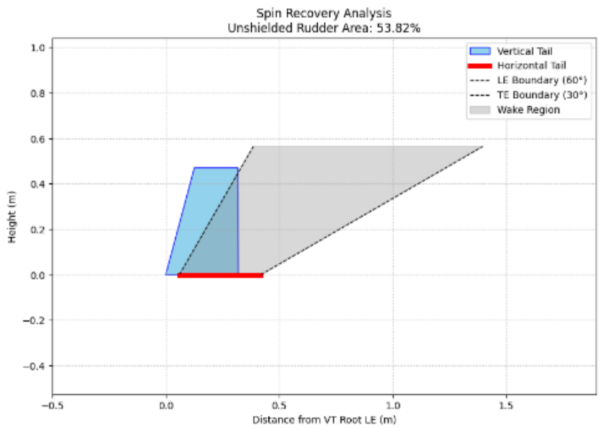

Figure 3.3: Spin Recovery criteria for vertical stabilizer

Vertical tail area:

$$S_v = \frac{b_w S_w V_V}{l_v} \tag{27}$$

Vertical tail span:

$$b_v = \sqrt{AR_v S_v} \tag{28}$$

Mean chord of the vertical tail:

$$\bar{c}_v = \frac{S_v}{b_v} \tag{29}$$

Vertical tail root chord:

$$C_{v,\text{root}} = \frac{3\bar{c}_v(1 + TR_v)}{2(1 + TR_v + TR_v^2)} \tag{30}$$

Vertical tail tip chord:

$$C_{v,\text{tip}} = TR_v \, C_{v,\text{root}} \tag{31}$$

The directional stability requirement of $0.05 < C_{n_\beta} < 0.4$ is also checked in here.

The static stability requirement of the UAV is fulfilled at the tail design modules.

3.1.2.9 Zero Lift Drag Calculation and Drag Polar

Once the basic UAV design is obtained the performance of the UAV is analysed. To estimate the zero-lift drag (C_{D_0}) of the UAV, an empirical approach is used. The component build-up method, accounting for skin friction of the components and interference factors is considered. Fuselage, wing, horizontal and vertical tail, landing gears and struts, high lift device are considered for the drag calculation [24].

$$C_{D_0} = K_C \cdot \sum CD_{0,\text{components}} \tag{32}$$

$$C_{D0,TO} = C_{D0} + C_{D0,HLD} \tag{33}$$

The overall drag polar is expressed as:

$$C_D = C_{D_0} + K.C_L{}^2 \tag{34}$$

Where the induced drag factor $K = {}^{1}/_{\pi.AR.e}$ with AR is the wing aspect ratio and e is the Oswald efficiency factor.

The framework executes the Athena Vortex Lattice (AVL, Details of AVL is described in **Error! Reference source not found.**) within, inputting calculated C_{D_0} and the basic UAV design. Using AVL, The total lift and drag coefficients are obtained for different angles of attacks. Once the data is extracted the drag-polar curve for the UAV is plotted.

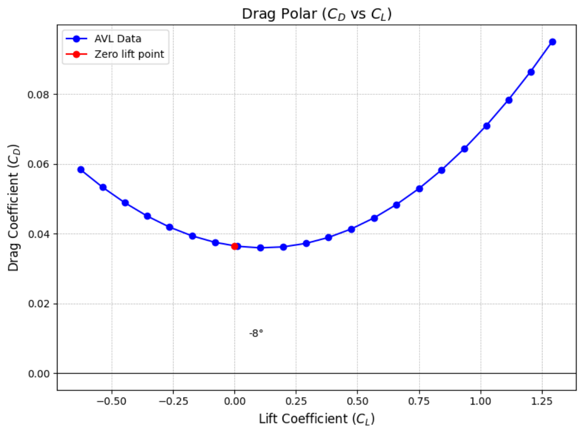

Figure 3.4: Drag polar curve

3.1.2.10 Performance Analysis

The chosen design point in the matching plot is checked against the updated configuration and the parameters as shown in Figure 3.5. If the design point is in feasible region, the design process is continued with the performance analysis. In this analysis, below parameters of the UAV is calculated.

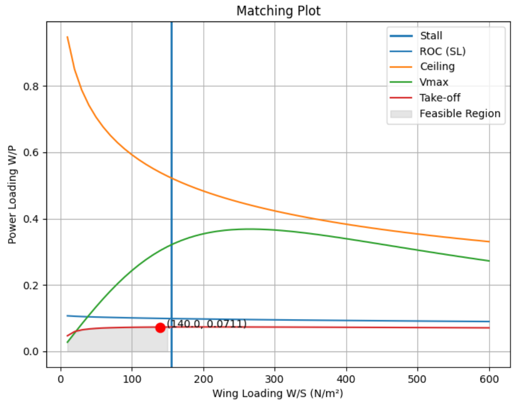

Figure 3.5: Matching plot for design point validation

Key flight performance,

- Maximum range - Distance the UAV can cover within one mission

$$R_{\max} = R_t^{1-n}\left(\frac{\eta_{\text{tot}}\, V_{\text{batt}}\, C_{\text{batt}}}{\frac{1}{\sqrt{\rho_c S}}\, C_{D0}^{1/4}\left(2W\sqrt{K}\right)^{3/2}}\right)^{n}\sqrt{\frac{2W}{\rho_c S}}\left(\frac{K}{C_{D0}}\right)^{1/4}\cdot 3.6 \tag{35}$$

- Maximum endurance – The time UAV can remain on air within one mission

$$E_{\max} = R_t^{1-n}\left(\frac{\eta_{\text{tot}}\, V_{\text{batt}}\, C_{\text{batt}}}{\frac{2}{\sqrt{\rho_c S}}\, C_{D0}^{1/4}\left(2W\sqrt{\frac{K}{3}}\right)^{3/2}}\right)^{n} \tag{36}$$

- Fastest Climb – Maximum rate of climb that the UAV can achieve

  Best ROC (Rate of Climb) speed

$$V_{\text{ROC}} = \sqrt{\frac{2W}{\rho_c S\sqrt{\frac{3C_{D0}}{K}}}} \tag{37}$$

  Climb angle:

$$\sin\gamma = -\frac{\sqrt{16K^2W^2 + \rho_c^2 S^2 V_{\text{ROC}}^4(1+4KC_{D0}) - 8PKS\rho_c\eta V_{\text{ROC}}} - \rho_c S V_{\text{ROC}}^2}{4KW} \tag{38}$$

  Maximum rate of climb:

$$\text{ROC}_{\max} = V_{\text{ROC}}\sin\gamma \tag{39}$$

- Steepest Climb – Maximum climb angle can be achieved by the UAV.

  Best-angle speed:

$$V_{\gamma_{\max}} = \frac{4KW^2}{P_{\max}\,\rho_c\,\eta\,S} \tag{40}$$

Climb angle:

$$\sin\gamma_{\max} = \frac{P\eta}{WV_{\gamma_{\max}}} - \frac{1}{2}\frac{\rho_c V_{\gamma_{\max}}^2 S C_{D0}}{W} - \frac{2KW}{\rho_c V_{\gamma_{\max}}^2 S} \tag{41}$$

Rate at steepest climb:

$$\mathrm{ROC_{steep}} = V_{\gamma_{\max}} \sin\gamma_{max} \tag{42}$$

- Take-off Distance

Ground roll:

$$S_G = \frac{m}{\rho_c S(C_{D,\mathrm{TO}} - \mu C_{L,\mathrm{TO}})} \ln\left[\frac{\frac{T}{W} - \mu}{\frac{T}{W} - \mu - \frac{k_{LO}^2(C_{D,TO} - \mu C_{L,TO})}{C_{Lmax}}}\right] \tag{43}$$

Rotation:

$$S_R = T_R V_R \tag{44}$$

Airborne (slant):

$$S_A' = \frac{W}{T_{\mathrm{air}} - D_{\mathrm{air}}}\left(\frac{\Delta V^2}{2g} + h_o\right) \tag{45}$$

Horizontal airborne distance:

$$S_A = \sqrt{S_A'^2 - h_o^2} \tag{46}$$

Total takeoff distance:

$$S_{\mathrm{TO}} = S_G + S_R + S_A \tag{47}$$

- Turn Performance – Tightest turn is considered here.

Corner speed: The velocity when the aircraft is turning with maximum bank angle having the maximum lift coefficient.

$$V^* = \sqrt{\frac{2n_{\max}W}{\rho_c S C_{L\max}}} \tag{48}$$

Minimum turn radius:

$$R_{\min} = \frac{V^{*2}}{g\sqrt{n_{\max} - 1}} \tag{49}$$

Bank angle:

$$\phi_{tt} = \cos^{-1}\left(\frac{1}{n_{\max}}\right) \tag{50}$$

are calculated.

3.1.2.11 Control Surface Design

After the performance of the UAV is satisfied, the framework moves to the control surface design module where it designs ailerons, elevator and rudder for desired maximum deflection of each control surfaces. The analytical workflow used to design control surfaces is derived from those presented in [2]. Design of each control surface is done by implementing an algorithm in Python based on standard procedures.

In all above calculations, effectiveness parameter for control surface design was approximated using the following graph given [2]. For more accurate results, data should be obtained for the relevant flow conditions experimentally (eg: wind tunnel testing).

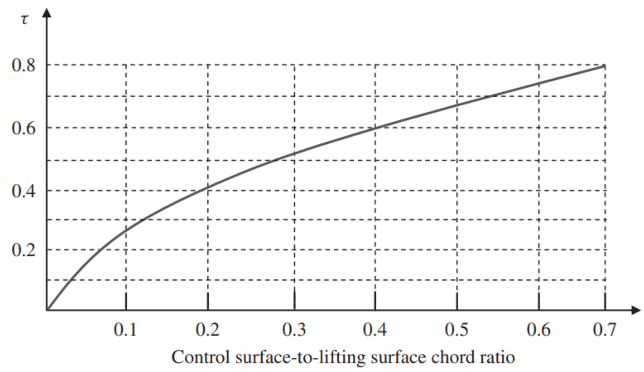

Figure 3.6: Control surface-to-lifting surface effectiveness parameter

Aileron Design

The aileron design was carried out by first defining the roll-control requirement for the critical flight phase, then fixing the aileron location along the span by considering the flap position, wing rear spar, and wing tip clearance. After that, the aileron chord ratio was selected and the effectiveness factor was obtained, before checking the maximum deflection limit, typically about ±25°, to avoid stall and loss of roll effectiveness. The main calculations were then based on the aileron rolling-moment derivative, the rolling-moment coefficient, the rolling moment, the steady-state roll rate, the bank angle reached at steady state, and the transient roll-rate/time-to-bank relations used to compare the predicted roll performance with the design requirement.

The key equations used in the procedure were the relations for from the aileron geometry and, the maximum-deflection rolling-moment coefficient, the aircraft rolling moment, the steady-state roll rate, the corresponding bank angle and the unsteady roll-rate equations until the required roll angle was achieved.

Partial derivative of coefficient of roll rate vs Aileron deflection:

$$C_{l_{\delta A}} = \frac{2C_{L_{\alpha w}}\tau C_r}{Sb}\left[\frac{y^2}{2} + \frac{2}{3}\left(\frac{\lambda - 1}{b}\right)y^3\right]_{y_i}^{y_o} \tag{51}$$

Rolling moment by incremental lift force generated by Ailerons:

$$L_A = 2\Delta L \cdot y_A \tag{52}$$

Steady-state roll rate:

$$P_{ss} = \sqrt{\frac{2 \cdot L_A}{\rho(S_w + S_h + S_{vt})C_{D_R} \cdot y_D^3}} \tag{53}$$

Bank angle when the roll rate become steady-state:

$$\phi_1 = \frac{I_{xx}}{\rho y_D^3 (S_w + S_h + S_{vt})C_{D_R}} \ln(P_{ss}^2) \tag{54}$$

Rate of roll rate:

$$\dot{\mathrm{P}} = \frac{\mathrm{P}_{\mathrm{ss}}^2}{2\Phi_1} \tag{55}$$

Elevator Design

The elevator design was carried out by following the systematic procedure given in [2], starting from the design requirements and then checking take-off rotation and trim capability against the aircraft geometry and flight conditions. The main idea was to select an elevator span and maximum deflection first, then evaluate the wing/fuselage lift, drag, and pitching moments during take-off rotation. After that, the required aircraft acceleration, desired horizontal tail lift, and tail lift coefficient were calculated for the most forward centre of gravity, since this is the most critical case. The elevator angle-of-attack effectiveness was then approximated, and the elevator-to-tail chord ratio was selected. The equations involved in calculations can be listed as follows.

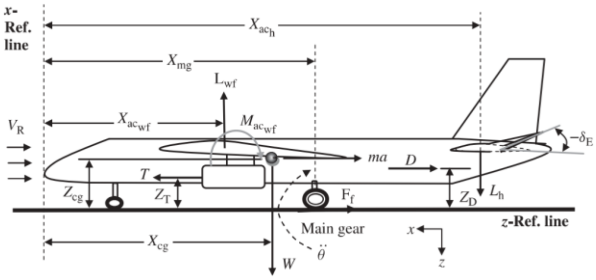

Figure 3.7: Distances to forces acting on aircraft from the nose

Horizontal tail lift during take-off rotation:

$$L_h = \frac{\left[L_{wf}\left(x_{mg} - x_{acwf}\right) + M_{acwf} + ma\left(z_{cg} - z_{mg}\right) - W\left(x_{mg} - x_{cg}\right) + D\left(z_D - z_{mg}\right) - T\left(z_T - z_{mg}\right) - I_{yy_{mg}}\ddot{\theta}\right]}{x_{ach} - x_{mg}} \quad (56)$$

Tail lift coefficient:

$$C_{L_h} = \frac{2L_h}{\rho V_R^2 S_h} \quad (57)$$

Tail angle of attack:

$$\alpha_h = \alpha + i_h - \varepsilon \quad (58)$$

Partial derivative of pitching moment coefficient with elevator deflection:

$$C_{m_{\delta E}} = -\,C_{L_{\alpha h}}\,\eta_h\,V_H\,\frac{b_E}{b_h}\,\tau_e \quad (59)$$

Partial derivative of lift coefficient with elevator deflection:

$$C_{L_{\delta E}} = C_{L_{\alpha h}}\,\eta_h\,\frac{S_h}{S}\,\frac{b_E}{b_h}\,\tau_e \quad (60)$$

Partial derivative of lift coefficient of horizontal stabilizer with elevator deflection:

$$C_{L_{h_{\delta E}}} = C_{L_{\alpha h}}\,\tau_e \quad (61)$$

Rudder Design

In accordance with the design procedures given in [2], rudder was designed to mainly satisfy cross-wind landing requirements. Initially, the maximum allowable cross-wind speed and the aircraft approach speed were determined according to regulatory requirements. The total airspeed under cross-wind conditions was calculated, followed by the estimation of the aircraft projected side area and its corresponding centre location. The side force generated by the cross-wind was subsequently evaluated. Appropriate rudder span and chord ratios relative to the vertical tail were selected, and the aircraft sideslip characteristics and control derivatives were calculated. Finally, rudder effectiveness and required deflection angles were assessed, and redesign considerations were introduced whenever the obtained values exceeded acceptable limits.

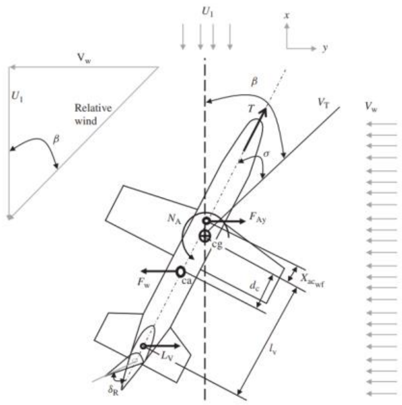

Figure 3.8: Crosswind effect on an aircraft

Main set of equations used in calculation are as follows.

Total Speed:

$$V_T = \sqrt{U_1^2 + V_W^2} \tag{62}$$

The force generated by the crosswind:

$$F_W = \frac{1}{2}\rho V_W^2 S_S C_{Dy} \tag{63}$$

σ and δ_R equation:

$$\frac{1}{2}\rho {V_T}^2 S\mathrm{b}\left[C_{n_o} + C_{n_\beta}(\beta - \sigma) + C_{n_{\delta_R}}\delta_R\right] + F_W d_c \cos\sigma = 0 \tag{64}$$

$$\frac{1}{2}\rho {V_W}^2 S_S C_{D_y} - \frac{1}{2}\rho {V_T}^2 S\left[C_{y_o} + C_{y_\beta}(\beta - \sigma) + C_{y_{\delta_R}}\delta_R\right] = 0 \tag{65}$$

$$\mathrm{C_{n_\beta}} = \mathrm{K_{f1}} \mathrm{C_{L_{\alpha_V}}}\left(1 - \frac{\mathrm{d}\sigma}{\mathrm{d}\beta}\right)\eta_\mathrm{V}\frac{\mathrm{l_{Vt}S_V}}{\mathrm{bS}} \tag{66}$$

$$C_{y_\beta} = \frac{\partial C_y}{\partial \beta} \approx C_{y_{\beta_V}} = -K_{f2}C_{L_{\alpha_V}}\left(1 - \frac{d\sigma}{d\beta}\right)\eta_V\frac{S_V}{S} \tag{67}$$

$$C_{y_{\delta_R}} = \frac{\partial C_y}{\partial \delta_R} = C_{L_{\alpha_V}}\eta_V\tau_R\frac{b_R}{b_V}\frac{S_V}{S} \tag{68}$$

Workflow of each control surface design is included in Appendix A.

3.1.2.12 Dynamic Stability Evaluation

The framework programmatically generates AVL input files, UAV geometry and operating condition (Refer Appendix D for the process within AVL). AVL is then executed as a subprocess within Python, with outputs parsed to extract:

(i) aerodynamic stability and control derivatives;

(ii) eigenvalues of the longitudinal and lateral-directional dynamic modes;

(iii) state-space system matrices [A] and [B].

$$\dot{x} = Ax + Bu \tag{69}$$

Where x – state vector, u- input vector.

The characteristics of both longitudinal and lateral-directional modes are calculated using the parsed eigenvalues. The exact eigenvalue for each mode is identified as below,

- If eigenvalue is real;
  - Most negative eigenvalue - Roll subsidence mode
  - Least negative eigenvalue - Spiral mode
- If eigenvalue is complex;
  - Most negative real value - Short period mode
  - Low negative real value with high complex value – Dutch roll mode
  - Low negative real value with low complex value – Phugoid mode

Table 3.1: Dynamic mode characteristics

Longitudinal Dynamics	Short period	Damping ratio ζ_{sp}
		Natural frequency ω_{sp}
	Phugoid	Damping ratio ζ_{ph}
		Natural frequency ω_{ph}
Lateral-directional Dynamics	Dutch roll	Damping ratio ζ_{d}
		Natural frequency ω_{d}
	Roll subsidence	Time constant τ_R
	Spiral	Time constant τ_s

After recognizing characteristics of each mode, the modal behaviours are calculated by eigen decomposition method and plotted for visualization of the modal response. Mathematically, the unforced linear system response to an initial condition x(0) is given by:

$$x(t) = V\, e^{\Lambda t}\, V^{-1}\, x(0) \tag{70}$$

Where:

- V is the matrix of eigenvectors.
- Λ is the diagonal matrix of eigenvalues.
- c = $V^{-1}\, x(0)$ represents the "modal participation factors" (how much the initial condition excites each specific mode).

3.1.2.13 Flying Quality Evaluation

The dynamic mode characteristics are then evaluated against the criteria specified in MIL-HDBK-1797 and MIL-F-8785C. UAVs falls into Class I aircraft classification. And Category B selected as surveillance is the primary goal of the design. Each mode is classified into Level 1, 2, or 3 flying qualities, providing a quantitative compliance assessment.

Table 3.2: Flying qualities for Class I, Category B aircraft

Flight Mode	Level 1 Criteria	Level 2 Criteria	Level 3 Criteria
Short Period	$0.30 < \zeta_{sp} < 2.00$	$0.20 < \zeta_{sp} < 2.00$	$0.10 < \zeta_{sp}$
Phugoid	$\zeta_{ph} > 0.04$	$\zeta_{ph} > 0$	-
Spiral	$T_{c_s} > 28.9$	$T_{c_s} > 11.5$	$T_{c_s} > 7.2$
Roll Subsidence	$T_{c_r} < 1.4$	$T_{c_r} < 3.0$	$6 < T_{c_r} < 10$
Dutch Roll	$\zeta_{dr} > 0.08$, $\zeta_{dr}\omega_{dr} > 0.15$, $\omega_{dr} > 0.5$	$\zeta_{dr} > 0.02$, $\zeta_{dr}\omega_{dr} > 0.05$, $\omega_{dr} > 0.5$	$\zeta_{dr} > 0$, $\omega_{dr} > 0.4$

[4], [5]

Since current flying quality standards are not suitable for an UAV, the trend of these flying quality standards is considered. Aircraft Classes for UAVs must account for requirements that will vary depending on expendability of the airframe and payload, mission performance objectives, potential for damage to people or property, and airspace integration. Without humans onboard, higher loss rates than piloted aircraft are acceptable.

3.1.2.14 Stability Augmentation

If the user is opting to modify the dynamic mode behaviour of the UAV, providing required data for Stability Augmentation System (SAS) Implementation is another key module regarding the framework. Stability Augmentation System is basically capable of improving the dynamic behaviour of the UAV to achieve better flying qualities [5] [25]. In most flight controllers, inbuilt SAS features are integrated to their control loops so that it automatically tunes the system during the flight to achieve better dynamic behaviour. However, there's no direct forward approach to tune the system to achieve required flying qualities for a certain mission.

The objective of this module is developing a SAS loop to tune for required flying qualities. In this application it is designed to mimic the cascaded control architecture in Arduplane Pitch damping tuning structure [39] so that it results could be verified and directly used with flight controller. However, the control structure can be manually added for other open source flight controllers in case if wanted.

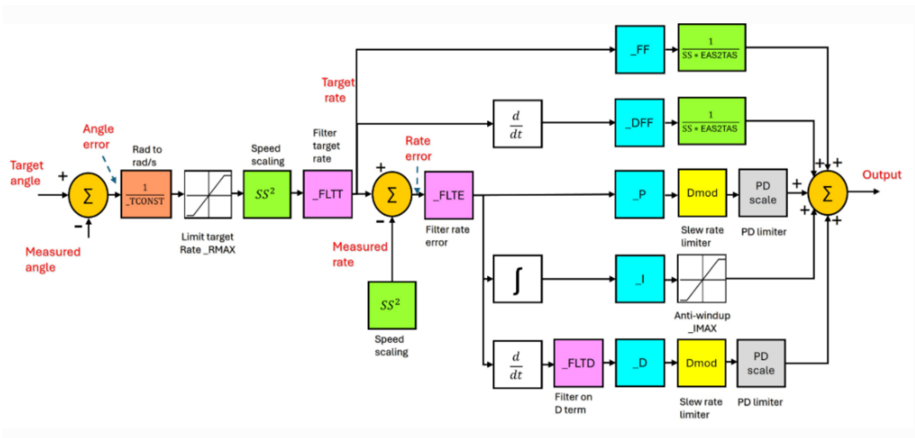

Figure 3.9 : ArduPlane pitch damper tuning structure

SAS is based on use of PID controllers for the tuning process. Control law for PID controller is as follows. [5] [25]

$$u(t) = K_p\, e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \frac{de(t)}{dt} \tag{71}$$

$$U(s) = \left(K_p + \frac{K_i}{s} + K_d s\right) E(s) \tag{72}$$

Theoretical tuning of PID controller is done by implementing the control loop in Python with control library and some other libraries.

SAS systems are designed separately for both longitudinal dynamic behaviour and lateral dynamic behaviour separately.

01. Pitch damper

In development of the Pitch damper, the theoretical transfer function for the pitch rate vs elevator input is used to model aircraft dynamics [5] [25].

$$\frac{q(s)}{\eta(s)} \equiv \frac{N_{\eta}^{q}(s)}{\Delta(s)} = \frac{k_q s\left(s+\left(\frac{1}{T_{\theta_1}}\right)\right)\left(s+\left(\frac{1}{T_{\theta_2}}\right)\right)}{\left(s^2+2\zeta_p\omega_p s+\omega_p^2\right)\left(s^2+2\zeta_s\omega_s s+\omega_s^2\right)} \tag{73}$$

Above transfer function is fed to the framework through dynamic analysis module of framework. Here, a cascaded structure which is commonly used in Ardupilot flight controllers is mimicked with a few assumptions.

- ❖ FF gain is assumed as given in [39].
- ❖ DFF gain is set to be 0.
- ❖ Speed Scalars and EAS2TAS factors are assumed to be 1 as the model is used for low altitude applications
- ❖ FLTT, FLTE filters are modelled as low-pass filters with adjustable frequency.

$$\mathrm{F(s)} = \frac{\omega_f}{s+\omega_f} \tag{74}$$

Control loop for the pitch damper is as *Figure 3.10*.

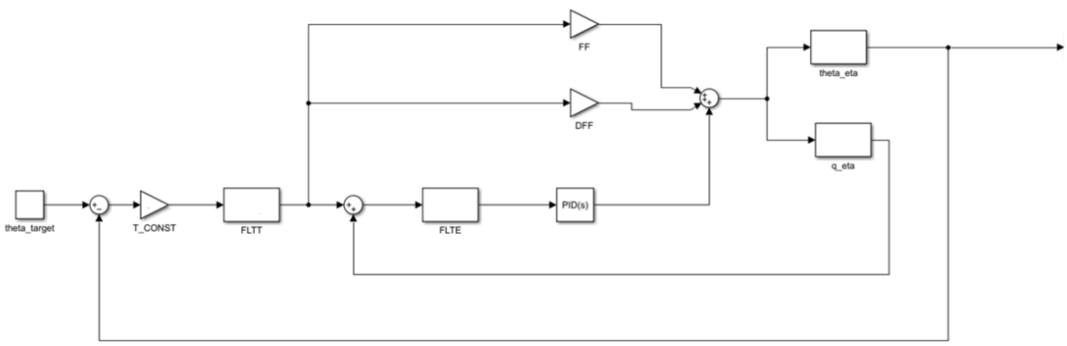

Figure 3.10:Cascaded Control structure for the pitch damper

Simplification of closed loop transfer function for the cascaded loop is attached under development of framework in APPENDIX A: Flowcharts of the Framework.

02. Roll/Yaw damper

In development of the roll damper, the theoretical transfer function for the roll rate vs aileron input is used to model aircraft dynamics [5] [25].

$$\frac{p(s)}{\xi(s)} \equiv \frac{N_{\xi}^{p}(s)}{\Delta(s)} = \frac{k_p s\left(s^2+2\zeta_\phi\omega_\phi s+\omega_\phi^2\right)}{\left(s+\left(\frac{1}{T_s}\right)\right)\left(s+\left(\frac{1}{T_r}\right)\right)\left(s^2+2\zeta_d\omega_d s+\omega_d^2\right)} \tag{75}$$

Control loop for the roll damper is as *Figure 3.11*.

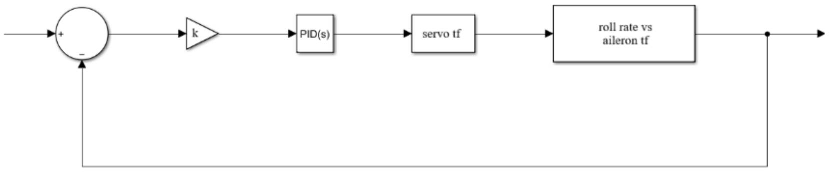

Figure 3.11: Control block diagram for the roll damper

Here, the control loop chosen for the Yaw damper is a preliminary structure. Tuning of Roll/Yaw damper cannot be done using the cascaded structure, because they are coupled dynamics. Problem still can be addressed with a precise state-space model which is out of the scope of the project and can be implemented as a future work.

In both situations a constant is set to be -1 to correct inherent negative response of aircraft transfer functions over control surface deflections.

Tuning Algorithm

The aircraft is represented in the Laplace domain using transfer functions specific for behaviour of each state variable under different system inputs. For longitudinal motion, the full transfer function preserves both the short-period and phugoid characteristics. For lateral-directional motion, the full model retains roll, Dutch roll, spiral mode and the integrator pole at the origin. By using the complete transfer function instead of a reduced-order approximation, the program evaluates the actual coupled closed-loop behaviour more realistically.

The tuning method is based on a cost function that penalizes deviation from the target modal values while also enforcing stability constraints. For example, cost function for longitudinal dynamics based on SPPO (Short Period Pitching Oscillation) dynamics is taken as follows.

$$cost(Kp, Ki, Kd) = \begin{cases} w_1\left[\dfrac{\zeta_{sp_{current}} - \zeta_{sp_{target}}}{\zeta_{sp_{target}}}\right]^2 + w_2\left[\dfrac{\omega_{sp_{current}} - \omega_{sp_{target}}}{\omega_{sp_{target}}}\right]^2 ;\ if\ stable\ Phugoid \\ w_1\left[\dfrac{\zeta_{sp_{current}} - \zeta_{sp_{target}}}{\zeta_{sp_{target}}}\right]^2 + w_2\left[\dfrac{\omega_{sp_{current}} - \omega_{sp_{target}}}{\omega_{sp_{target}}}\right]^2 + w_3\left[\dfrac{T_{2_{current}} - T_{2_{target}}}{T_{2_{target}}}\right]^2 ;\ if\ unstable\ Phugoid \end{cases} \quad (76)$$

Here, w_i represents the weights given to each flying quality in the equation which are set to be constant considering the significance of each quality. Here, subscript *sp* represents short period mode and T2 is considered for the Phugoid mode.

Here, the above cost function is set to evaluate only when the mode flying qualities lies outside the required boundaries of flying qualities. For example, error for Short period damping ratio is introduced into the equation only when it is deviated from $0.3< x <2.0$ range considered for level 1 flying qualities. Normal acceleration consideration wasn't introduced to the cost function to reduce the complexity. It is expected that won't matter for the UAV as there is no any human pilots inside it.

So, in the tuning algorithm, cost function will be continuously evaluated for the same control system with acceptable range of gain values set to different combinations within considered boundaries and the combination of constants when the cost function is minimum will be calculated.

This makes the optimization both performance-oriented and physically meaningful. The implementation uses the Python 'control' library for transfer-function manipulation, feedback connection and pole analysis, while 'SciPy' is used for global and local optimization. These are established research-grade libraries, widely used in academic control studies, which makes the workflow reliable, reproducible and suitable for use. Flow chart of tuning program and other Flow charts regarding algorithm are attached in APPENDIX A: Flowcharts of the Framework.

3.1.3 Framework Workflow and Iteration Logic

The overall workflow of the framework can be sectioned into 3 phases. In the first phase, the UAV sizing and geometric definition is established. It spans from mission requirements to landing gear design. The second phase is based on performance verification. It ensures that the aerodynamic configuration is consistent with mission requirements. At the last phase, which includes the control surface sizing, dynamic stability evaluation and flying quality assessment determines that the configuration achieves the required dynamic behaviour.

3.2 Geometry and Inertia Measurement - Experimental Setup

3.2.1 Measurement Objectives and Requirements

The development of a flight dynamic modeling framework for a fixed-wing UAV, and the reliability of the results obtained from it, heavily relies on the accuracy of the physical input parameters used to represent the aircraft. In order to validate the developed framework, the predictions given by the model must be verified by real flight data obtained through experimental testing. For this purpose, the UAV designed and developed here is utilized in this study. Consequently, it is essential to determine the physical parameters of this UAV through experimental methods to ensure that the model accurately reflects the actual system.

Among these parameters, both the geometric characteristics and the mass moments of inertia play a crucial role in determining the aerodynamic behavior and dynamic response of the vehicle. Geometric parameters such as wingspan, chord distribution, control surface dimensions, and their relative positioning directly affect the estimation of aerodynamic coefficients. In simple manner, inertia properties govern the aircraft's resistance to rotational motion about its principal axes and significantly affect its stability and control characteristics. Any inaccuracies in these inputs can propagate through the modeling framework, leading to noticeable discrepancies between predicted and actual flight behavior.

Even though initial input parameter values are obtained from design models, CAD configurations, or analytical approximations in order to develop the model, those methods may not capture real world variations due to manufacturing tolerances, assembly misalignments, and material inconsistencies. Small geometric deviations may affect aerodynamic characteristics considerably, while non-uniform mass distribution can lead to errors in inertia estimations. Therefore, relying solely on theoretical or design-based values may limit the reliability and predictive capability of the developed framework.

To overcome these limitations, experimental validation becomes an essential step. Geometry verification ensures that the actual UAV dimensions and configurations used in testing align with those assumed in the model, thereby improving the accuracy of aerodynamic predictions. Likewise, experimental determination of mass moments of inertia, typically through pendulum-based methods, enables the capture of true inertial properties of the assembled UAV system with the deviations from the original design mentioned above. By integrating experimentally verified geometric data and inertia values into the framework, the overall model fidelity is significantly enhanced.

This combined validation approach reduces uncertainty and strengthens confidence in the framework's ability to predict real flight behavior. Consequently, it provides a reliable foundation for analyses, including stability assessment, control system design, and performance evaluation, ensuring that the developed model can be effectively applied in both research and practical UAV applications.

Therefore, purpose of this procedure is to determine and verify some of the key physical parameters of the selected fixed-wing UAV in an experimental manner, specifically its geometric characteristics and mass moments of inertia, to enhance the reliability of the developed flight dynamic modeling framework. This validation ensures that the inputs provided to the model accurately represent the real UAV, thereby allowing a meaningful comparison between predicted and actual flight behavior.

3.2.2 Geometry Measurement Procedure and Instrumentation

Accurate geometric representation of the UAV is essential for reliable aerodynamic modelling and validation of the flight dynamic framework. Therefore, a systematic measurement procedure was implemented to obtain the actual dimensions and spatial configuration of the UAV. The objective of this process was to capture the true geometry of the assembled UAV and minimize mismatches between the physical model and the parameters used in analysis.

3.2.2.1 Instrumentation

A combination of precision measuring tools and a custom designed and fabricated measurement setup was used to achieve both accuracy and repeatability. Conventional measuring instruments such as steel measuring tapes and Vernier callipers were used for direct dimensional measurements, providing sufficient resolution for both large-scale and small-scale features. However, due to the complexity of the UAV geometry and the need to maintain alignment consistency across multiple measurements, a dedicated 3D measurement frame constructed using aluminium extrusion profiles was utilized as the primary reference system.

This frame, consisting of rigid 2040 aluminium extrusions, was designed to establish a stable planar reference grid around the UAV. Laser modules were mounted on the frame to project straight reference lines, enabling precise alignment of the UAV axes and facilitating coordinate-based measurements.

Figure 3.12 The measurement system designed with planner laser modules for each direction in space

The core concept is based on the use of three mutually intersecting planar laser modules at once, each representing a reference plane aligned with one principal coordinate directions. These planes are oriented to correspond with the longitudinal, lateral, and vertical axes of the defined measurement frame. When projected, each laser module generates a thin plane of light that extends across the measurement region, effectively acting as a visual and geometric reference surface.

The intersection of any two planes forms a line, while the intersection of three independent planes defines a unique point in three-dimensional space. The laser pointers can move along the frame and from the readings on the millimetre scale on the frame, the location of the laser pointer can be identified which will be used as the relevant coordinate in the direction. This principle is utilized to determine the spatial coordinates of specific points on the UAV.

Additional alignment tools such as spirit levels and plumb lines were used to verify horizontal and vertical orientation, respectively. This integrated instrumentation approach ensured that both global dimensions and local geometric features could be captured with a high degree of accuracy.

3.2.2.2 Measuring Procedure

1. **Calibration of the instrument**

Prior to conducting any measurements, the vertical and horizontal planar laser modules

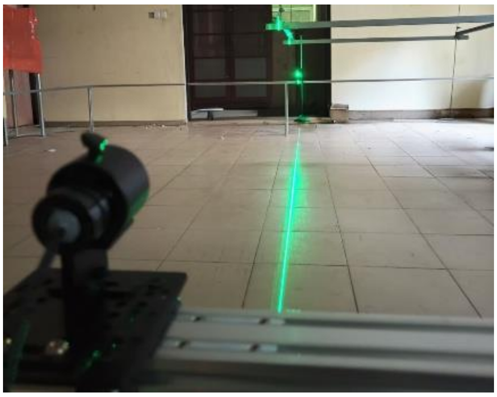

Figure 3.13: Calibrating the laser modules

were carefully calibrated to establish a reliable reference coordinate system. Calibration was performed using distant vertical and horizontal reference lines in order to minimize angular alignment errors. Increasing the calibration distance improves the sensitivity and accuracy of alignment, since even small angular deviations become more noticeable over longer distances.

The tiled floor pattern available in the laboratory environment was used as the primary floor coordinate reference system. The straight grout lines between tiles provided convenient horizontal and longitudinal reference axes for laser alignment and positioning.

To establish the UAV longitudinal reference axis, the two vertical planar laser modules positioned on either side of the measurement frame were aligned such that their projected laser planes overlapped precisely. The intersection of these aligned laser planes produced a single straight reference line along the floor. The nose and tail sections of the UAV were then aligned with this reference line, ensuring that the aircraft longitudinal axis remained parallel to the measurement frame.

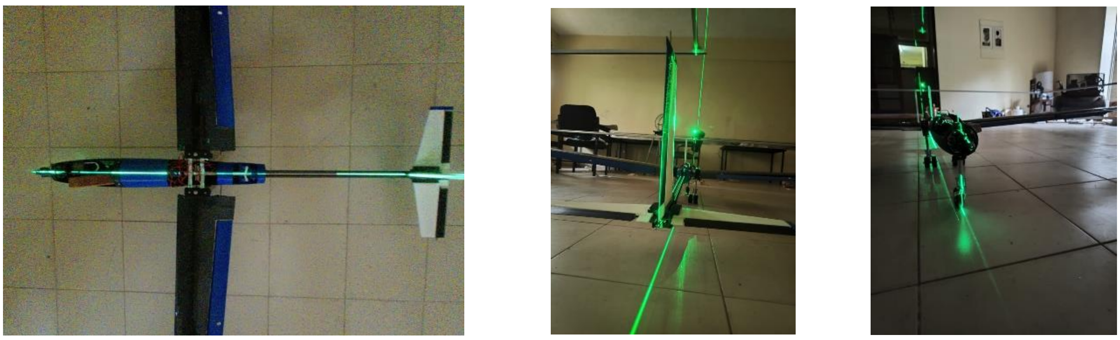

Figure 3.14: Aligning the UAV with the instrument frame

This alignment procedure provided a consistent geometric reference for all subsequent measurements. In addition, it enabled the identification of geometric asymmetries, manufacturing deviations, or structural defects relative to a known baseline reference. Without such a reference axis, obtaining repeatable and reliable measurements would be difficult due to possible aircraft misalignment within the measurement setup.

2. **Measuring Horizontal Distances**

Horizontal distance measurements were conducted using the vertical planar laser modules. Since the projected laser beam possesses a finite thickness, one specific edge of the laser line was selected as the measurement reference edge throughout the entire measurement process. Maintaining a consistent reference edge minimized errors associated with beam thickness variation.

To obtain a measurement, the selected reference edge of the vertical laser beam was first aligned with the starting point of the feature to be measured. The corresponding position on the horizontal scale was then recorded. Subsequently, the laser module was carefully translated along the measurement direction until the same reference edge aligned with the ending point of the feature. The final scale reading was recorded.

The horizontal distance between the two points was determined by calculating the difference between the initial and final scale readings.

Figure 3.15: Measuring Wingspan-

3. **Measuring Vertical Distances**

Vertical distance measurements followed the same methodology used for horizontal measurements. However, in this case, the horizontal planar laser modules were utilized instead of the vertical lasers.

The reference edge of the horizontal laser beam was aligned with the initial measurement point, and the corresponding vertical scale reading was recorded. The laser module was then translated vertically until the same beam edge aligned with the second measurement point. The difference between the two recorded readings provided the required vertical distance.

4. Measuring Angles

Angular measurements were determined using geometric relationships and basic trigonometric principles. Initially, two distinct points located along the inclined feature were selected. The horizontal and vertical distances between these two points were then measured using the procedures previously described. Once these distances were obtained, the angle of inclination was calculated using trigonometric relationships.

For example, if the vertical distance between the two points is represented by $\Delta y$and the horizontal distance is represented by Δx, the angle θcan be determined using:

$$\theta = \tan^{-1}\left(\frac{\Delta y}{\Delta x}\right) \tag{77}$$

This approach enables accurate measurement of wing incidence angles, dihedral angles, sweep angles, and other inclined geometric features of the UAV.

3.2.2.3 Target Geometric Parameters for Measurement

To accurately define the UAV geometry for integration into the flight dynamic modeling framework, a set of critical geometric parameters was identified for experimental measurement. These parameters were selected based on their direct influence on aerodynamic behavior, control effectiveness, and overall aircraft configuration.

The target parameters to be measured are categorized as in Table :

Table 3.3: Target parameters of the UAV to measure

Wing Geometry	• Wingspan • Root chord • Tip chord • Wing incidence angle • Dihedral angle
Flap Geometry	• Flap span • Flap chord

Horizontal Tail (H-tail) Geometry	• Distance between wing and horizontal tail (tail moment arm) • Horizontal tail span • Horizontal Tail Root Chord • Horizontal tail tip chord • Horizontal tail incidence angle • Horizontal tail sweep angle
Vertical Tail (V-tail) Geometry	• Distance between wing and vertical tail • Vertical tail span • Vertical Tail Root Chord • Vertical tail tip chord • Vertical tail sweep angle
Control Surfaces	• Elevator span • Elevator chord • Rudder span • Rudder chord

3.2.3 Moment of Inertia Measurement Technique

The determination of the mass moments of inertia of the UAV was carried out using the pendulum oscillation method, which is a well-established experimental technique for large and irregular bodies such as aircraft. This method is based on relating the period of oscillation of a suspended body to its moment of inertia about a defined axis.

Fundamental Principle

For a pendulum undergoing small-amplitude oscillations, the motion can be described by [26]:

$$I\frac{d^2\theta}{dt^2} + b\theta = 0 \tag{78}$$

where:

I= moment of inertia about the axis of oscillation

b= constant depending on pendulum configuration

θ= angular displacement

The resulting period of oscillation is:

$$T = 2\pi\sqrt{\frac{I}{b}} \tag{79}$$

Rearranging:

$$I = \frac{bT^2}{4\pi^2} \tag{80}$$

This equation forms the basis for determining the moment of inertia from experimentally measured oscillation periods.

Two different pendulum configurations were employed to determine the inertia about the principal axes of the UAV:

3.2.3.1 Bifilar Torsion Pendulum (Z-axis)

For rotation about the vertical axis, a bifilar suspension system was used, where the UAV is suspended using two parallel filaments.

The moment of inertia is given by:

$$I = \frac{WT^2A^2}{16\pi^2 l} \tag{81}$$

where:

W= weight of the system

A= distance between filaments

l= length of filaments

T= oscillation period

This configuration ensures that the axis of rotation passes through the center of gravity.

Figure 3.16: The bifilar torsional pendulum setup

3.2.3.2 Compound Pendulum (X and Y axes)

For rotation about horizontal axes, the UAV was mounted on a compound pendulum setup, where oscillation occurs about a fixed axis offset from the center of gravity.

The moment of inertia about the center of gravity is obtained using:

$$I = \frac{WT^2L}{4\pi^2} - ML^2 \tag{82}$$

where:

L= distance between center of gravity and axis of oscillation

M= mass of the system

This formulation accounts for the shift between the oscillation axis and the centre of gravity.

Figure 3.17: Compound pendulum setup

3.2.3.3 Effect of Ambient Air

In practical conditions, the UAV oscillates in air rather than in a vacuum. Therefore, the measured inertia corresponds to a virtual moment of inertia**,** which includes additional effects due to the surrounding medium [26].

The measured inertia can be expressed as:

$$I_v = I_s + I_e + I_A \tag{83}$$

where:

I_s= moment of inertia of the structure

I_e= inertia of entrapped air

I_A= additional inertia due to surrounding air

The additional inertia arises from the momentum imparted to the surrounding air during motion, commonly referred to as the added mass effect**.**

3.2.3.4 Determination of True Moment of Inertia

The required true moment of inertia of the UAV is obtained by removing the additional inertia effects:

$$I = I_v - I_A \tag{84}$$

The additional inertia $I_A$is evaluated based on geometric approximations of the UAV components and experimentally derived coefficients, considering contributions from major components such as wings, fuselage, and tail surfaces.

3.2.3.5 Use of Multiple Pendulum Lengths

To improve accuracy and eliminate unknown parameters such as additional mass effects, the UAV was oscillated using two different pendulum lengths. This results in two simultaneous equations:

$$I_v = \frac{WT_1^2 L_1}{4\pi^2} - (V\rho + M_A)L_1^2 \tag{85}$$

$$I_v = \frac{WT_2^2 L_2}{4\pi^2} - (V\rho + M_A)L_2^2 \tag{86}$$

Solving these equations allows direct determination of the virtual moment of inertia and correction terms.

3.2.3.6 UAV Inertia Measurement Procedure

1. The UAV was prepared by securing all loose components and maintaining the aircraft in the intended flight configuration. The total mass of the UAV was measured accurately.
2. The centre of gravity (CG) of the UAV was determined prior to the swinging tests, since the inertia values were required about the CG axes.
3. A rigid suspension frame was assembled to ensure oscillation about a single well-defined axis only. Flexible support was avoided to minimize measurement errors.
4. The moments of inertia about the $X$and Ybody axes were determined using the compound pendulum method.

- The UAV was suspended such that the oscillation axis was parallel to the required body axis.

- The pendulum length L, defined as the distance between the suspension axis and the CG, was measured.
- A small angular displacement of less than approximately 2°was applied.
- The UAV was released carefully without applying any additional force.
- The time required for 50 or more oscillations was recorded.
- The oscillation period $T$was calculated.
- The procedure was repeated using a second pendulum length for verification purposes.

The moment of inertia was calculated using:

$$I = \frac{WT^2L}{4\pi^2} - ML^2 \tag{87}$$

5. The moment of inertia about the Zbody axis was determined using the bifilar pendulum method.

- The UAV was suspended using two vertical strings positioned at equal spacing.
- The string length $l$and spacing between the strings Awere measured.
- A small angular displacement about the vertical axis was applied.
- The UAV was released carefully.
- The time required for 50 or more oscillations was recorded.
- The oscillation period $T$was determined.

The moment of inertia was calculated using:

$$I_{zz} = \frac{WT^2A^2}{16\pi^2 l} \tag{88}$$

6. Multiple trials were conducted for each axis, and the measured values were averaged to improve accuracy. Trials exhibiting wobbling, coupled oscillations, or irregular motion were discarded.
7. During the experiments, special care was taken to:

- measure the actual pendulum length from the suspension axis to the CG,
- maintain small oscillation amplitudes,
- avoid external disturbances during release,
- ensure oscillation about only one axis at a time, and
- maintain consistent SI (Système International: International System of Units) units throughout the calculations.

8. The final moments of inertia about the X, Y, and Zbody axes were calculated and tabulated.

3.2.4 Uncertainty and Error Analysis Approach

Experimental measurements are associated with unavoidable uncertainties arising from instrument resolution, alignment errors, environmental effects, and human observation. Therefore, an uncertainty analysis was carried out for both geometric and inertia measurements to quantify the reliability of the experimentally obtained parameters.

For all direct linear measurements, the uncertainty was taken as the least count of the corresponding measuring scale or instrument. Since geometric dimensions were determined using differences between two measured positions, the uncertainty of a measured distance was calculated using standard uncertainty propagation principles.

For a measured distance:

$$q = x - y \tag{89}$$

The uncertainty of the distance measurement becomes:

$$\delta_q = \sqrt{(\delta_x)^2 + (\delta_y)^2} \tag{90}$$

where:

δ_x= uncertainty of the initial reading
δ_y= uncertainty of the final reading

Since both readings were obtained using the same measurement scale, the uncertainty was primarily governed by the least amount of the scale together with the finite thickness of the laser beam. Maintaining a consistent reference edge of the laser line throughout all measurements minimizes systematic uncertainty caused by beam thickness variation.

The fractional uncertainty of each measured dimension was evaluated using:

$$\frac{\delta_q}{q}$$

where $q$represents the experimentally measured geometric parameter.

For angular measurements such as wing incidence angle, sweep angle, and dihedral angle, the angle was calculated from measured horizontal and vertical distances using trigonometric relationships. Therefore, the uncertainty of the calculated angle depends on the uncertainties associated with both Δx and Δy. Since the angle was obtained from multiple measured quantities, uncertainty propagation was applied considering the uncertainties of the measured distances.

For inertia measurements, uncertainties mainly originated from oscillation period measurements, pendulum length measurements, filament spacing measurements, center of gravity location estimation, and damping effects caused by surrounding air. Since the inertia equations involve multiplication, division, and squared terms, fractional uncertainty propagation methods were used.

For the bifilar pendulum equation:

$$I = \frac{WT^2A^2}{16\pi^2 l} \tag{91}$$

the fractional uncertainty of the calculated inertia was determined using:

$$\frac{\delta_I}{I} = \sqrt{\left(2\frac{\delta_T}{T}\right)^2 + \left(2\frac{\delta_A}{A}\right)^2 + \left(\frac{\delta_l}{l}\right)^2} \tag{92}$$

where:

δ_T= uncertainty in oscillation period measurement
δ_A= uncertainty in filament spacing measurement
δ_l= uncertainty in filament length measurement

Similarly, for the compound pendulum method:

$$I = \frac{WT^2L}{4\pi^2} - ML^2 \tag{93}$$

The uncertainty was evaluated by propagating the uncertainties associated with oscillation period T, pendulum length L, and measured mass M.

To minimize random errors in inertia measurements, multiple oscillation cycles were recorded and averaged when determining the oscillation period. In addition, measurements were repeated several times, and the average values were used for final calculations. The use of two different pendulum lengths further reduced uncertainty associated with additional air inertia effects and unknown correction parameters.

3.3 Sensitivity Analysis of Dynamic Mode Characteristics

3.3.1 Purpose of Sensitivity Analysis

The aircraft dynamic modes describe how an aircraft responds to a disturbance from its steady flight condition. These disturbances can be due to wind gusts, pilot input, changes in throttle, or small changes in attitude and speed. For a conventional fixed-wing aircraft, the main dynamic modes can be divided into two longitudinal modes and three lateral-directional modes as described in Table 3.4: Aircraft dynamic modes [4] [5].

Table 3.4: Aircraft dynamic modes

Longitudinal Modes	Short period
	Phugoid
Lateral-Directional Modes	Dutch roll
	Roll subsidence
	Spiral

These modes determine whether the aircraft returns to its stabilized condition, oscillates for a period or diverges from its original condition after a disturbance. The main characteristics can be described by frequency and damping ratio for oscillatory modes and time constant for non-oscillatory modes.

Flying qualities depend strongly on how the UAV responds to a disturbance. If the dynamic modes are well damped, the UAV can recover smoothly without excessive oscillation. If the damping is low, the aircraft may continue to oscillate, making it difficult to control. If a mode is unstable, the motion may grow with time and reduce flight safety. Therefore, analyzing dynamic modes is essential when designing a UAV, because it helps determine whether the aircraft will be easy to fly, stable during manoeuvres, and suitable for experimental flight testing.

Aircraft stabilization can be achieved using two main methods [4] [5].

1. Active stabilization
2. Passive stabilization

Active stabilization refers to a closed loop control system in which on board sensors such as gyroscope and accelerometers continuously monitor the aircraft motion and use corrective control algorithms to actuate the control surfaces of the aircraft (Aileron, elevator and rudder) to maintain the desired flight conditions.

In passive stabilization, the aircraft does not rely only on continuous corrective control inputs. Instead, the inherent aerodynamic and geometric characteristics of the aircraft itself provides a natural stabilizing tendency. The aircraft geometry is designed in such a way that it naturally tends to return toward stable flight after a disturbance [4] [29] [30].

Small UAVs usually have limited payload capacity, limited onboard power, and limited space for complex systems. If the airframe is less stable, the flight controller must continuously apply corrective control inputs to maintain stable flight, increasing control surface actuator activity, power consumption and workload for the flight controller. Apart from that, active systems highly depend on sensors, wirings, power supply and actuators. Any error in sensor measurements, controller tuning, signal transmission or actuator response can affect the stability of the UAV making active stabilization less reliable.

Based on this requirement, this study focused on understanding how changes in aircraft geometry influence the characteristics of the major flight modes. Parameters such as wing geometry, horizontal and vertical tail volume ratio, tail arm length, center of gravity location, and aerofoil selection can significantly affect the aerodynamic stability derivatives, which then affect the aircraft eigenvalues and modal characteristics. Typically, most of these are interdependent variables. Changing one parameter can affect several others, and the combined effect of multiple parameter changes on the aircraft's modal characteristics is not always obvious. As a result, identifying which parameters or combinations have the greatest influence on stability and dynamic response can be challenging. But analyzing the reduced order models and determining the most influential parameters out of all parameters reduces the number of parameters and parameter combinations that need to be adjusted when tuning the UAV's modal characteristics. In this way, the design can be improved to achieve better passive flying qualities during the design phase. Using a trial-and-error approach for changing major geometric features such as wing characteristics and tail characteristics after the fabrication of the UAV can be difficult, expensive and time consuming. This allows evaluating several design alternatives using modelling and simulation.

3.3.2 Reduced-Order Aircraft Dynamic Models

The dynamic behaviour of a fixed wing aircraft can be described by the six degrees of freedom equations of motion. The six degrees of freedom consist of three translational motions along the x, y, and z axes, and three rotational motions about these axes, namely roll, pitch, and yaw [4] [5].

However, for sensitivity analysis, the full six-degree-of-freedom model is usually more complex than necessary. Therefore, the equations are linearized about a steady trimmed flight condition and separated into longitudinal and lateral-directional motion. From these linearized equations, reduced-order models are obtained by retaining only the dominant states that describe each flying mode.

In a full-order model, the effect of a geometric parameter may be distributed across many coupled states, making interpretation difficult. In reduced-order form, the relationship between geometry, stability derivatives, eigenvalues, damping ratio, and natural frequency becomes easier to explain. Therefore, reduced-order models were selected because they are simple, computationally efficient, and suitable for identifying dominant trends in UAV flying-mode characteristics.

As reduced order models explain, dynamic mode characteristics of an aircraft are functions of stability derivatives. Stability derivatives describe how aerodynamic forces and moments change when the aircraft experiences small changes in motion variables such as angle of attack, pitch rate, sideslip angle, roll rate, yaw rate, or control surface deflection.

These stability derivatives are extracted from the AVL simulations of the aircraft design. AVL uses Vortex Lattice Method (VLM) where it discretizes the aircraft lifting surfaces into rectangular panels. A bound vortex is placed at the quarter chord of each panel to model the lifting effect. Kutta condition is used to ensure that no flow passes through the lifting surfaces to determine the vortex strengths [29].

3.3.2.1 State Space Representation

The aircraft dynamics can be represented using a linear state-space model [4] [5] [31]:

$$\dot{x} = Ax + Bu \tag{94}$$

$$y = Cx + Du \tag{95}$$

For modal analysis, the eigenvalues of the system matrix can be calculated as,

$$\det(\lambda I - A) = 0 \tag{96}$$

For oscillatory modes, eigen values have the form of

$$\lambda = \sigma \pm j\omega_d \tag{97}$$

For non-oscillatory modes,

$$\lambda = \sigma \tag{98}$$

The natural frequency can be derived by,

$$\omega_n = \sqrt{\sigma^2 + {\omega_d}^2} \tag{99}$$

The damping ratio is,

$$\zeta = \frac{-\sigma}{\omega_d} \tag{100}$$

3.3.2.2 Short-Period Mode

Short period mode is a longitudinal oscillatory mode, mainly involving angle of attack and pitch rate. It occurs when the aircraft experiences a pitch disturbance such as an elevator input or a wind gust which changes the angle of attack. Here the aircraft angle of attack oscillates while the forward speed remains constant throughout the period as shown in the Figure 3.18: Short period oscillations

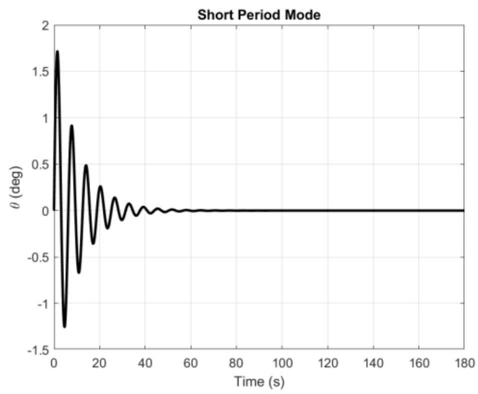

Figure 3.18: Short period oscillations

The main equations involved in this are for the frequencies and damping ratio as shown below [29].

$$\omega_{sp} = \frac{1}{2}\sqrt{-M_q^2 - 4u_0 M_w} \tag{101}$$

$$\zeta_{sp} = -\frac{\frac{1}{2}M_q}{\sqrt{-u_0 M_w}} \tag{102}$$

3.3.2.3 Phugoid Mode

Phugoid mode is a slow longitudinal oscillation involving the energy exchange between the kinetic and potential energy. In this mode, the aircraft climbs and loses velocity, then descends and gains velocity and the cycle repeats. The angle of attack remains constant and pitch angle changes in sin waveform throughout the period. This can be visualized using the Figure 3.19: Phugoid mode.

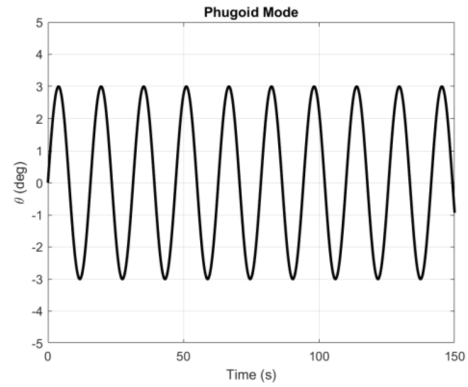

Figure 3.19: Phugoid mode

The frequency and the damping ratio can be modelled using [4] [5] [31]:

$$\omega_p = \sqrt{2}\frac{g}{u_0} \tag{103}$$

$$\zeta_p = \frac{1}{\sqrt{2}}\frac{C_D}{C_L} \tag{104}$$

3.3.2.4 Dutch Roll Mode

The Dutch roll is a lateral-directional oscillation involving coupled yawing and rolling motion, typically featuring very strong lateral stability combined with weak directional stability. When the aircraft experiences a sideslip disturbance, the aircraft yaws to one side. Then rolls in the same direction. Due to strong lateral stability, roll is corrected and overshoots the levelling point. Then the aircraft yaws to the other side and the cycle repeats. Common reduced order Dutch roll characteristics can be represented as[31] [32],

$$\omega_{dr} = \sqrt{u_0 N_v + Y_v N_r - N_v Y_r} \tag{105}$$

$$\zeta_{dr} = -\frac{Y_v + N_r}{2\sqrt{u_0 N_v + Y_v N_r - N_v Y_r}} \tag{106}$$

3.3.2.5 Roll Subsidence

Roll subsidence is a non-oscillatory lateral mode that describes how fast the roll rate of the aircraft decays after a roll disturbance or aileron input. Due to the non-oscillatory behaviour of this mode, it is represented by a single real eigenvalue which can be used to determine the time constant. Reduced order models use roll rate as the main variable to describe this motion. It can be visualized as shown in Figure 3.20: Roll subsidence mode

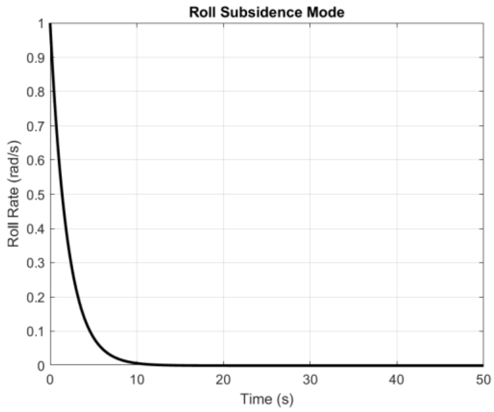

Figure 3.20: Roll subsidence mode

The time constant can be modelled as [4], [29]:

$$\tau_r = \frac{-1}{L_p} \tag{107}$$

3.3.2.6 Spiral Mode

Spiral is a slow lateral-directional mode that describes the long-term tendency of the aircraft after a roll or heading disturbance. Typically, the spiral can be weakly stable or weakly unstable. This means that the divergence can be very slow, and the pilot or the on board flight controller can usually correct it. Since this is a non-oscillatory type mode, the time constant can be modelled as below [29].

$$\tau_s = N_r - N_v \frac{L_r}{L_v} \tag{108}$$

3.3.3 Parametric Variation

Prior to the sensitivity study, baseline UAV design was created first. The baseline design was used as a realistic small fixed-wing UAV design that could be fabricated and tested for the later experimental validation of the developed framework and the dynamic characteristics. So, all parametric variations were performed with respect to this baseline UAV configuration.

As described by the reduced order models, each characteristic depends upon aircraft stability derivatives. Each stability derivative of the aircraft is driven by a set of geometric parameters in the UAV configuration. Therefore, the parameter variation process was focused on the geometric variables that were theoretically expected to have a significant effect on the longitudinal and lateral-directional modes. The main geometric parameters that affect these stability derivatives can be listed as shown in Table and Table . [5] [31] [32].

Table 3.5: Geometric parameters influencing longitudinal motion

Stability Derivative	Affecting Parameters
C_L and C_D	Main wing aerofoil
	Horizontal tail aerofoil
	Vertical tail aerofoil
	Fuselage shape
M_q	Horizontal tail aerofoil
	Horizontal tail efficiency
	Horizontal tail volume ratio
	MAC (Mean Aerodynamic Chord)
M_w	Main wing aerofoil
	Horizontal tail aerofoil
	CG location
	Aerodynamic center (AC) location
	Horizontal tail volume ratio
	Main wing aspect ratio

Table 3.6: Geometric parameters influencing lateral-directional motion

Stability Derivative	Affecting Parameters
L_p	Main wingspan
	Main wing aspect ratio
	Main wing sweep angle
	Main wing taper ratio
	Main wing dihedral angle

	Main wing aerofoil
	Vertical position of the wing
N_r and Y_r	Vertical tail volume ratio
	Vertical tail aspect ratio
	Main wing sweep angle
	Main wingspan
	Fuselage side area
N_β and Y_β	Vertical tail volume ratio
	Fuselage side area distribution
	Main wing sweep angle
	Vertical position of the wing
L_r	Main wing sweep angle
	Main wing dihedral angle
	Main wingspan
	Main wing aspect ratio
	Main wing taper ratio
	Vertical tail volume ratio
	Vertical position of the wing
L_β	Main wing dihedral angle
	Vertical position of the wing
	Main wing sweep angle
	Vertical tail volume ratio
	Main wingspan
	Main wing aspect ratio

In this parametric variation approach, one geometric parameter was selected and varied within a typical practical design range, while the remaining parameters were kept unchanged as much as possible. However, aircraft geometric parameters are not completely independent. A change in one parameter can influence several other related design quantities. For example, changing the wing aerofoil can affect lift, drag, pitching moment, and trim condition. Therefore, instead of manually changing only one parameter value in isolation, the developed framework was used to update the complete aircraft configuration in a consistent manner.

For each parameter variation, the framework recalculated the resulting feasible UAV configuration while respecting the overall design constraints. This ensured that each generated configuration remained practically meaningful and suitable for analysis. The aim was to isolate the effect of the selected parameter as much as possible, while still maintaining realistic and feasible aircraft geometry.

After each updated configuration was generated, the AVL analysis was re-executed to calculate the corresponding aerodynamic stability derivatives. These updated derivatives were then used in the reduced-order dynamic models to evaluate the dynamic mode characteristics.

This procedure was repeated for each selected geometric parameter and for each value within its practical range. By comparing the modal characteristics of the modified configurations with the baseline configuration, the effect of each parameter on the dynamic response was quantified.

3.3.4 Normalized Sensitivity Calculation

After evaluating the dynamic modal characteristics for each geometric parameter variation, the sensitivity of parameter was quantified using a normalized sensitivity. The purpose of using normalized sensitivity was to compare the influence of different geometric parameters on a common basis. Since the selected parameters have different units and magnitudes, direct comparison using ordinary gradients would not provide a fair measure of influence. For example, wing sweep angle is measured in degrees, tail volume ratio is dimensionless, and wing or tail aerofoil changes affect aerodynamic characteristics in a different way. Therefore, a normalized method was required to express the relative effect of each parameter.

In this study, the modal characteristic was represented by Y, while the geometric parameter was represented by X. The modal characteristics Y refers to the damping ratio (ζ), natural frequency (ω_n), or time constant (τ) depending on the mode that is being analyzed.

Normalized sensitivity S_{norm} can be expressed by [31],

$$S_{norm} = \frac{\Delta Y / Y_0}{\Delta X / X_0} \tag{109}$$

Where X_0, Y_0 are corresponding baseline values.

For a discrete parameter variation used in this study, this equation can be written as [31],

$$S_{norm} = \frac{(Y_i - Y_0)/Y_0}{(X_i - X_0)/X_0} \tag{110}$$

Here, X_i, Y_i represent the modified geometric parameter and modified modal characteristics respectively.

The percentage changes in the parameter and the modal characteristics can also be expressed as [31],

$$\%\Delta X = \frac{(X_i - X_0)}{X_0} \times 100 \tag{111}$$

$$\%\Delta Y = \frac{(Y_i - Y_0)}{Y_0} \times 100 \tag{112}$$

Therefore, the normalized sensitivity can also be interpreted as.

$$S_{norm} = \frac{\%\Delta Y}{\%\Delta X} \tag{113}$$

This means that the normalized sensitivity represents the percentage change in the dynamic modal characteristics caused by one percent change in the selected geometric parameter.

Since multiple discrete values were analysed for each geometric parameter, a logarithmic-gradient method was used to estimate the overall sensitivity trend. For each parameter case, the natural logarithm of the parameter value and the corresponding modal response was calculated. Then the values of ln(Y) were then plotted against ln(X). this representation was used because the gradient of this curve gives the normalized sensitivity as shown below[31],

$$S_{norm} = \frac{dln(Y)}{dln(X)} \tag{114}$$

When the relationship between ln(Y) and ln(X) was approximately linear, a first order linear regression was fitted to the discrete data points as,

$$\ln(Y) = m \times ln(X) + c$$

Where m is the gradient of the fitted curve which is equal to the normalized sensitivity of discrete data points. This method allowed all analysed discrete parameter values to be considered together instead of calculating sensitivity using only two points. Therefore, the calculated sensitivity represented the overall trend of the modal characteristics within the selected feasible parameter range.

The constant gradient method was only used when the ln(Y) and ln(X) relationship showed an approximately linear trend. If the ln(Y) and ln(X) was non liner, then a single sensitivity value is not sufficient to represent the full parameter range as the influence of the parameter could vary at different points within the design range.

Because of this reason, only the parameter ranges that produced approximately linear trends were considered for the main methodological interpretation and design fine tuning. This was done because a linear log-log relationship indicates that the modal characteristic changes in a predictable and consistent manner with respect to the selected geometric parameter.

In contrast, when the $\ln(Y)$ versus $\ln(X)$ relationship is strongly nonlinear, the influence of the geometric parameter changes across the design range. In that situation, a single sensitivity coefficient may be misleading because the parameter may have a strong effect in one region and a weak or opposite effect in another region. Such behavior is less suitable for simple design-stage fine-tuning because the response is not represented accurately by one constant gradient.

Therefore, the methodology focused mainly on linear sensitivity trends, as these provide a more reliable basis for predicting and adjusting the UAV dynamic characteristics. Parameters that showed clear linear behavior in the log-log sensitivity plots were treated as more suitable candidates for passive stability tuning. This allowed the design process to focus on geometric variables whose effects on damping ratio, natural frequency, or time constant could be estimated more confidently within the feasible design range.

3.4 UAV Development for Flight Data Extraction for Validation

3.4.1 Purpose of UAV development

The main purpose of developing the UAV was to create an experimental platform that could be used for extracting dynamic characteristics to be compared with the predicted dynamic characteristics predicted by the framework. As mentioned previously, the AVL outputs stability derivatives from the simulation. While reduced order models use stability derivative data for theoretically predicting dynamic characteristics, actual flight testing is required to observe how the aircraft behaves in practical condition. This stage was important because in the actual conditions, the flying conditions can be affected by many factors that are not fully represented in the theoretically predictions. This includes wind disturbances, structural flexibility, manufacturing errors, sensor noise, actuator delay and pilot inputs. By extracting this practical behavior, expected dynamic characteristics can be compared with the actual dynamic characteristics.

3.4.2 Baseline UAV Design and Fabrication

Using the developed framework, a baseline conventional lightweight fixed wing UAV was designed and fabricated for flight testing and flight data extraction. See

APPENDIX C : UAV Development and Fabrication for more details on fabrication [32].

3.4.3 Flight Controller and Avionics Integration

An ArduPilot Mega (APM) 2.8 flight controller was integrated with UAV mainly for initial flight stabilization and flight data recording purposes. The flight controller was configured for a fixed wing aircraft using ArduPilot firmware allowing the UAV to operate with different flight modes, suitable for initial flight testing and flight data recording [35] [36]. The APM 2.8 flight controller was mounted inside the fuselage to be coincide with the center of gravity of the UAV. Simply, the flight controller receives the pilot commands from the transmitter through the receiver. The servos and the motor ESC (Electronic Speed Controller) are connected to the flight controller. The wiring diagram can be shown as in Figure 3.21.

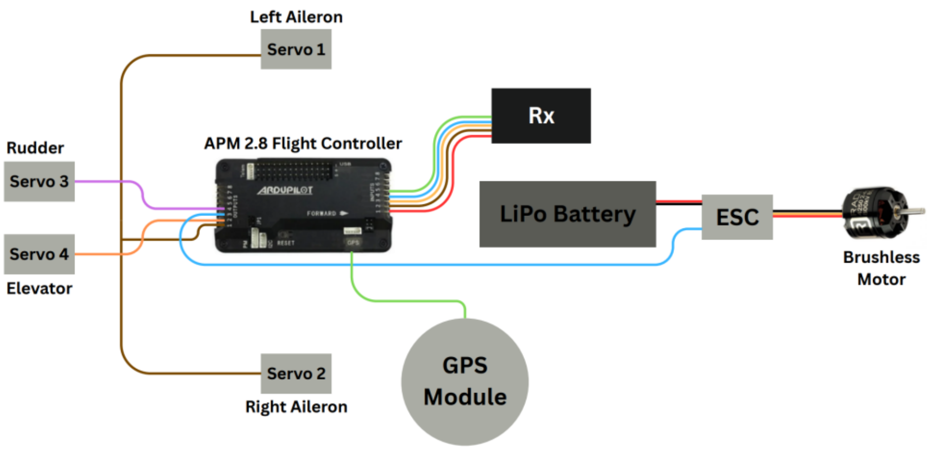

Figure 3.21: Electronics wiring diagram

Here, the aileron channel only had a single output from the flight controller. So, Y harness was used to give the single PWM (Pulse Width Modulation) signal to both servo motors actuating left and right aileron. The UAV has total of five channels. Three channels were connected to the control surfaces; aileron, elevator and rudder. Another single channel was connected to the electronic speed controller which controls the brushless motor. Last channel was connected for changing the flight mode. Three position switch in the transmitter was setup to switch between Manual mode, FBWA (Fly-By-Wire A) mode and Stabilize mode [35].

Manual mode: Pilot inputs from the transmitter are sent directly to the control surfaces with no flight controller corrections. Therefore, manual mode is used to collect the dynamic characteristics data since it allows the natural response of the UAV to be observed with no interference from the active stabilization. This is crucial for evaluating the flight mode characteristics physically.

FBWA: This stands for Fly-By-Wire A mode. In this mode, the flight controller provides stabilized control for pitch and roll. Still the pilot is capable of controlling the aircraft, but the flight controller limits the roll and pitch angles to safer levels. The throttle remains manually controlled. This mode is useful during initial flight testing as it prevents excessive pitch or bank angles.

Stabilize mode: In this mode also, the aircraft is manually controlled, but the flight controller provides stabilization assistance. When the pilot releases the control sticks, the flight controller levels the wings and attempts to bring the aircraft back to a stable attitude. This is done by using the sensor feedback from the IMU (Inertial Measurement Unit) to measure the roll and pitch motion while applying the corrective aileron and elevator inputs.

Since the UAV was newly fabricated, there could be uncertainties and errors due to manufacturing imperfections, center of gravity location, control surface settings, wind disturbances, and propulsion effects. The flight controller helps compensate for these by actively correcting undesired roll and pitch motions. So, these FBWA and Stabilize modes were used to improve safety and controllability during early stage of flying the UAV.

The inbuilt function of APM 2.8 flight controller was also used as the main onboard data logger. These logs can later be downloaded using Mission Planner interface when the UAV is grounded. The recorded data includes roll angle, pitch angle, yaw angle, roll rate, pitch rate, yaw rate, acceleration, altitude, GPS (Global Positioning System) position, and ground speed [36]. These parameters are crucial for analyzing both longitudinal and lateral-directional response.

3.4.4 Ground Testing and Pre-Flight Verification

Prior to the airborne operations, the ground test of the UAV was carried out to verify airworthiness and readiness of the UAV and identify any mechanical system level deficiencies:

- Flight path linearity
- Dynamic wing imbalances in static and ground roll condition
- Imbalances in wings while moving forward
- The actuation of control surfaces

- Full thrust effects on the UAV
- Tail section ground clearance

Figure 3.22: Initial ground testing

As Figure 3.22 illustrates, the UAV was tested on ground prior to the flight. The linearity of the UAV path during ground roll was conducted. Without any rudder deflections, the path taken by the UAV was observed for asymmetric rolling resistance, misalignment of landing gears or counter moment effects from the motor. It should not have any self-turning and follow on a straight path and any observed deviations was corrected. While in static and motion, the wing roll movement was observed for any collision between ground and wing tip caused by the uneven suspension from landing gears or by the flow over the wing. The actuation of the control surfaces was checked with the transmitter signals. The correct and consistent response of each surface for respective transmitter channel was confirmed prior to flight. Full throttle static test was conducted to observe any structural integrity of the motor mount, vibration induced reactions. The clearance between tail section and the ground was observed throughout the ground roll for any collision while on motion.

3.4.5 Flight Test Procedure

In this phase, flight tests were conducted to collect experimental data for evaluating the longitudinal dynamic response of the developed UAV. Typically, a fixed wing aircraft has both longitudinal and lateral-directional flying modes. The longitudinal modes include the short period mode and the phugoid mode which are primary related to pitch attitude, forward speed and altitude variations. In comparison, lateral-directional modes involve coupled roll, yaw and side slip behavior. For accurate characteristic identification of these modes requires careful excitation of aileron and rudder control inputs, side slip angle measurements and separation of coupled roll and yaw responses. Since small UAVs are highly affected by wind gusts, pilot corrections and sensor errors, complex lateral-directional mode identification becomes difficult. So, only longitudinal dynamic characteristics were selected for the assessment in this study.

To avoid and minimize the atmospheric disturbances on the dynamic response, the flight test was conducted during the calm wind conditions. Each flight was initiated by manual take-off and gradual transition to a stable, level cruising flight in trim conditions of the UAV where no control inputs were required by the pilot. Once the pilot informed about the stable flight, the dynamic excitation is established by sudden and sharp deflection of elevator, where it can be considered as a pulse input [4]. These disturbances can allow both short period and phugoid dynamic modes to be developed and decay over time. The excitations and damp behavior of those dynamic modes in the free motion of the UAV were recorded in the onboard flight controller. The full dynamic response of the UAV was captured continuously throughout each manoeuvre by the onboard ArduPilot flight controller, which logged inertial measurement unit data, GPS-derived position and velocity, airspeed, and control surface commands at the configured sampling frequency. Multiple repeat runs of the elevator pulse input were performed across the flight to provide redundant datasets and to assess the repeatability of the observed dynamic response. The logged flight data were subsequently downloaded for post-flight processing and system identification analysis, as described in the results section.

4. Results and Discussion

4.1 UAV Design Framework

The framework was applied in three distinct contexts. First, a complete worked example of the full framework workflow applied to a notional design is presented to demonstrates the sequential function of each module and a comprehensive capability of the framework. It is also used as the baseline design to develop a methodological approach to achieve desired flying qualities by fine tuning UAV geometry. Second, the design data for the modified UAV configuration that was adopted for fabrication and used as the experimental platform for framework validation is generated using the framework. It is noted that for the modified UAV, the framework was not run from the initial modules as we have adapted an UAV and kept the existing wing and skeleton, the framework modules from static stability evaluation onwards were applied to characterise its dynamic properties. Third, framework was applied to an existing RC aircraft – Blu-Baby to generate its closest geometry and aerodynamic and dynamic behaviour results to be compared with the flight extracted data.

4.1.1 Notional Design Case | Full Workflow Demonstration

This demonstrates the sequential workflow of the framework and the extracted data from it that can be used to analyze the dynamic behavior and the control augmentation data of the UAV. Design data of wing, fuselage propulsion and landing gears are mentioned in APPENDIX D : Framework data.

Horizontal Tail Design

Table 4.1: Horizontal tail design data

Aerofoil	NACA 0012
Tail area	0.220 m²
Span	0.811 m
Root chord	0.315 m
Tip chord	0.220 m
Mean chord	0.270 m
Aspect ratio	3
Taper ratio	0.7
LE sweep	12.09°
Tail volume coefficient	0.6
Tail arm	1.175 m
Tail incidence	+0.20°
Downwash angle, ε	3.95°
Tail operating angle of attack	−2.72°
Tail lift coefficient, CL,h	−0.1596
CL,α,h	3.477 /rad

The resulting longitudinal static stability characteristics are presented in Table 4.6.

Table 4.2: Longitudinal Stability data

Neutral point, NP	58% MAC
Centre of gravity	20% MAC
Static margin	38% MAC
Cm,α	−1.5423 /rad
Cm,0	0.2411
Trim angle of attack	8.96°

Vertical Tail Design

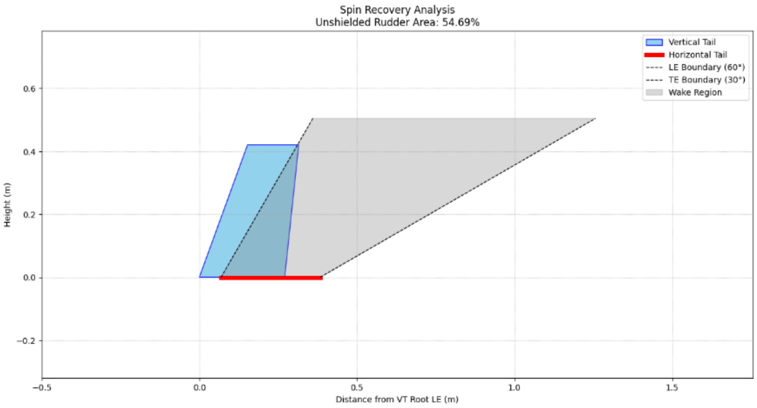

Figure 4.1: Spin recovery of the vertical tail

Table 4.3: Vertical tail design data

Aerofoil	NACA 0008
Tail area	0.090 m²
Span	0.421 m
Root chord	0.276 m
Mean chord	0.221 m
Aspect ratio	1.9
Taper ratio	0.6
LE sweep	20.0°
Tail volume coefficient	0.03
Tail arm	1.085 m
Cn,β	0.0638 /rad
Unshielded area ratio	54.69%

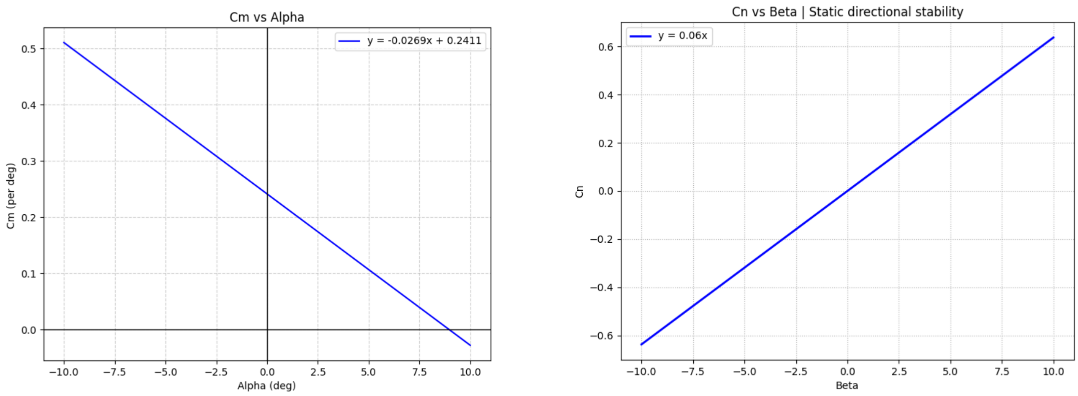

Figure 4.2: Static Stability – Cm vs angle of attack and Cn vs sideslip angle

Zero-Lift Drag Calculation

The drag contributions are summarized in Table .

Table 4.4: Drag contribution of components

Wing	0.0088

Fuselage	0.0013
Horizontal tail	0.0014
Vertical tail	0.0005
Landing gear	0.0016
High-lift devices (HLD)	0.0258
Total clean, CD0,AC	0.0342
Total take-off, CD0,TO	0.0599

AVL analysis for drag polar gives Cd_0 as 0.0396.

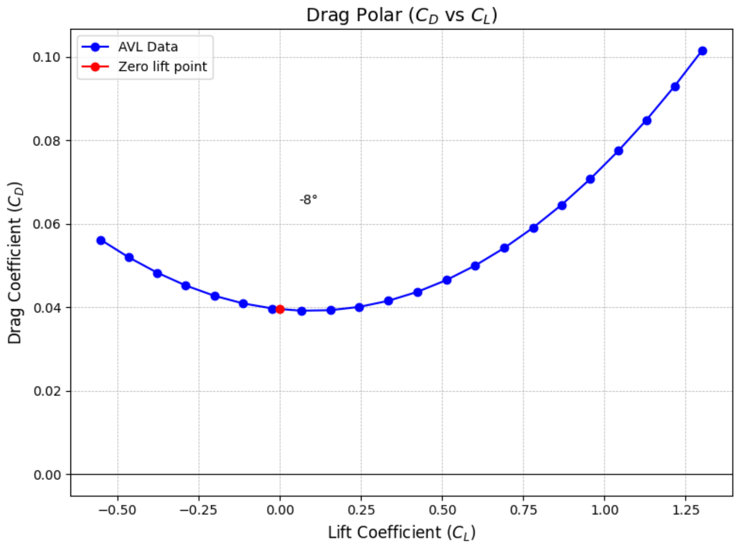

Figure 4.3: Drag polar curve plotted for the design

Performance Analysis

The performance results are presented in Table .

Table 4.5: Performance data of the design

Performance Parameter	**Required**	**Achieved**
Cruise speed	19 m/s	18.10 m/s
Maximum speed	25 m/s	36.27 m/s
Stall speed	≤ 14 m/s	14.56 m/s
Maximum Range	-	309.22 km
Maximum Endurance	-	5.8615 h
Speed for Max ROC	-	13.20 m/s
Angle for Max ROC	-	32.00 deg
Maximum Rate of Climb	-	6.99 m/s
Speed for Steepest Climb	-	2.84 m/s
Maximum Climb Angle	-	90.00 deg
ROC at Steepest Climb	-	2.84 m/s
Total Takeoff Distance	-	63.35 m
Corner Speed	-	27.40 m/s
Minimum Turn Radius	-	22.81 m

Bank angle for tightest turn	-	73.40 deg

Control Surface Design

Aileron Design

The ailerons were sized to deliver a 45° bank angle within 1.7 seconds, meeting the roll responsiveness requirement for the surveillance mission category.

Table 4.6: Aileron design details

Inboard position, bai/b	0.7
Outboard position, bao/b	0.98
Chord ratio, ca/c	0.2
Aileron chord	0.08 m
Aileron span	0.42 m
Rolling moment derivative, Cl,δa	0.0731 /rad
Hinge position	80% MAC

Elevator Design

The elevator was sized to satisfy the pitch angular acceleration requirement during take-off rotation, which demands a minimum pitch angular acceleration of 12.5 deg/s² to rotate the aircraft to the take-off attitude within the available ground roll.

Table 4.7: Elevator design details

Span ratio, bE/bH	0.9
Chord ratio, cE/cH	0.0852
Elevator span	0.730 m
Elevator chord	0.023 m
Elevator area, SE	0.020 m²
Control effectiveness, τE	0.12
Trim elevator deflection	9.94°
Maximum deflection	−25° (up)
Hinge position	91% MAC

Rudder Design

The rudder was sized to maintain directional control under a crosswind of 5 m/s, which represents the critical lateral control sizing condition.

Table 4.8: Rudder design details

Chord ratio, CR/CV	0.15
Span ratio, bR/bV	0.7
Rudder span	0.294 m
Rudder chord	0.033 m
Rudder area, SR	0.010 m²
Control effectiveness, τR	0.34
Required deflection for crosswind	29.24°
Maximum deflection	30.0°

Hinge position	85% MAC

Dynamic Stability Evaluation

Longitudinal State-Space Model:

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} -0.0871 & 0.2644 & 0.1833 & -9.81 \\ -1.051 & -3.350 & 17.69 & 0 \\ -0.01396 & -1.070 & -2.540 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} -0.002668 \\ -0.03277 \\ -0.1661 \\ 0 \end{bmatrix} \eta$$

Lateral-Directional State-Space Model:

$$\begin{bmatrix} \dot{v} \\ \dot{p} \\ \dot{r} \\ \dot{\phi} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} -0.1841 & -0.4205 & -18.74 & 9.81 & 0 \\ -0.7725 & -8.553 & 3.256 & 0 & 0 \\ 2.066 & -1.608 & -3.234 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} v \\ p \\ r \\ \phi \\ \psi \end{bmatrix} + \begin{bmatrix} -0.02670 & -0.02167 \\ -1.579 & -0.01939 \\ 0.3781 & 0.3619 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$

The eigenvalues of the longitudinal and lateral-directional system matrices were extracted and the dynamic mode characteristics computed, as presented in Table .

Table 4.9: Dynamic stability evaluation of the design

Mode	**Eigenvalue**	**ωn (rad/s)**	**ζ**	**Tc (s)**
Short Period	$-2.944 \pm 4.327i$	5.25	0.561	-
Phugoid	$-0.012 \pm 0.620i$	0.62	0.02	-
Dutch Roll	$-1.800 \pm 6.424i$	6.67	0.27	-
Roll Subsidence	-8.333	-	-	0.12
Spiral	0.11	-	-	-9.12

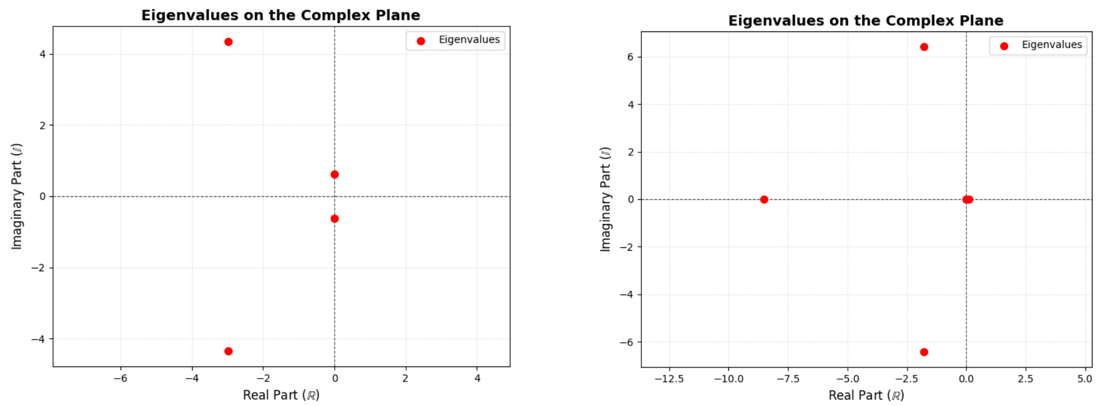

Figure 4.4: Eigen values of longitudinal and lateral-directional modes of the deisgn

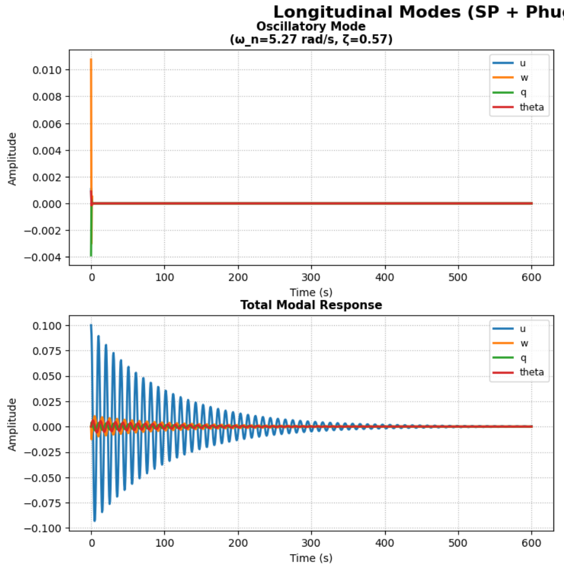

Figure 4.5: Modal response of longitudinal mode for initial condition of u=0.1

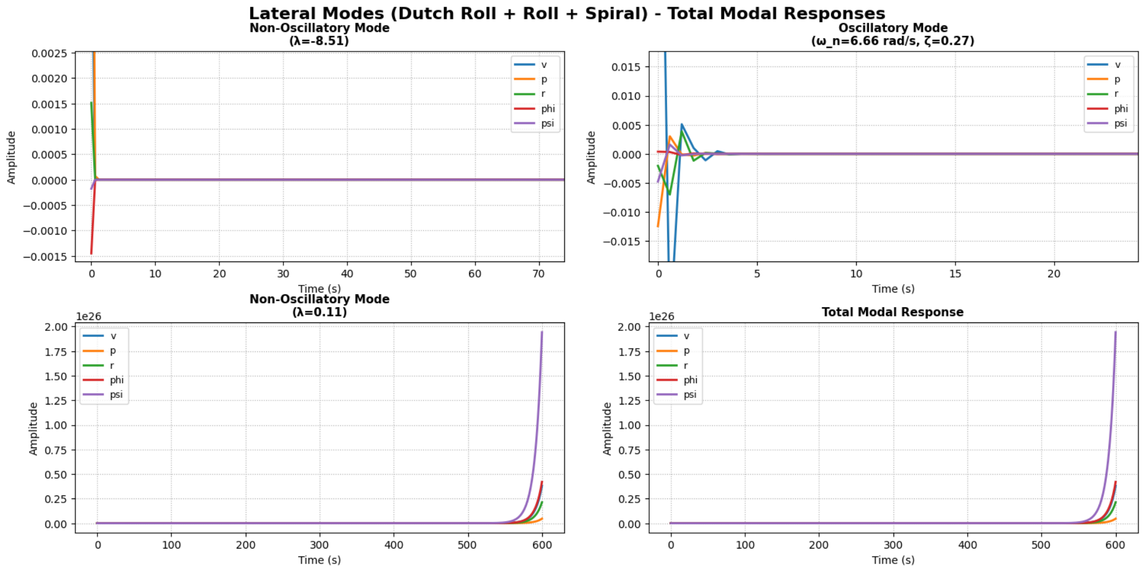

Figure 4.6: Modal response of lateral mode for initial condition of v=0.1

Flying Quality Evaluation

The compliance assessment of flying qualities is presented in Table 4.16.

Table 4.10: Flying quality assessment of the design

Mode	Parameter	Level 1 Requirement	Predicted	Level
Short Period	ζ	0.30 – 2.00	0.561	Level 1
Phugoid	ζ	> 0.04	0.0160	Level 1
Dutch Roll	ζ	> 0.08	0.2710	Level 2
Dutch Roll	ωn	> 0.5 rad/s	6.6656 rad/s	Level 1
Dutch Roll	$\zeta * \omega n$	> 0.15	1.799 rad/s	Level 1

Roll Subsidence	Tc	< 1.4 s	0.1175 s	Level 1
Spiral	Tc	> 28.9 s	−9.12 s	Unstable

4.1.2 Modified UAV Configuration - Framework Application for Validation

The developed framework is used verify the existing wing design of the UAV. Additionally, the horizontal tail, vertical tail, propulsion system has been designed using the developed framework. Performance analysis, dynamic stability evaluation has also been conducted using framework. Significant data for UAV design calculated from framework is listed below. For Detailed design data representation refer APPENDIX D : Framework data.

The flight test results for this configuration and due to the challenges encountered during trimming and achieving stable flight with this newly fabricated UAV, the flight data obtained were insufficient for a complete system identification, as discussed in Chapter 4.3.

4.1.3 Blu-Baby RC Aircraft | Framework Applied for Verification and Aerodynamic and Dynamic Stability Result Extraction

For Detailed design data representation refer APPENDIX D : Framework data. The flight test results for this configuration are discussed in Chapter 4.3.

4.1.4 Stability Augmentation Module Results

The stability augmentation system was tested using two UAV transfer function models which are theoretically obtained by dynamic analysis module of framework. The selected flying quality targets were,

iii) short-period natural frequency: 1 < < 10 rad/s

iii) short-period damping ratio: 0.3 < < 2.0

iii) minimum phugoid doubling time: T_2 = 55 s for an unstable phugoid mode.

Table 4.11:Tuned gain values for two UAV models

Model	Kp	Ki	Kd	TCONST	FF	Main result
Developed UAV model	0.3195	0	0.00625	0.2203	0.15	Stabilized successfully
Blu-baby RC model	0.05	0	0.08	10.0	0.15	Not effectively improved

The developed UAV model (glider) could be brought into the required stable region using acceptable Kp, Kd and TCONST values. Its short-period frequency and damping ratio remain within the target range while the phugoid instability was treated to the have required doubling time (T2 = 55s). This shows that the selected SAS feedback path has enough control authority for this model. Parameters before and after tuning are mentioned in appendix D.

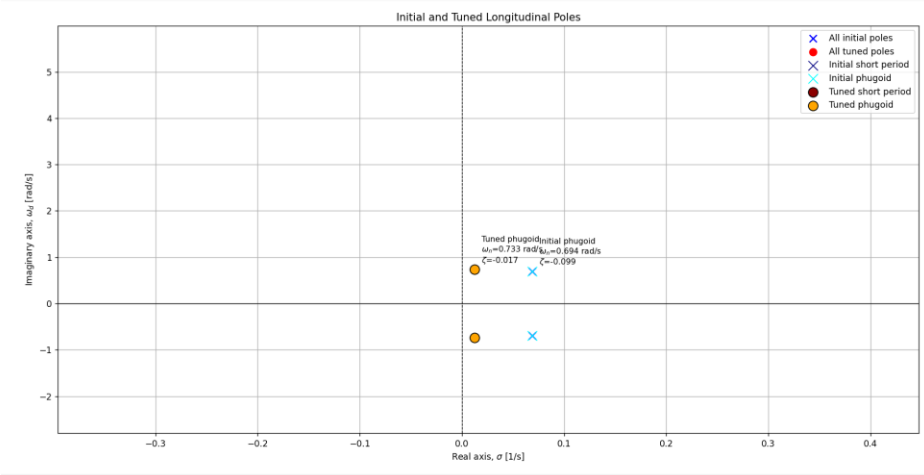

Figure 4.7: Inherent vs tuned Phugoid poles for developed UAV

However, the Blu-baby RC model couldn't be brought into the required region using acceptable gains (appendix D). Here, gains cannot be increased beyond the acceptable ranges, as it may cause control surface actuators to get saturated which may cause loss of control of UAV. Although the damping values are stable, the short-period natural frequency remains far above the maximum target level. This indicates that the aircraft has very fast inherent pitch dynamics, and the SAS loop mainly changes the response slightly instead of reshaping the mode.

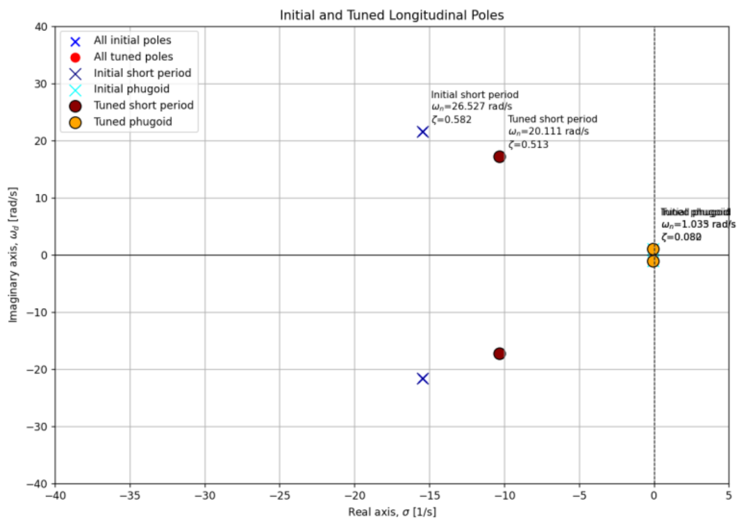

Figure 4.8:Inherent vs tuned poles on complex plane for Blubaby RC model

In practical autopilot systems, K_p and K_d are usually kept in decimal ranges because actuator deflection, sensor scaling and rate feedback are highly sensitive. A large K_p can cause overshoot, oscillation, servo saturation and excessive elevator movement. A large K_d can amplify gyro noise and create high-frequency servo vibration. Therefore, very high gains may make the simulated response look stronger, but in real flight they can excite unmodelled dynamics, reduce robustness and even make the UAV unstable.

4.1.5 Discussion of Framework Effectiveness and Limitations

The results taken from framework demonstrates the capability for a full design synthesis and standalone function of each module as it can be used at an intermediate stage of a design process. The framework predictions are based entirely on analytical estimation methods and the experimentally measured geometric and inertial properties. The fidelity of these predictions is inherently limited by the assumptions embedded in the derivative estimation methods, including the use of linear aerodynamic theory and empirical correction factors. A quantitative assessment of the accuracy of these predictions, through direct comparison with flight-identified model parameters, follows with the completion of the flight test campaign and system identification analysis.

4.2 Geometry and Inertia Measurement

4.2.1 Measured Geometric Properties of the UAV

The following features of the UAV were measured from the fabricated measurement system.

Table 4.12: Measured geometric parameters of the UAV using the measurement system

Component	Parameter	Value	Unit
Wing Geometry	Wingspan	2530	mm
	Root chord	188	mm
	Tip chord	115	mm
	Wing incidence angle-root-Right	9.95	degrees
	Wing incidence angle-tip-Right	11.309	degrees
	Wing incidence angle-root-Left	8.769	degrees
	Wing incidence angle-tip-Left	9.865	degrees
	Dihedral angle	3.48	degrees
Flap Geometry	Flap span-Right	686	mm
	Flap span-Left	686	mm
	Flap chord	56	mm
Horizontal Tail Geometry	Wing-to-horizontal-tail distance / tail moment arm	754	mm
	Horizontal tail span	448	mm
	Horizontal-tail root chord	117	mm
	Horizontal-tail tip chord	73	mm
	Horizontal-tail incidence angle	2.353	degrees
	Horizontal-tail sweep angle	11	degrees
Vertical Tail Geometry	Wing-to-vertical-tail distance	754	mm
	Vertical tail span / height	267	mm
	Vertical-tail root chord	158	mm
	Vertical-tail tip chord	123	mm
	Vertical-tail sweep angle	13.37	degrees
Control Surfaces	Elevator span-Right	170	mm
	Elevator span-Left	169	mm
	Elevator chord	65	mm
	Rudder span	232	mm
	Rudder chord	38	mm

Apart from these flight dynamics related measurements, few more measurements were checked to identify the deviations of the actual aircraft from the design. The main deviations were observed in the tail boom section where it sags due to the weight of the tail, changing the effective tail incident angle when moving on the ground (during the takeoff). The sagging angle of the tail section was found to be 0.614˚ when the aircraft is kept horizontal on the ground effectively reducing the tail incident angle to -2.086˚. To fix this deviation, a cable support from the vertical tail to the fuselage was attached. By tensioning this cable, the sagging effect was minimized.

Also, the difference between the root and tip incident angles of the wings of the UAV indicates that there is a slight twist in the wings which implies the manufacturing limitations and the problems which can rise during real-world applications.

4.2.2 Measured Inertia Properties

As the first and one of the most important inertia properties, the center of gravity location of the UAV was determined by using the three-point weighing method for the horizontal location and the tilting method to determine the vertical location.

With usage of the fabricated compound pendulum and the bifilar torsional pendulum, the mass moments of inertia about the three principal axes of the both the UAV fabricated in University of Moratuwa and the “33-inch Blu-Baby Trainer Model” were determined in an experimental manner.

4.2.2.1 Position of the Centre of Gravity of the UAV

1. **Horizontal Position of the Center of Gravity**

For the determination of the horizontal location of the center of gravity of the UAV, a simple but well know reliable method was used, which is the three point weighing method. The UAV is supported on three points to the ground, and these points carry the weight of the UAV to the scales placed on the leveled floor. Three identical and calibrated weighing scales were used in the procedure with 0.1g accuracy. Three landing gears of the UAV, two for the rear and one for the front were used as the supporting points and the results are as follows.

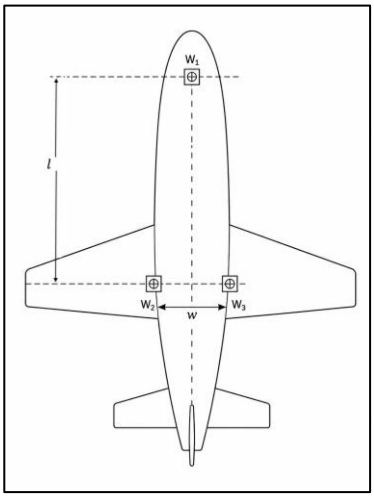

Figure 4.9: Top view of the supported UAV

Table 4.13: Readings from the Cg determining setup

W1	217.2	g
W2	1256.4	g
W3	1407.5	g
l	435	mm
w	115	mm

Therefore, the X position relative to the rear landing gear axis:

$$P_x = 435 \times \frac{217.2}{217.2 + 1407.5 + 1256.4} = 32.79\, mm\, forward.$$

The Y position relative to the rear-left landing gear:

$$P_x = 115 \times \frac{1407.5}{1407.5 + 1256.4} = 60.76\, mm\, rightward.$$

This proves that the symmetry existing in the CAD models does not remain as it is when it comes to real-world fabrication due to manufacturing limitations and non-uniform material distribution. However, these results were in an acceptable range to conduct a test flight.

2. Vertical position of the Center of Gravity

In this procedure, the same supporting method was used, with a tilted floor to get the effect of the vertical position of the center of gravity into the scale readings. A rigid wooden board was used as the tilted floor, which the deflection due to the weight of the UAV and the board can be neglected. The floor was tilted in three different angles for both nose up direction and nose down direction. The angle was calculated considering the length of the board and the elevation of the other end of the board. The calculation was done considering the moments of the forces acting on the UAV around points A and B in the figure below.

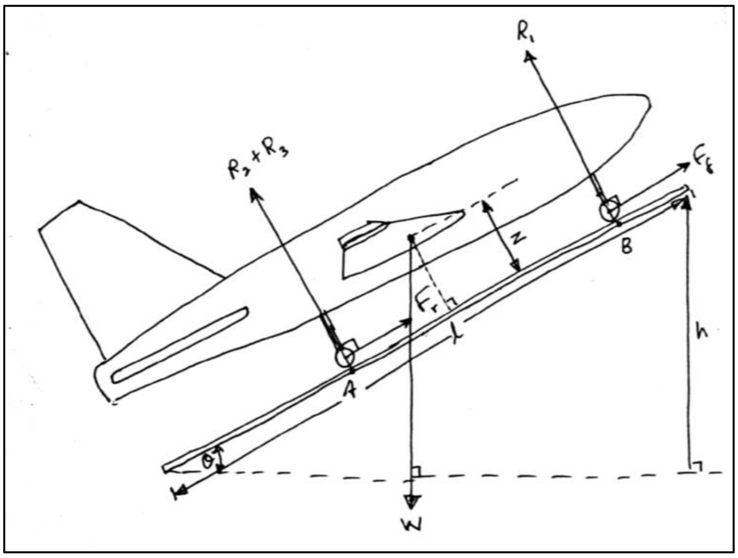

Figure 4.10: Tilted UAV on the scales

The results were obtained as follows.

Table 4.14: Reaction forces and the tilt angle for each configuration

	R1 (g)	R2 (g)	R3 (g)	Theta (Degrees)	
1	409.8	1402.6	1452.7	2.39585	nose up readings
2	377	1408.1	1472.2	3.853933	
3	366.8	1462.3	1425.7	4.466078	
4	509.8	1330	1420	2.724925	Nose down readings
5	532.3	1286	1435.1	3.712734	
6	539.7	1223.5	1486.8	4.37187	

Table 4.15: Calculation of the vertical Cg position of the UAV

		h (m)	R1 (g)	R2 (g)	R3 (g)	X (m)	Avg (m)	Final_avg (m)
nose up readings	h1	0.051	0.4098	1.4026	1.4527	0.14832	0.151361	**0.152434**
	h2	0.082	0.377	1.4081	1.4722	0.155069		
	h3	0.095	0.3668	1.4623	1.4257	0.150694		
Nose down readings	h1	0.058	0.5098	1.33	1.42	0.151746	0.153507	
	h2	0.079	0.5323	1.286	1.4351	0.159516		
	h3	0.093	0.5397	1.2235	1.4868	0.149259		

Therefore, the Vertical position of the center of gravity of the UAV fabricated in the University of Moratuwa was found to be 15.2434 cm above ground. For the detailed calculation refer to the Appendix E section.

Same way the center of gravity location for the "33-inch Blu-Baby Trainer Model" was determined. In this case since the aircraft did not have any landing gears, external support structures were used and their weights were subtracted.

4.2.2.2 Virtual Moment of Inertia about the X-Axis

The virtual moment of inertia about the longitudinal body axis (I_{Vx}), was determined using the compound pendulum setup. The calculation was done for two different setups with a longer suspension and a shorter one. The calculated values obtained from the short and long pendulums were $0.698301886 \text{ kg m}^2$ and $0.69377871 \text{ kg m}^2$, respectively. Therefore, the average virtual moment of inertia about the X-axis was obtained as:

$$I_{Vx} = 0.696040298 \; kg \; m^2$$

The reliability of the results can be verified through a simultaneous equation system for the two pendulum settings, without considering for the additional mass and the volume, treating Ixx and the summation of additional mass and added volume as the only two unknowns. This provides a value of,

$$I_{Vx} = 0.712358 \; kg \; m^2$$

The difference between the two obtained values was approximately 2.3%, which provides a reasonable agreement and thus confirms the reliability of experimental data and procedure.

The additional moment of inertia about the X-axis is primarily contributed by the main wing and horizontal tailplane. The total additional moment of inertia was found to be:

$$I_{Ax} = 0.04439797 \; kg \; m^2$$

Thus, the true mass moment of inertia about the X-axis can be calculated as:

$$I_{Tx} = I_{Vx} - I_{Ax} = 0.696040298 - 0.04439797 = 0.651642328 \; kg \; m^2$$

For the detailed calculations, refer to the Appendix section VI.

4.2.2.3 Virtual Moment of Inertia about the Y-Axis

The virtual moment of inertia about the y axis was also determined using the same procedure. The short and long pendulum configurations were used here as well and the values were $0.471067647 \text{ kgm}^2$ and $I_{V\text{y}} = 0.4708091 \text{ kgm}^2$, respectively. The average value was therefore calculated as:

$$I_{Vy} = 0.470938373\ kg\ m^2$$

The simultaneous-equation verification method gave:

$$I_{Vy} = 0.472401034\ kg\ m^2$$

The difference between the two values was only (0.31%), showing very strong agreement between the two approaches.

The additional moment of inertia about the Y-axis is mainly produced by the fuselage and horizontal tail surface. The total additional moment of inertia was calculated as:

$$I_{Ay} = 0.002471549\ kg\ m^2$$

Therefore, the true moment of inertia about the Y-axis was obtained as:

$$I_{Ty} = I_{Vy} - I_{Ay} = 0.470938373 - 0.002471549 = 0.468466825\ kg\ m^2$$

For the detailed calculations, refer to the Appendix section VI.

4.2.2.4 Virtual Moment of Inertia about the Z-Axis

The virtual moment of inertia about the vertical body axis (z) was determined using the bifilar pendulum method. Considering the obtained data through the experiment the virtual mass moment of inertia about the z axis was calculated as,

$$I_{vz} = 0.869206691\ kg\ m^2$$

The additional inertia contribution about the Z-axis was determined by considering the vertical projected areas of the fuselage and tail surfaces. The resulting additional inertia was,

$$I_{Az} = 0.004350343\ kg\ m^2$$

Hence, the true moment of inertia about the Z-axis was calculated as:

$$I_{Tz} = I_{Vz} - I_{Az} = 0.869206691 - 0.004350343 = 0.864856348\ kg\ m^2$$

4.2.2.5 Summary of Final Moments of Inertia

The final corrected mass moments of inertia of the fabricated UAV in the university are summarised in Table below.

Table 4.16: Calculated mass moments of inertia for the fabricated UAV

Body Axis	**Virtual Moment of Inertia** (kg m^2)	**Additional Moment of Inertia** (kg m^2)	**True Moment of Inertia** (kg m^2)
X-axis (I_{xx})	0.696040298	0.04439797	**0.651642328**
Y-axis (I_{yy})	0.470938373	0.002471549	**0.468466825**
Z-axis (I_{zz})	0.869206691	0.004350343	**0.864856348**

The moments of inertia for the "33-inch Blu-Baby Trainer Model" are shown below.

Table 4.17: Calculated mass moments of inertia for the 33-inch Blu-Baby Trainer Model

Body Axis	**Virtual Moment of Inertia** (kg m^2)	**Additional Moment of Inertia** (kg m^2)	**True Moment of Inertia** (kg m^2)
X-axis (I_{xx})	0.050323679	0.001343024	**0.048980656**
Y-axis (I_{yy})	0.020144596	0.000505606	**0.019638989**
Z-axis (I_{zz})	0.153906911	0.000100645	**0.153806267**

The highest moment of inertia was obtained about the Z-axis, followed by the X-axis and Y-axis. This can be expected because yaw motion about the vertical axis affects the total distributed mass of the wings, tail section and the fuselage with relatively larger displacements from the axis. The comparatively lower value of inertia about the y axis indicates that the aircraft has less resistance to pitching motion than to rolling or yawing motion. This was observed during the initial test flights of the UAV as well, where the UAV was very responsive for elevator inputs and the flaperon inputs, while resisting to yaw and turn during flight, leading to uncontrollable situations.

Overall, the close agreement between the short and long-pendulum configurations confirms that the experimental procedure produced consistent and reliable inertia estimates for use in the UAV dynamic model.

4.2.3 Uncertainty Quantification and Error Bounds

The experimental measurements that are being taken in this research are subject to unavoidable uncertainties arising from instrument resolution, laser-line thickness, alignment errors, oscillation damping of bearings, timing errors from the videos, and assumptions used in the inertia correction models. Therefore, an uncertainty analysis must be carried out for the geometric parameters gained through the laser measurement setup and experimentally determined mass moments of inertia to assess the reliability of the results.

4.2.3.1 Uncertainty in Laser-Based Geometric Measurements

The main geometric dimensions of the UAV, including wingspan, chord lengths, tail dimensions, control-surface dimensions, and distances between aerodynamic surfaces, were measured using a laser measuring device fabricated during the research. The uncertainty relevant to each direct reading was defined by the smallest measurement (least count) or stated accuracy of the measuring device.

For a dimension obtained as the difference between two measured locations,

$$\mathrm{q} = \mathrm{x} - \mathrm{y}$$

the combined uncertainty was calculated using the root-sum-square method:

$$\delta_q = \sqrt{(\delta_x)^2 + (\delta_y)^2}$$

where (δ_x) and (δ_y) are the uncertainties related to the first and last readings, respectively.

Since the same laser measurement arrangement was used for both readings, the uncertainty in a measured length was influenced by:

- Resolution or least count of the measuring scale
- Finite thickness of the projected laser beam
- Difficulty in identifying the exact starting edge or reference point of the UAV
- Laser misalignment relative to the measurement direction
- Parallax error when reading the reference scale
- Variation in the aircraft position during measurement.

The laser line does not represent a perfectly sharp geometric point. Therefore, its finite width introduces uncertainty when locating edges, especially on curved surfaces such as the fuselage, wing leading edge, and tail surfaces. This uncertainty was reduced by maintaining the same side of the laser beam as the reference through all measurements.

Also, keeping a bright surface behind the measuring feature and keeping the room lighting level low helps to identify the laser beam and the UAV intersections better.

The laser measuring device had an accuracy of ($\pm 2\ mm$). The reference scale had a least count of (1 mm), and the laser line thickness was approximately (3 mm). The laser line was consistently referenced using the same edge; therefore, half of the beam thickness was considered as the beam-position uncertainty.

For one laser-based position measurement,

$$\delta_x = \sqrt{(2)^2 + (1)^2 + \left(\frac{3}{2}\right)^2}$$

$$\delta_x = \pm 2.69 \text{ mm}$$

A geometric dimension was obtained as the difference between two position measurements. Therefore,

$$\delta_q = \sqrt{(\delta_x)^2 + \left(\delta_y\right)^2}$$

$$\delta_q = \sqrt{(2.69)^2 + (2.69)^2}$$

$$\boxed{\delta_q = \pm 3.81 \text{ mm}}$$

Therefore, the measured values can be presented as follows.

Table 4.18: Corrected geometric parameters considering the error bound

Parameter	**Measured value (mm)**	**Uncertainty**
Wingspan	2530	($\pm$3.8 mm)
Root chord	188	($\pm$3.8 mm)
Tip chord	115	($\pm$3.8 mm)
Flap span, right	686	($\pm$3.8 mm)
Flap span, left	686	($\pm$3.8 mm)
Flap chord	56	($\pm$3.8 mm)
Tail moment arm	754	($\pm$3.8 mm)
Horizontal-tail span	448	($\pm$3.8 mm)

Horizontal-tail root chord	117	($\pm$3.8 mm)
Horizontal-tail tip chord	73	($\pm$3.8 mm)
Vertical-tail span	754	($\pm$3.8 mm)
Vertical-tail root chord	267	($\pm$3.8 mm)
Vertical-tail tip chord	158	($\pm$3.8 mm)
Elevator span, right	123	($\pm$3.8 mm)
Elevator span, left	170	($\pm$3.8 mm)
Elevator chord	169	($\pm$3.8 mm)
Rudder span	65	($\pm$3.8 mm)
Rudder chord	232	($\pm$3.8 mm)

For all calculated angles, a tolerance of

$$\boxed{\pm 1.0^\circ}$$

was used. This covers the propagated effect of the (3.8 mm) distance uncertainty in the horizontal and vertical measurements used to calculate the angles.

Therefore, the measured angles may be reported as:

Table 4.19: Corrected angles considering the error bound

Angular Parameter	**Measured Value**	**Uncertainty**
Wing incidence angle at right wing root	9.95°	$\pm$1.0°
Wing incidence angle at right wing tip	11.309°	$\pm$1.0°
Wing incidence angle at left wing root	8.769°	$\pm$1.0°
Wing incidence angle at left wing tip	9.865°	$\pm$1.0°
Wing dihedral angle	3.48°	$\pm$1.0°
Horizontal tail incidence angle	2.353°	$\pm$1.0°
Horizontal tail sweep angle	11.0°	$\pm$1.0°
Vertical tail sweep angle	13.37°	$\pm$1.0°

4.2.4 Uncertainty in Inertia Measurements

Every mass measurement was done using weighing scales with an accuracy of:

$$\boxed{\delta m = \pm 0.1 \text{ g} = \pm 0.0001 \text{ kg}}$$

All pendulum lengths, filament lengths, filament spacing values, and centre-of-gravity locations were measured with an uncertainty of:

$$\boxed{\delta L = \delta A = \delta l = \delta x_{CG} = \pm 1 \text{ mm} = \pm 0.001 \text{ m}}$$

The data logging frequency of the flight controller used to determine the oscillation period was 50Hz, therefore,

$$\boxed{\delta T = \pm 0.0004 \text{ s}}$$

The uncertainty in the true moments of inertia was calculated by considering the uncertainties in mass, period, pendulum length, centre-of-gravity position, and additional inertia corrections.

The final experimentally determined moments of inertia, including their estimated uncertainties, are:

Table 4.20: Corrected mass moments of inertia of the UAV considering error bound

Axis	**True moment of inertia**	**Absolute uncertainty**	**Percentage uncertainty**
X-axis	$I_{xx} = 0.6516 \text{ kg m}^2$	$\pm 0.0024 \text{ kg m}^2$	± 0.37
Y-axis	$I_{yy} = 0.4685 \text{ kg m}^2$	$\pm 0.0006 \text{ kg m}^2$	± 0.13
Z-axis	$I_{zz} = 0.8649 \text{ kg m}^2$	$\pm 0.0024 \text{ kg m}^2$	± 0.28

Accordingly, the final values may be reported as:

$$\boxed{I_{xx} = 0.6516 \pm 0.0024 \text{kgm}^2}$$

$$\boxed{I_{yy} = 0.4685 \pm 0.0006 \text{kgm}^2}$$

$$\boxed{I_{zz} = 0.8649 \pm 0.0024 \text{kgm}^2}$$

The uncertainty in I_{zz} is mainly influenced by the filament spacing because it is in the equation as a squared term in the bifilar pendulum equation. The uncertainty in I_{xx} is comparatively higher than I_{yy}, mainly due to the additional moment of inertia correction for the wing, which depends strongly on measured chord and span dimensions.

4.2.4.1 Overall Reliability of Results

The measured UAV geometry and inertia properties were sufficiently reliable for use in the dynamic model. Direct linear dimensions had an uncertainty of ± 3.8 mm, while calculated angular parameters were reported with a conservative tolerance of $\pm 1.0°$.

The final moments of inertia were obtained as:

$$I_{xx} = 0.6516 \pm 0.0024 \text{ kg m}^2$$
$$I_{yy} = 0.4685 \pm 0.0006 \text{ kg m}^2$$
$$I_{zz} = 0.8649 \pm 0.0024 \text{ kg m}^2$$

The corresponding uncertainty levels were below 0.4%, indicating good reliability of the pendulum experiments. The agreement between alternative pendulum configurations, within 2.3%for the X-axis and 0.31%for the Y-axis, further supports the reliability of the results.

4.2.5 Comparison Between Experimental and CAD-Based Moments of Inertia

Table 4.21: Comparison Between Experimental and CAD-Based Moments of Inertia

Axis	Experimental value, kg m^2	CAD value, kg m^2	Difference relative to experimental value
I_{xx}	0.6516	0.5165	20.7% lower
I_{yy}	0.4685	0.7113	51.8% higher
I_{zz}	0.8649	0.2137	75.3% lower

There are considerable differences between the experimental and CAD-derived inertia values, specially about the Y and Z-axes. The CAD model predicts a lower roll inertia, a significantly higher pitch inertia, and a substantially lower yaw inertia compared with the experimental results.

These differences are usually expected because CAD-based mass properties depend strongly on modelling assumptions. The CAD model does not fully represent the actual mass distribution of the fabricated UAV, which was effortlessly observed during the measurement procedure and the inertia determining procedure, including internal components such as batteries, flight controller, wiring, servos, fasteners, payloads, adhesive material, reinforcement members, and local variations in material thickness. Even small deviations in the location of these components can produce large changes in the mass moment of inertia because inertia depends on the square of the distance from the relevant axis.

In contrast, the experimental pendulum tests were performed using the fabricated UAV which was fly-ready and therefore included the real mass distribution of the complete aircraft. The experimental method always accounts for the electronic components, structural joints, manufacturing variations, and mass offsets that may not be represented accurately in the CAD model. In addition, the experimental inertia values showed low propagated uncertainty and good agreement between alternative pendulum configurations. Therefore, although the CAD results are useful for preliminary estimations and preliminary developments, the experimentally obtained moments of inertia can be considered more reliable for the final six-degree-of-freedom dynamic model of the UAV.

4.2.6 Discussion of Measurement Setup Adequacy

The whole measurement setup can be considered adequate for determining the geometric properties, centre of gravity location, and mass moments of inertia of the UAV. The laser-based measurement arrangement provided a practical method for measuring the aircraft dimensions without requiring complete dismantling of the UAV. Although the directly measured dimensions had an estimated uncertainty of ± 3.8 mm, this uncertainty was small relative to the main aircraft dimensions, such as the wingspan and tail moment arm. Also, it provides more accurate values of the actual aircraft dimensions rather than they are being separately measured when disassembled. Therefore, the setup was sufficiently accurate for defining the overall fabricated geometry of the UAV.

The use of weighing scales with an accuracy of 0.1g together with 1 mm accuracy for measuring distances provided adequate accuracy for determining the centre of gravity location. The three-point support arrangement allowed the reaction forces at the supports to be measured and used to calculate the longitudinal and lateral CG positions. Careful steps were taken to maintain a stable UAV position and minimise movement during the weighing process due to natural wind flows.

For inertia measurements, the compound pendulum provided a good reliability for determining the moments of inertia about the X and Y-axes, while the bifilar pendulum arrangement was suitable for determining the Z-axis inertia. The use of flight-controller data sampled at 50Hz avoided stopwatch reaction-time error. Measuring the time over 50 oscillation cycles reduced the period uncertainty to approximately ± 0.0004 s, which contributed to lowering the calculated uncertainty in the final inertia values.

The use of two pendulum lengths for the compound pendulum tests seems to have further improved confidence in the results. The agreement between the alternative analytical calculations was within (2.3%) for the X-axis and (0.31%) for the Y-axis, indicating that the setup produced consistent results.

However, some limitations remained and must be noted for the future considerations. Small misalignments of the pendulum axis, damping due to air resistance and the bearing friction, uncertainty in the exact centre of gravity location when placing the UAV on the oscillating frame, and simplified additional-inertia corrections could affect the final values. Despite these limitations, the low propagated uncertainties and the consistency between repeated configurations indicate that the measurement setup was adequate and reliable for obtaining experimental inertia properties of the UAV.

4.3 UAV Development and Flight Data Extraction for Validation

4.3.1 Data Acquisition System and Pre-processing

In this project, flight data were collected to compare the dynamic responses obtained from the developed UAV model and the Blu-baby RC model. Initially, the APM 2.8 flight controller was selected as both the flight controller and the main flight data recorder. Its inbuilt inertial measurement system was used to record flight parameters such as ground velocity, altitude, pitch angle, roll angle, yaw angle, angular rates, and translational accelerations. These data were stored as .bin log files and were used for post-flight analysis.

However, during the testing of the Blu-baby RC model, limitations of the APM 2.8 logging system were identified. The sampling rate was not sufficient to capture the short-period mode accurately, since the model produced rapid pitch responses. In addition, the internal flash memory capacity of the APM 2.8 was limited to 4 MB. Therefore, only about 5 to 7 minutes of flight data could be recorded after arming, which was not sufficient to obtain useful data after the aircraft stabilization period.

To overcome these limitations, a separate data acquisition system was designed, implementing Arduino nano, external IMU sensor and SD card module. A Kalman filter was also used to filter out the raw pitch and pitch rate data output from the IMU sensor. As a result, rapid changes in pitch behavior and pitch angle could be captured more clearly at a sampling rate around 30Hz. An auxiliary toggle switch was also connected to mark the timestamp at which the sudden elevator deflection was given. This reduced ambiguity during post-processing and allowed the required perturbation window to be isolated accurately.

4.3.2 Flight Tests for Data Extraction

Flight tests were carried out using two UAV models: the developed UAV and the Blu-baby RC model. Initially, the developed UAV was tested at university grounds.

Figure 4.11: During the UAV flight testing

However, these attempts were unsuccessful due to the limited take-off distance, restricted flying area, requirement for quick turns, and some roll instability issues observed during flight. Several test attempts were made after the damaged UAV components had been rebuilt. But none of the attempts succeeded. Therefore, the stable Blu-baby RC model was selected for the data extraction process after its design behavior had been validated using the developed dynamics framework. With the integration of the separate data acquisition system, useful flight data were successfully obtained from the Blu-baby RC model.

Figure 4.12: During the Blu-baby flight testing

4.3.3 Post Processing

A dual method analysis was employed during post processing to verify the modal frequencies of aircraft's dynamic response. In order to get both frequency and the damping estimation, Prony method was used. It is highly effective in extracting oscillatory characteristics from a short transient signal. Paring it with FFT, which provides highly robust direct identification of the dominant frequency peaks gives solid comparative baseline as the cross validation against Prony method. This redundancy minimizes the errors occur due to curve fitting and elevates the accuracy of the result to be compared with the analytical results. The details about the FFT and Prony method are attached in the APPENDIX F: Flight Data Extraction.

An example for extracted data around the doublet region is shown in Figure 4.13.

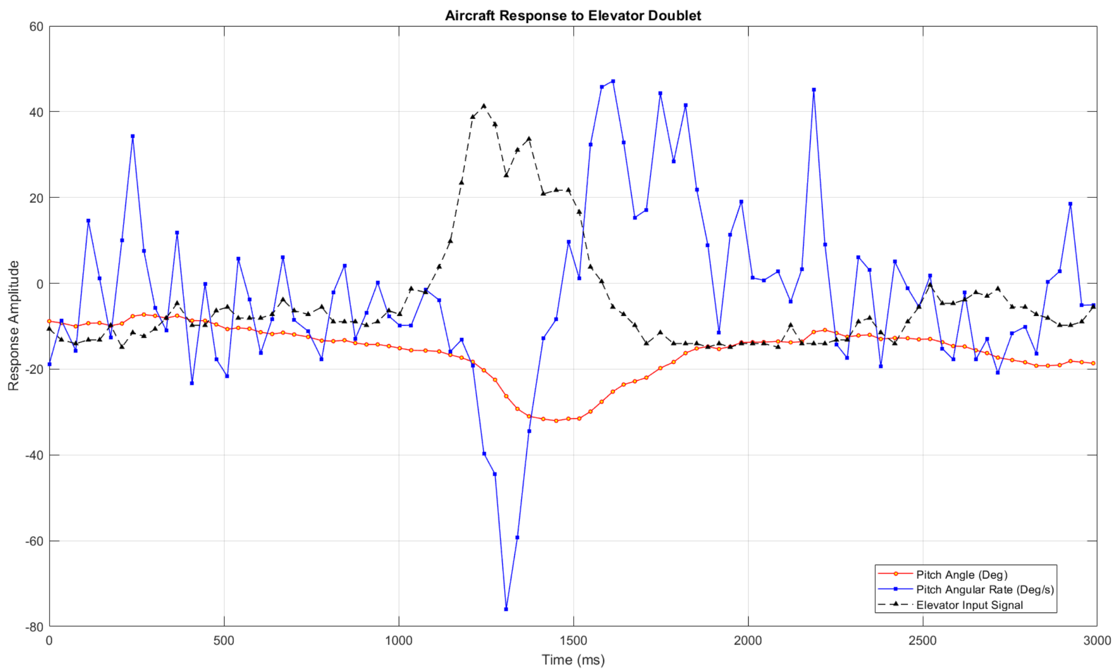

Figure 4.13: Extracted flight data around doublet input

Observing the flight data, doublet responses were identified and saved as .csv files for post processing. An example trimmed doublet pitch rate response of the aircraft can be show as in Figure 4.14.

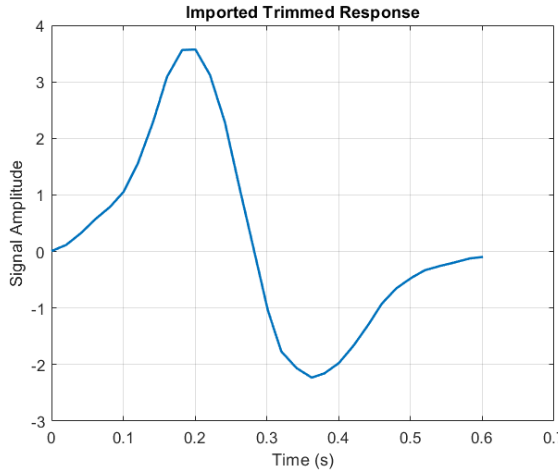

Figure 4.14: Pitch rate response of the aircraft

During the flight testing, it was observed that the aircraft didn't show a clear phugoid oscillation behavior after an elevator doublet was applied. The aircraft returned to the equilibrium position rapidly, withing approximately 0.5s. Since the RC aircraft used for the flight data extraction was highly stable, the long period phugoid response could not be clearly captured through the pitch angle and pitch rate data. This is mainly due to the very low frequency in phugoid mode and longer period compared to the short period response, making it difficult to identify within the available flight data. Therefore, during the post-processing the flight data, only short period mode was considered in the aircraft dynamic response. An assumption was made that the response due to elevator doublet is dominated by short period mode. As a result of that, Prony's method was applied with model order of 2, representing one complex conjugate pole pair corresponding to the short period oscillation.

4.3.3.1 MATLAB Analysis Results

The frequency and damping coefficient were determined using two flight data parameters, pitch angle response and pitch rate response. The resulting

Table 4.22: Pitch rate analysis results

Doublet Number	Characteristics of SPPO Mode				
	Frequency (Hz)		Frequency (rad/s)		Damping Ratio
	FFT	Prony	FFT	Prony	Prony
1	1.6105	2.3502	10.1191	14.7667	0.0839
2	1.1927	1.8468	7.4940	11.6038	0.1378
3	1.6685	2.0465	10.4835	12.8585	0.0626
4	1.6144	2.4581	10.1436	15.4447	0.2484
5	1.7825	1.4741	11.1998	9.2620	0.7929
Mean Value	1.5737	2.0351	9.8880	12.7872	0.2651

Table 4-.4.23: Pitch analysis results

Doublet Number	Characteristics of SPPO Mode				
	Frequency (Hz)		Frequency (rad/s)		Damping Ratio
	FFT	Prony	FFT	Prony	Prony
1	1.4006	1.5812	8.8002	9.9350	0.2046
2	1.8863	1.1900	11.8520	7.4770	0.0981
3	1.7825	1.7702	11.1998	11.1225	0.2052
4	1.2839	1.1448	8.0670	7.1930	0.2454
5	1.5659	1.4504	9.8388	9.1131	0.3607
Mean Value	1.5838	1.4273	9.9516	8.9681	0.2228

The main reason for the difference between the frequency calculated from FFT and that calculated from the Prony technique lies in their underlying assumptions. FFT gives us the dominant frequencies in the chosen signal segment, while Prony determines the modal frequency based on a sinusoidally damped signal. Hence, FFT is a spectral peak picking approach, while Prony is a time domain parametric identification approach.

In this flight test, the aircraft response due to the elevator doublet input was a short, damped transient rather than a continuous periodic signal. This makes FFT less accurate because FFT assumes that the analyzed signal is stationary and periodic within the selected time window. Since the short-period response decays rapidly with a short period of time, only a single oscillation cycle is available in the data. This reduces the frequency resolution of the FFT and causes the dominant spectral peak to shift from the actual modal frequency.

Prony's method directly approximates the damped modal frequency from the time domain response. On the other hand, the FFT uses the frequency domain. Since the response was assumed to be dominated by the short-period mode, a second-order Prony model was used to represent one complex conjugate pole pair. This allows Prony's method to capture the damped oscillatory behaviour more directly than FFT. For this reason, the Prony frequency is expected to be closer to the actual short-period modal frequency, specially for short transient responses.

The deviations between the pitch and pitch-rate results occur because the two signals represent different aspects of the aircraft response. Pitch angle shows the overall attitude motion, while pitch rate represents the faster rotational response. Since pitch rate is the derivative of pitch angle, it is more sensitive to high-frequency short-period motion and noise which is ideal for this study. Therefore, the Prony method can identify a higher frequency from pitch-rate data, while pitch-angle data may give a smoother response with lower frequency and higher damping. The close FFT mean values show that both signals captured the same dominant short-period behaviour, but the Prony results differ because the method is more sensitive to signal quality, trimming window, and decay characteristics.

Considering all above facts, Prony's method with pitch rate response was selected for the comparison between the AVL predicted results and experimental data.

4.3.3.2 Comparison of MATLAB Analysis Results with Experimental Data

Table 4.24: Comparison between experimental data and simulated data

AVL Predicted			Experimental Data		
Frequency (Hz)	Frequency (rad/s)	Damping Ratio	Frequency (Hz)	Frequency (rad/s)	Damping Ratio
4.2224	26.5302	0.5821	2.0351	12.7869	0.1233

As shown in Table 4.24, significant deviation was observed between the AVL-predicted short-period characteristics and the experimentally extracted values. This deviation is expected because AVL is based on a linear, inviscid aerodynamic model and assumes ideal aircraft geometry, rigid-body motion, accurate mass properties, and small disturbance flight about a trim condition. In actual flight, the response is affected by several additional factors such as incorrect or approximate mass moment of inertia values, slight CG mismatch, structural flexibility, servo delay, control surface free play, atmospheric disturbances, sensor noise, and uncertainty in the actual trim speed. The real aircraft also includes viscous drag effects, propeller slipstream effects, and nonlinear aerodynamic behaviour, which are not fully captured by the AVL model.

The lower experimental damping ratio implies that there is less damping effect on pitch in the real aircraft than was estimated. This may be due to reduced effectiveness of horizontal tail, lower dynamic pressure of tail in actual conditions, delayed response of elevator, or inaccuracies related to estimation of pitch moment of inertia. Also, the values of experimental frequency and damping ratio were determined by short transient response using Prony's method, which consisted of only few cycles.

Therefore, the results of AVL analysis should be regarded as an ideal prediction, while the flight test results show the real dynamic characteristics of the fabricated aircraft.

Comparison of two sets of data provides valuable information about uncertainty of modelling, but exact agreement should not be expected without precise measurements of mass properties, geometry modelling, actuators modelling, and identification on flight tests.

4.4 Parametric Sensitivity Analysis and Fine Tuning of UAV Flight Dynamics

This chapter identifies the key geometric and mass parameters that influences the stability derivatives and examines how these derivative variations affect the five primary dynamic modes. Based on these relationships, a systematic methodology can be established to fine tune the UAV configuration to move the dynamic poles towards more stable regions of the S-plane and achieve improved flying qualities.

4.4.1 Parametric Dependencies of Stability Derivatives

The flight dynamic characteristics of the UAV are influenced by its geometric and mass properties, which can be directly matched into its stability derivatives. Based on the fundamental meaning of each derivative, primary inside and outside variables are summarized below.

4.4.1.1 Phugoid Mode

Phugoid is an oscillatory mode with a frequency and a damping ratio. This mode is mainly governed by the Xu and Zu derivatives which represents the drag and lift variations due to forward velocity perturbations. Therefore, any changes to the total lift and drag of the aircraft influences these derivatives.

Xu mainly depends on how the forces in X direction, which are the thrust and the drag of the aircraft vary with forward velocity perturbation. The drag coefficient involves in this since when the velocity increases, the drag increases as well. The phugoid damping is strongly connected with the drag coefficient as most of the energy dissipation is done due to drag.

Zu is mainly governed by the lift force and the weight of the aircraft. Lift can be expressed as,

$$L = \frac{1}{2}\rho U^2 S C_L$$

Where S is the wing planform area and U is the forward velocity of the aircraft. The lift directly varies with the dynamic pressure term. Typically for an aircraft, compressibility effects are considered where the lift is affected by the mach number (M). Therefore, a lift related parameter is introduced as,

$$C_{L_u} = \frac{M^2}{1 - M^2} C_{L_0}$$

Where, M represents the mach number and C_{L0} represents the reference lift coefficient. However, the variation of lift due to mach number effect can be neglected for this study of small UAVs, as compressibility effects are not considered for low Reynolds number cases.

Considering all mentioned facts, the mainly influencing parameters can be listed down as,

- Main wing aerofoil lift slope
- Main wing aspect ratio
- CL/CD of the total aircraft
- Main wing sweep angle
- Main wing incident angle
- Mass of the aircraft

4.4.1.2 Short Period Mode

Short period is also an oscillatory mode, and the characteristics are strongly governed Mq and Mw derivatives. These are also known as pitch damping derivative and pitch stiffness derivative.

Mq derivative represents the pitching moment generated due to the change in pitch rate. When the aircraft pitches, the horizontal stabilizer experiences an additional effective angle of attack caused by the rotational motion of the tail section. This leads to lift variation, which is acted as a strong restoring pitch moment about the center of gravity of the aircraft. Mainly, a moment is governed by a force and an arm length. So, the Mq derivative is influenced by,

- Horizontal stabilizer aerofoil lift slope
- Horizontal tail volume ratio (VH)
- Horizontal stabilizer effectiveness parameter
- Horizontal stabilizer incident angle

Out of these, VH is an important parameter since it represents the overall control authority of the horizontal tail. Here, horizontal tail volume ratio can be expressed as,

$$V_H = \frac{S_t l_t}{S\bar{c}}$$

A larger horizontal stabilizer area (St) increases the aerodynamic forces acted upon the tail section, while the tail arm (lt) increases both pitch rate induced angle of attack and pitching moment. Therefore, lt has a stronger influence to this stability derivative. Apart from these, the pitch moment of inertia (Iyy) of the aircraft determines how strongly the aircraft responds to a given pitching moment.

M_w represents the pitching moment generated due to the change in vertical velocity or angle of attack. This is closely related to the longitudinal static stability of the aircraft. This derivative can also be expressed as the slope of the Cm vs α curve (Total moment coefficient of the aircraft around the center of gravity), which is used to define the longitudinal static stability. A more negative slope means more negative M_w leading to stronger restoring moment and increase in short period frequency. This is affected by,

- Center of gravity location (Xcg)
- Aerodynamic center location (Xac)
- Horizontal tail volume ratio (VH)
- Main wing aspect ratio
- Main wing aerofoil lift slope
- Horizontal stabilizer aerofoil lift slope

The separation of X_{cg} and X_{ac} represents the static margin of the aircraft. Moving the center of gravity forward, typically increases the restoring pitching moment and makes the M_w more negative.

The main wing characteristics come into play with the induced angle of attack of the tail section, due to the downwash. Mainly, downwash can be expressed as,

$$\varepsilon = \varepsilon_0 + \frac{d\varepsilon}{d\alpha}\alpha$$

Where. ε_0 is the downwash at zero aircraft angle of attack and $\frac{d\varepsilon}{d\alpha}$ is the downwash gradient. It can also be represented as,

$$\frac{d\varepsilon}{d\alpha} \approx \frac{2C_{L_\alpha}}{\pi(AR_w)}$$

This induced downwash angle changes the effective horizontal stabilizer angle, which affects the moment from the tail section.

4.4.1.3 Roll Subsidence Mode

This is a non-oscillatory mode. Therefore, it only has one real eigen values, so it is characterized only by a roll time constant. This mainly depends on Roll damping derivative Lp which describes the change in rolling moment due to perturbation in roll rate of the aircraft. This depends on,

- Main wing aerofoil lift slope
- Main wing taper ratio
- Main wing twist

During the rolling motion, one wing experiences additional vertically upward velocity component and the other wing experiences an additional vertically downward velocity component. This increases the effective angle of attack of the downward rolling wing and vice versa. This change in effective angle of attack creates a lift change between the two wng halves, restoring the rolling moment back to its equilibrium. So, main wing aerofoil lift slope is a key parameter to this variable.

Taper ratio of the wing changes the lift distribution along the spanwise direction. Twist of the wing also changes the effective angle of attack along the spanwise direction. Both of these govern how the lift distributes along the wing during a roll.

Rolling moment inertia (Ixx) is the most important mass property that affects the roll subsidence. For a same rolling moment, the aircraft with larger rolling inertia produces a slower response and higher rolling time constant. Therefore, reducing the Ixx allows the aircraft to damp roll disturbances more quickly.

4.4.1.4 Dutch Roll Mode

This is an oscillatory mode involving coupled yawing and sideslip motion with a frequency and a damping ratio. Mainly, lateral force (Y) and directional moment (N) play a major role in determining the characteristics.

The derivative $N\beta$ represents the yawing moment due to change in sideslip angle. This is a crucial parameter as it behaves like the directional stiffness of the aircraft. When the aircraft experiences a side slip angle, the vertical stabilizer produces a restoring moment. Specially, these moments governed by the vertical tail arm and the produced lift force by the vertical stabilizer. Due to this, vertical tail volume ratio (Vv) is a key parameter in determining this derivative. Vv can be expressed as,

$$V_v = \frac{S_v l_v}{Sb}$$

Where, b is the wingspan l_v is the vertical tail arm and S_v is the vertical stabilizer area. Apart from this, vertical tail efficiency represents how efficiently the vertical tail operates in the flow field. If the vertical stabilizer is placed in the wake region of the main wing or horizontal tail wing, the effectiveness of it reduces drastically. The vertical stabilizer lift slope also determines the magnitude of the lateral force generated.

Due to the side wash effects, the flow that reaches the vertical stabilizer is not exactly equal to the total aircraft side slip angle. The effective angle of attack can be expressed using the side wash derivative as,

$$\beta_{effective} = \beta - \frac{d\sigma}{d\beta}\beta$$

Nr derivative represents the change in yawing moment due to yaw rate perturbation. This mainly acts as the yaw damping derivative. During a yaw, the vertical stabilizer experiences an additional lateral velocity component due to the rotation. The force produced by this opposes the yawing rate.

So, both Nβ and Nr are basically governed by,

- Vertical tail arm
- Side wash derivative
- Vertical stabilizer aerofoil lift slope
- Vertical tail efficiency

The side force derivative Yβ represents the change in side force due to change in side slip angle. Apart from the vertical stabilizer, the fuselage side area also contributes to the side force, during a side slip. This is commonly related to the fuselage volume, wing reference area and span where wingspan is used to non dimensionalize the values.

The side force derivative Yr represents the change in side force due to change in yaw angle of the aircraft. When the aircraft yaws, te vertical stabilizer experiences additional force due to yaw rate and the vertical tail arm. Therefore, this derivative strongly depends

- Vertical tail arm

- Vertical stabilizer area
- Vertical tail efficiency
- Vertical stabilizer aerofoil lift slope

Apart from all these derivatives, yawing moment of inertia (Izz) also contributes to the Dutch roll characteristics. The higher the Izz value, lower the rate of response to a yawing disturbance.

4.4.1.5 Spiral Mode

Spiral is a slow, non-oscillatory mode with the time constant as the main characteristic of the motion. A stable spiral mode requires the spiral related pole to be in the left side of the s-plane. The main derivatives responsible for spiral mode are Lβ, Nβ, Nr and Lr together with the yaw moment of inertia (Izz).

The derivative Lβ represents the rolling moment change due to a sideslip angle perturbation. This is mainly associated with the dihedral angle of the main wing. When the aircraft is subjected to a sideslip angle, the positive dihedral angle of the wing creates different effective angle of attacks between the two wings, creating a restoring moment to level the wings up. Main wing taper angle, twist and main wing aerofoil lift slope also contribute to this restoring moment, influencing the Lβ value.

Nβ is the change in yawing moment due to a side slip angle perturbation. Because of the longer distance between the center of gravity and the vertical stabilizer, the force acted upon it creates a larger moment. As a result of that, vertical stabilizer characteristics mainly determine this derivative. Those characteristics include parameters such as,

- Vertical tail volume ratio
- Vertical tail arm
- Vertical stabilizer area
- Vertical tail efficiency
- Vertical stabilizer aerofoil lift slope

The derivative Nr represents the yawing moment change due to yaw rate perturbations. During yawing, the vertical tail experiences an additional lateral velocity component, which acts as a restoring force. Due to this effect, this derivative is also driven by the vertical stabilizer characteristics that are mentioned above.

Lr derivative means the change in rolling moment due to yaw rate perturbation. When the aircraft yaws, the outer wing is exposed to a higher relative velocity due to the wing sweep, creating a lift imbalance between two wings. This unbalanced force acts as a rolling moment. Therefore, Lr is affected by,

- Main wing sweep angle
- Main wing aerofoil lift slope
- Main wing twist
- Main wing taper

Apart from all these derivatives, yawing moment of inertia (Izz) also determines the spiral mode characteristics. A larger Izz leads to slow response to the yawing moments and vice versa.

Spiral mode is associated with the vertical stabilizer. Due to the side wash, the effective angle of attack that it experiences is different than the fuselage. Therefore, the sidewash derivative is also a key contributor to the characteristics of spiral mode.

4.4.2 Sensitivity Analysis Results

This section presents how changes in selected geometric parameters influence the aerodynamic stability derivatives and, how the combined variation of these derivatives affects the dynamic characteristics of flying modes. The results can be presented by plots, showing the variations of each modal characteristics with respect to a change in selected geometric design parameter. The trend of each case can be categorized into linear or nonlinear. For the linear behaviour, a sensitivity value is calculated to quantify the rate of change of modal characteristics with respect to the selected geometric parameter. All the relevant tables and graphs are presented in VII below.

The numerical sensitivity values obtained in this study are specific to the used UAV configuration, trim condition, and inertia properties. It should be used primarily as a trend learning tool, rather than as universal design constants that can be applied to any aircraft configuration. To develop more general design guidelines, the same sensitivity analysis should be performed for many UAV configurations with different wing geometries, tail geometries, mass distribution properties and operating conditions. By comparing the results from this large database of configurations, recurring trends and how much a particular geometrical parameter is sensitive to modal characteristic can be identified. Therefore, the study in this provides useful insights into the relationship between the modal characteristics and geometric parameters, while a broader conclusion requires analysis of wider set of UAV designs

4.4.2.1 Phugoid Mode Sensitivity

As the AVL doesn't calculate the Xu and Zu derivatives, eigen values approach was used to determine the frequency and damping ratio of phugoid mode. The main wing aspect ratio, main wing sweep and horizontal tail sweep were taken for this analysis.

The main wing aspect ratio was varied from 7 to 9 in this study, and it showed a linearly increasing trend in phugoid frequency. 1% change in the aspect ratio resulted in 0.2429% increment in the phugoid frequency. How ever, the damping ratio didn't show a strong linear relationship with the main wing aspect ratio as seen from the graph.

The main wing sweep angle was then varied from 0 degrees to 30 degrees and none of modal characteristics showed a linear trend. In contrast, the horizontal tail sweep angle variation affected both phugoid damping ratio and frequency in an exponential way.

4.4.2.2 Short Period Mode Sensitivity

For this study, main wing aerofoil lift slope, sweep angle, aspect ratio, horizontal tail efficiency, lift slope, sweep and horizontal tail volume ratio were chosen as parameters.

Horizontal tail volume ratio was the most effective parameter in short period sensitivity analysis. Both frequency and damping ratios showed a predictable linear trend. The study showed that, 1% change in horizontal tail volume ratio resulted in 1.03% and 0.68% increment in frequency and damping ratio respectively.

Both horizontal tail and main wing sweep angle influenced both frequency and damping ratio to behave in an exponential way as shown in the graph.

According to the results obtained, horizontal tail lift slope showed a linear, but relatively low sensitivity towards the characteristics. 1% increment in horizontal tail lift slope resulted 0.004% and 0.001% decrement in frequency and damping ratio respectively. All the other parameters showed a non-linear behaviour as seen from the graphs.

4.4.2.3 Roll Subsidence Mode Sensitivity

Main wing lift slope, aspect ratio and dihedral angle were selected as parameters for this study. Since the roll subsidence is a non-oscillating mode, it is characterized by the time constant. None of them showed a linear trend between the parameter and the time constant. The wing dihedral angle showed approximately parabolic trend between time constant and dihedral angle. As the time constant is driven by only a single stability derivative, this clearly shows the geometric parameter effects that are mentioned in the previous section are non-linear.

4.4.2.4 Dutch Roll Mode Sensitivity

Parameter sweeps of vertical tail volume ratio and main wing aspect ratio were considered for this. The vertical tail volume ratio sweep showed an initial increment of 2.43% in frequency and 1.03% in damping ratio for a 4% increment of vertical tail volume ratio. This agrees with the expected physical behaviour mentioned before, since increasing vertical tail volume ratio increases directional stiffness and yaw damping However, this trend was discontinued after the initial parameter variation and both frequency and damping ratio remained constant afterwards.

The aspect ratio sweep resulted in highly non-linear trend in both frequency and damping ratio plots.

4.4.2.5 Spiral Mode Sensitivity

Main wing dihedral angle, vertical tail volume ratio, main wing aspect ratio and main wing lift slope were considered for the spiral sensitivity analysis.

The dihedral angle sweep produced noticeably clear sensitivity results. Increasing the wing dihedral angle from 0 degrees to 8 degrees reduced the spiral time constant from approximately 40.2s to 3.0s. This represents a reduction of more than 92%, showing that dihedral angle is the most dominant geometric parameter for improving spiral stability in this UAV configuration. The relationship between dihedral angle and spiral time constant was nonlinear. The largest improvement occurred at low dihedral angles, and the improvement became smaller as dihedral angle increased. This justified the physical explanation of wing dihedral effect as mentioned in previous section.

The vertical tail volume ratio sweep indicated a small increment in spiral time constant from approximately 3.64s to 3.81s when Vv increased from 0.025 to 0.026. After this point, the time constant remained nearly constant for several cases with increasing Vv.

The main wing aspect ratio sweep showed a generally increasing spiral time constant, from 3.48s at an aspect ratio of 7 to 7.13s at an aspect ratio of 9. However, the trend was nonlinear and showed some variation between cases.

The main wing lift slope sweeps also showed nonlinear behaviour. The spiral time constant varied from approximately 3.64s to 8.42s and then reduced to about 4.83 s. This indicates that changing the wing aerofoil or lift slope affects several coupled lateral directional derivatives simultaneously. Therefore, it should not be treated as a simple linear tuning parameter for spiral stability.

4.4.3 Methodological Approach to Fine-Tuning Dynamic Modes

In order to enhance the flying characteristics of the UAV, the dynamic modes cannot be arbitrarily tuned. Hierarchical tuning has to be implemented since any modification made to the geometry will impact several modes of flight. The tuning process must start with the modes that have the highest impact on aircraft stability and controllability, followed by the coupled lateral directional modes, and then end with the least important modes.

4.4.3.1 Step 1: Fine tune short period mode

The objective is to achieve suitable short period frequency and damping ratio by improving pitch stiffness and pitch damping of the aircraft. Since short period is mainly governed by pitch stiffness derivative (Mw) and pitch damping derivative (Mq), the most effective parameters to modify are those that increase the stabilizing contribution of the horizontal tail and improve the pitching moment response of the aircraft.

1. Increase horizontal tail volume ratio (Vh): This is the most effective and predictive parameter for short period tuning in the AVL results. Increasing Vh increased both the short period frequency and damping ratio. This can be done by,
    - Increasing horizontal tail arm length
    - Increasing horizontal stabilizer area
    - Reducing main wing reference area if possible
    - Reducing mean aerodynamic chord length
2. Move the center of gravity forward: A forward CG increases static margin and makes the pitching moment slope more stabilizing. This improves pitch stiffness and helps stabilize the short period mode. Important to note that, excessive forward CG should be avoided because it increases the elevator control input to keep the aircraft at trimmed condition, increasing the drag excessively.
3. Reduce the pitch moment inertia (Iyy): Physically, this can be done by moving the heavy components of the UAV such as batteries and payload closer to the aircraft

CG. This allows the aircraft to respond more quickly to stabilizing pitching moments.

Important to note that increasing the horizontal tail sweep of the UAV as the AVL analysis revealed a decrement in the short period frequency and damping ratio. Therefore, horizontal tail sweep cannot be used as a main tuning parameter for improving short period performance.

4.4.3.2 Step 2: Fine tune spiral mode

The main objective is to reduce the spiral time constant or move the spiral pole further into the stable region (Left side) of the s-plane. This can be achieved by the modifications below.

1. Increase the wing dihedral angle: This is the most efficient way of tuning spiral mode according to AVL results obtained. The increase in dihedral led to decrease in the spiral time constant.
2. Use effective dihedral devices: Due to the limitation of increasing the dihedral beyond certain angles, winglets, and wing tip dihedrals can be used.
3. Reduce yaw moment of inertia: This can be achieved by placing heavy components closer to the aircraft center line.
4. Adjust wing sweep: The wing sweep influences the lateral stability and might aid in enhancing the performance during the spiral mode. However, it also influences the Dutch roll effect, longitudinal effects, and aerodynamics. Therefore, it should be used as a secondary tuning parameter.

Spiral mode and Dutch-roll mode must be tuned together. An increment in dihedral angle increases spiral stability but may affect Dutch roll stability.

4.4.3.3 Step 3: Fine tune Dutch roll mode

Dutch roll characteristics are mainly governed by, vertical stabilizer parameters, side force generation and yaw inertia. This can be fine-tuned by using the methods mentioned below.

1. Increase vertical-tail volume ratio: This improves the directional stability and yaw damping of the aircraft. This can be done by.
    - Increasing vertical stabilizer area
    - Increasing vertical tail arm

2. Improve vertical stabilizer efficiency: The vertical stabilizer should be placed away from highly disturbed wing or fuselage wake. However, the placement of the vertical stabilizer should be carefully done considering the spin recovery effects.
3. Change the vertical stabilizer aerofoil: Aerofoils with higher lift slopes increases the side force generation when the aircraft is subjected to side slip angle. This helps in improving the directional stability and yaw damping.

4.4.3.4 Step 4: Fine tune roll subsidence mode

This is mainly controlled by roll damping derivative Lp. Since this mode is non oscillatory, it should be assessed using roll time constant. The objective is to achieve fast roll rate decay by reducing the roll time constant.

1. Increase dihedral angle: Increment in dihedral angle showed a clear trend of parabolic reduction in time constant of roll subsidence. This is due to the linearization of aircraft motion equations and assumptions done in the process of achieving the reduced order model for roll subsidence. Therefore, practically dihedral angle is a good initial parameter to be used for fine tuning the roll time constant.
2. Reduce roll moment of inertia (Ixx): This is the most practical way to improve roll subsidence. Heavy components should be placed closer to the fuselage centreline instead of near the wing tips.
3. Modify wing aerofoil: Since the roll damping mainly happens due to the lift generation from the main wing, the characteristic of main aerofoil is crucial in deciding the roll damping characteristics. Aerofoils with higher lift slope generates more restoring moments.
4. Adjust wing taper ratio and twist: Both wing taper and twist affect the spanwise lift distribution of the wing.

4.4.3.5 Step 5: Fine tune phugoid mode

The main objective of this is maintain positive damping ratio and an acceptable low frequency long period response. Mainly phugoid characteristics are associated with exchange between kinetic and potential energy. This gives lift, drag, velocity and mass top priority for fine tuning. However, during the flight test validation phase, the phugoid mode was barely visible to the flight pitch response data due to the significantly lower frequency. Even though the AVL simulated results influenced the characteristics of phugoid mode, practically, the aerodynamic changes rarely affect the characteristics. If critical, the damping ratio can be increased by the following methods.

1. Adjust drag related geometry: This is a trade-off between the flying quality and the efficiency of the aircraft. As the phugoid is strongly related to energy dissipation, aerodynamic drag acts as a damper.
2. Adjust lift related parameters: Mainly, the main wing aerofoil can be changed.

5. Conclusions and Future Work

5.1 Conclusions

The flight dynamic and control framework demonstrates the capability of connecting the UAV design with dynamic behavior of the airframe and the stability augmentation logic simultaneously. This approach enables the feasible iterative design procedure before prototyping providing financial benefits. From design stage of empennage section to evaluation of performance, static and dynamic stability criteria and stability augmentation the python based modular framework work through mathematical approaches and follow well established guidelines to determine UAV geometric parameters.

The proposed fine-tuning methodology provides a systematic way of improving the UAV dynamic characteristics by linking each flight mode to its dominant stability derivatives and corresponding geometric or mass parameters. The short-period mode should be treated as the priority because it directly affects longitudinal stability and controllability, mainly through horizontal-tail volume, CG location, and pitch inertia. The spiral and Dutch-roll modes must then be tuned together because both are strongly coupled through lateral-directional derivatives, especially dihedral effect and vertical-tail contribution. Roll subsidence can be refined by reducing roll inertia and improving the wing lift distribution, while phugoid tuning should be considered as a secondary adjustment because it is a slow mode and is less sensitive to practical geometric changes. Overall, this approach ensures that design modifications are not made arbitrarily but are guided by the physical relationship between aircraft geometry, stability derivatives, and pole movement in the s-plane. However, every modification must be rechecked using the complete eigenvalue analysis, since improving one mode may degrade another due to aerodynamic and inertial coupling.

5.2 Limitations of the Study

The UAV developed has been used to retrieve flight data, but difficulties arose as it consumes lot of flying time to get used to the control authority and the sensitivity of the control surfaces. Furthermore, to achieve stable flight conditions to observe the longitudinal behavior after a disturbance, the UAV should be trimmed properly by having reasonably high number of flying tests. The bottleneck occurred as UAV has being damaged after failed attempts of flying tests and requires high amount of time for the repair and get it to a flying condition.

5.3 Recommendations for Future Work

The Validation of the framework can be furtherly improved by solid system identification process using the logged data of tests flights. Utilizing the synchronized input-output flight data, parameter estimation algorithms will be applied to derive formal mathematical models, specifically second-order transfer functions and state-space representations of the short-period mode. Extracting these empirical aerodynamic derivatives will not only validate theoretical stability estimations but also provide the highly accurate, foundational plant model required for the future development and simulation of closed-loop active flight control systems. Furthermore, use of state space model for Yaw damper tuning is also recommended.

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I. APPENDIX A: Flowcharts of the Framework

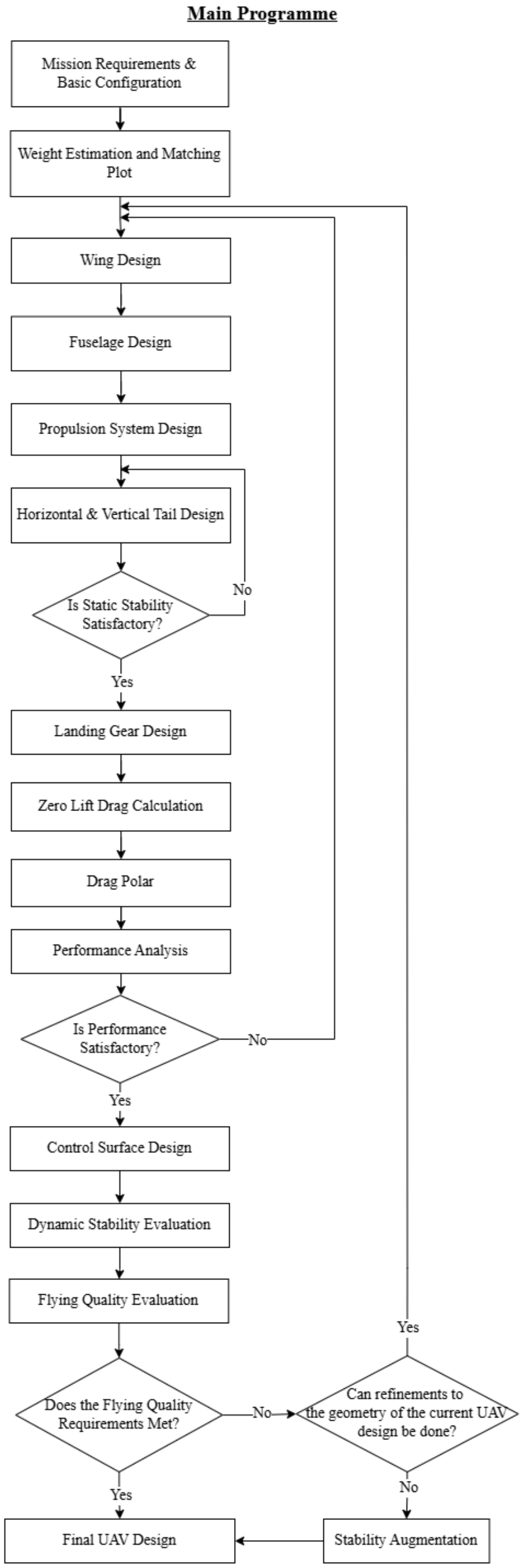


Figure I.1: Mian flowchart of the framework

Horizonal and vertical tail

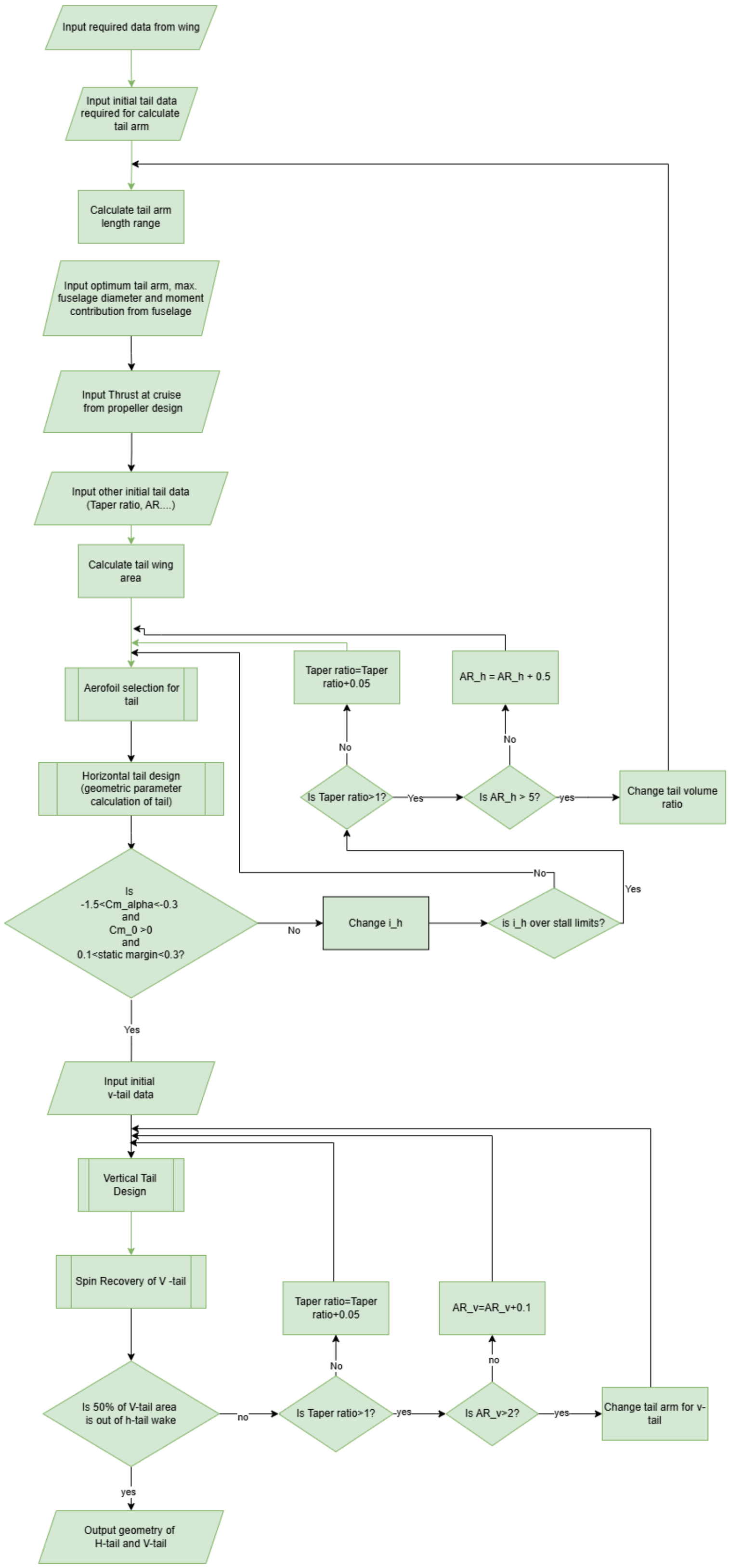


Figure I.2: Flowchart of the horizontal and vertical tail design

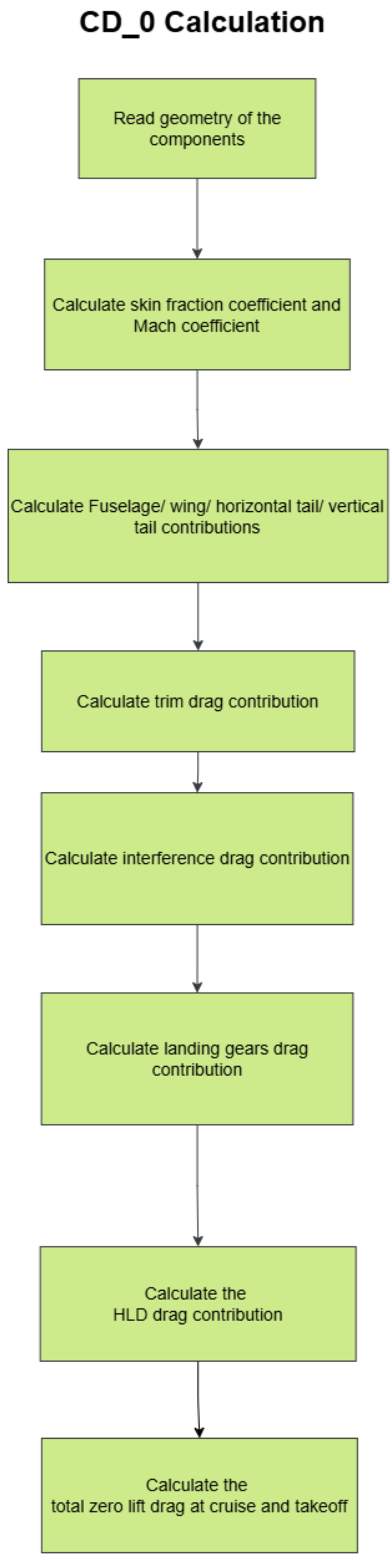


Figure I.3: Flowchart of the CD0 calculations

Performance analysis

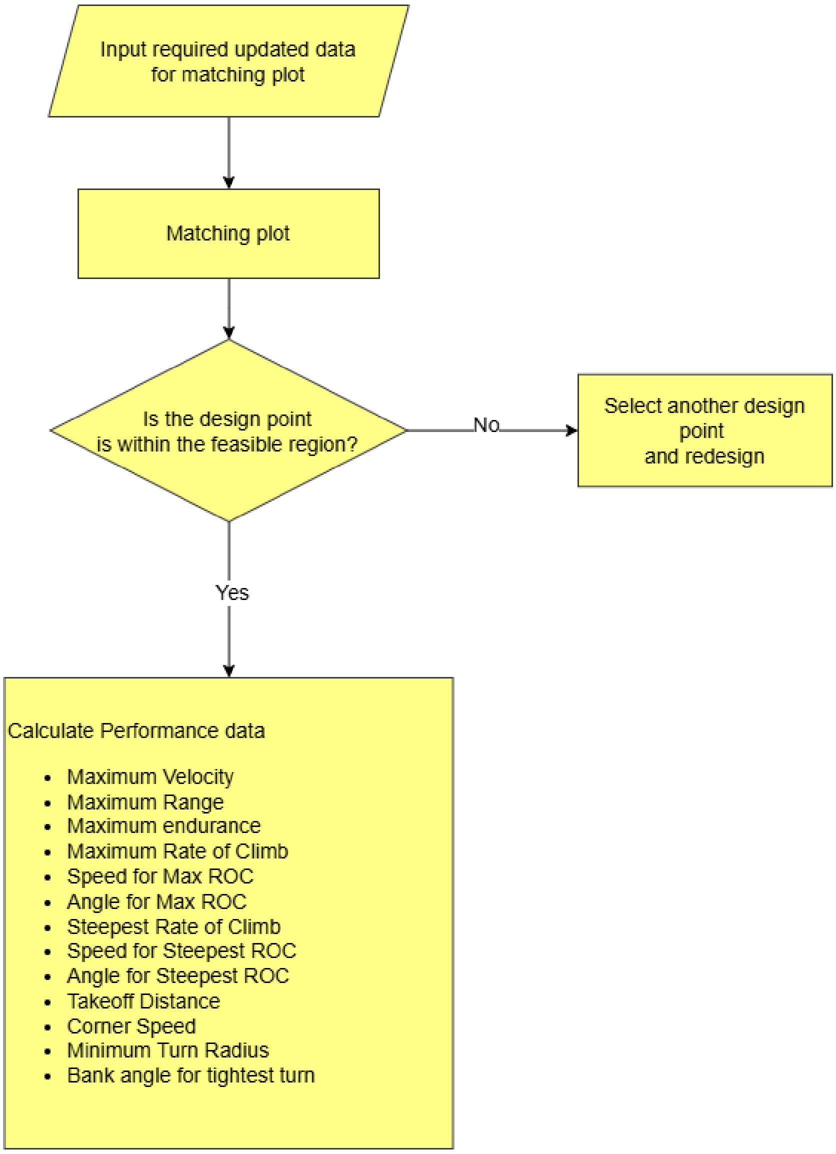


Figure I.4: Flowchart of the performance analysis calculations

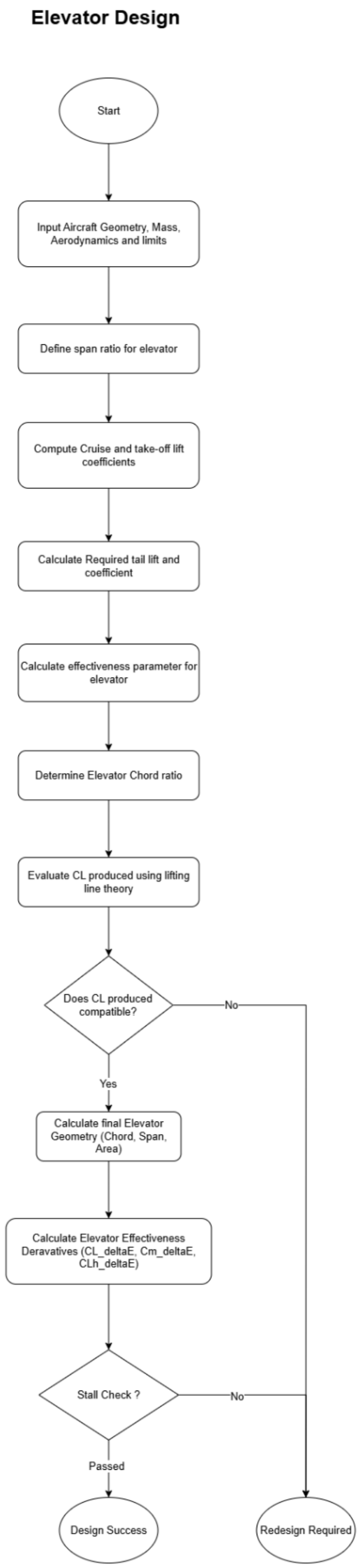


Figure I.5: Flowchart of the elevator design

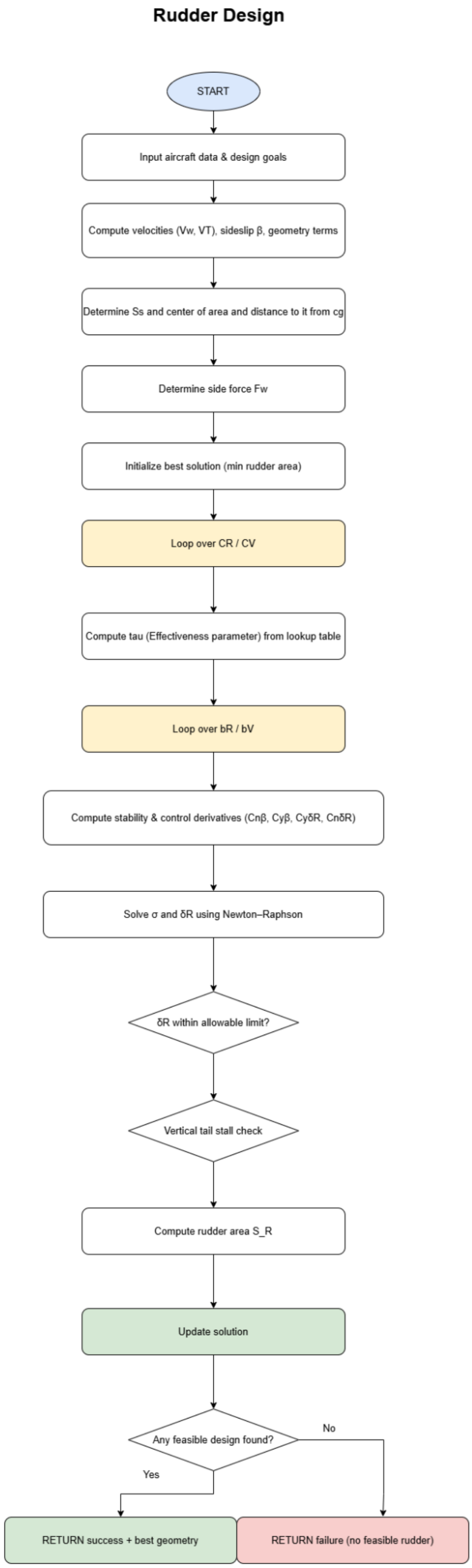


Figure I.6: Flowchart of the rudder design

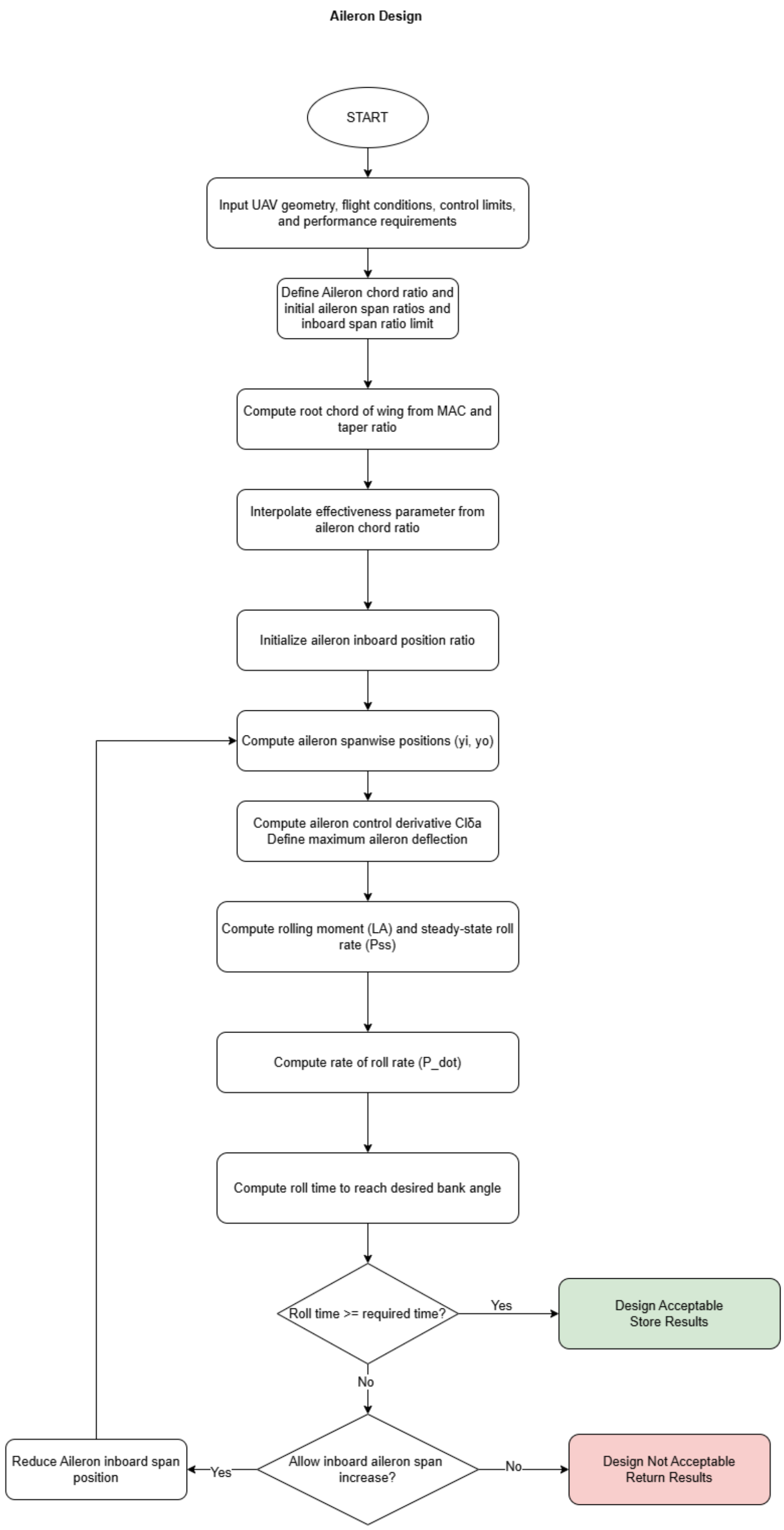


Figure I.7: Flowchart of the aileron design

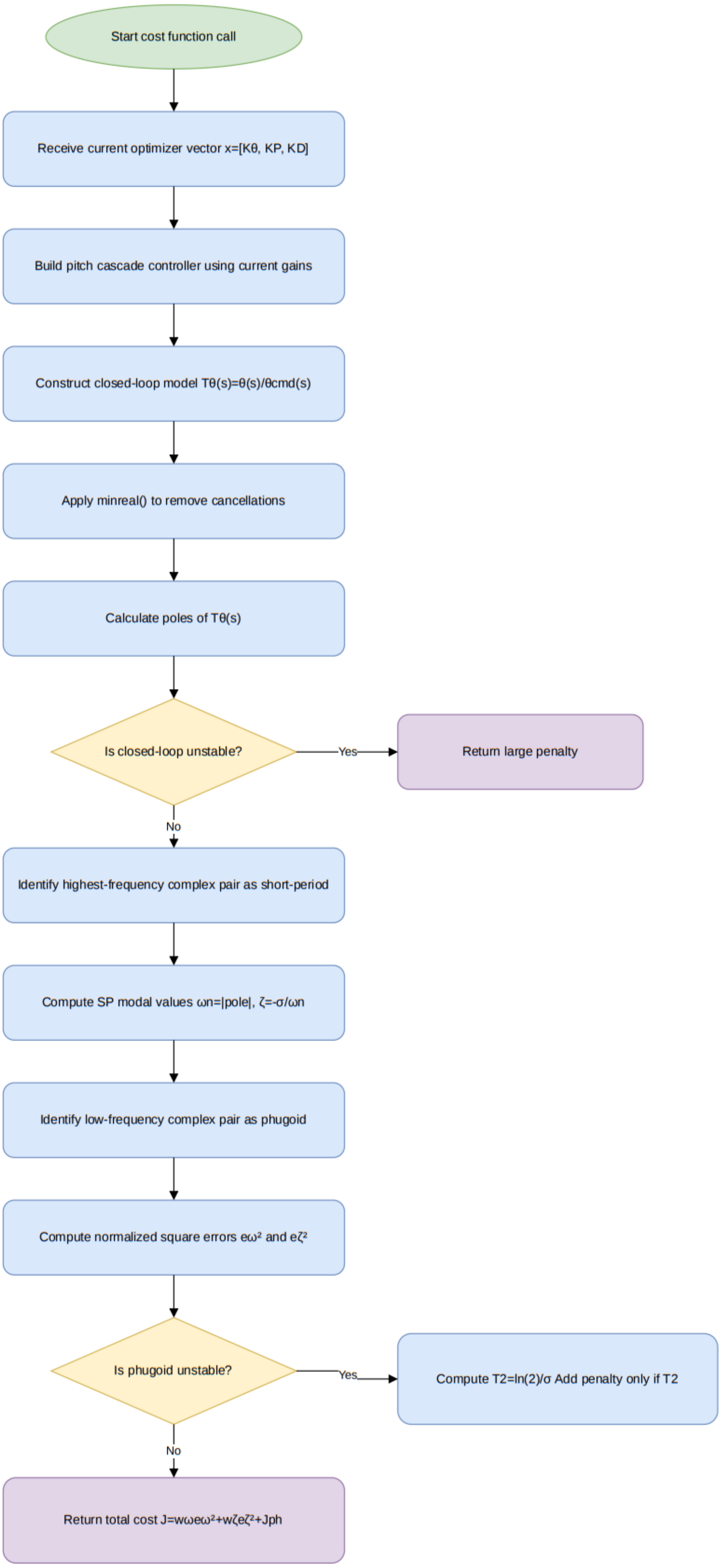


Figure I.8: Flowchart of cost function in pitch damper

Pitch Cascade Transfer Function Derivation

Pitch attitude Transfer function:

$$G_{\theta}(s) = \frac{\theta(s)}{\delta_e(s)}$$

Pitch attitude and Pitch-rate relation:

$$q(s) = s\theta(s)$$

Pitch-rate transfer function:

$$G_q(s) = \frac{q(s)}{\delta_e(s)} = sG_{\theta}(s)$$

Pitch attitude error:

$$e_{\theta}(s) = \theta_{cmd}(s) - \theta(s)$$

Outer pitch attitude loop:

$$q_{cmd}(s) = K_{\theta}F_{LTT}(s)e_{\theta}(s)$$

Relation between K_{θ} and PTCH2SRV_TCONST:

$$K_{\theta} = \frac{1}{PTCH2SRV\_TCONST}$$

Target-rate low-pass filter:

$$F_{LTT}(s) = \frac{\omega_{LTT}}{s + \omega_{LTT}}$$

$$\omega_{LTT} = 2\pi f_{LTT}$$

Inner pitch-rate error:

$$e_q(s) = q_{cmd}(s) - q(s)$$

Pitch-rate PID feedback controller:

$$C_e(s) = SS^2\left[K_P F_{LTE}(s) + \frac{K_I}{s} + K_D s F_{LTD}(s)\right]$$

Error and derivative filters:

$$F_{LTE}(s) = \frac{\omega_{LTE}}{s + \omega_{LTE}}$$

$$F_{LTD}(s) = \frac{\omega_{LTD}}{s + \omega_{LTD}}$$

Feedforward and derivative-feedforward terms:

$$C_{FF} = \frac{SS}{EAS2TAS} K_{FF}$$

$$C_{DFF}(s) = SS^2\, K_{DFF}\, s\, F_{LTD}(s)$$

Combined commanded-rate controller block:

$$H(s) = C_e(s) + C_{FF} + C_{DFF}(s)$$

Elevator command from cascade controller:

$$\delta_e(s) = H(s) q_{cmd}(s) - C_e(s) q(s)$$

General closed-loop pitch transfer function:

$$\frac{\theta(s)}{\theta_{cmd}(s)} = \frac{G_\theta(s) H(s) K_\theta F_{LTT}(s)}{1 + G_\theta(s) H(s) K_\theta F_{LTT}(s) + G_q(s) C_e(s)}$$

Final simplified tuning transfer function:

- For the tuning case,

$$K_I = 0$$

$$K_{DFF} = 0$$

$$SS = 1$$

$$EAS2TAS = 1$$

therefore,

$$C_e(s) = K_P F_{LTE}(s) + K_D s F_{LTD}(s)$$

$$H(s) = K_P F_{LTE}(s) + K_D s F_{LTD}(s) + K_{FF}$$

Thus, the final transfer function used for tuning is,

$$\frac{\theta(s)}{\theta_{cmd}(s)} = \frac{G_\theta(s)[K_P F_{LTE}(s) + K_D s F_{LTD}(s) + K_{FF}] K_\theta F_{LTT}(s)}{1 + G_\theta(s)[K_P F_{LTE}(s) + K_D s F_{LTD}(s) + K_{FF}] K_\theta F_{LTT}(s) + G_q(s)[K_P F_{LTE}(s) + K_D s F_{LTD}(s)]}$$

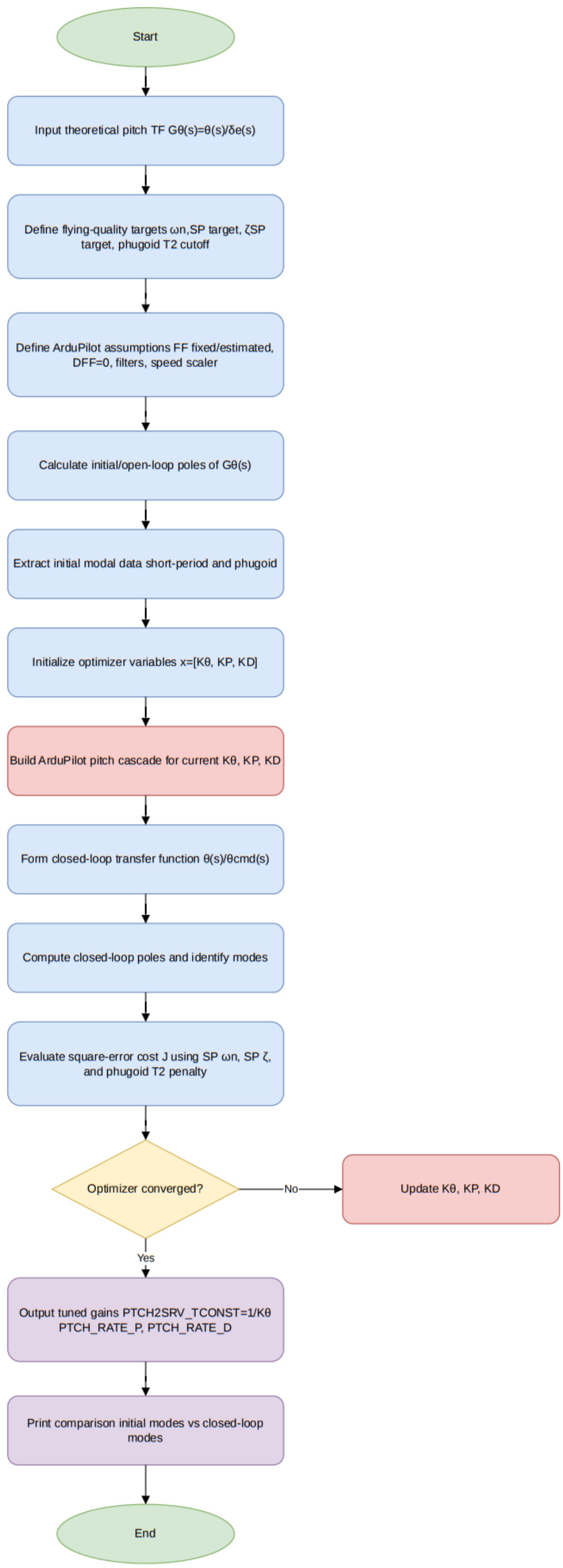


Figure I.9: Flow chart of tuning program

II. APPENDIX B: Nomenclature

V_H - Horizontal tail volume ratio
S_H - Horizontal tail area
l_H - Horizontal tail moment arm
S - Wing reference (planform) area
$\bar{c}$ - Wing mean aerodynamic chord (MAC)
C_{LC} - Cruise lift coefficient
W_{ac} - Aircraft weight
$\rho_{ceiling}$ - Air density at ceiling altitude
V_{cruise} - Cruise airspeed
S_w - Wing reference area
$C_{m_f,\mathrm{trim}}$ - Trim pitching moment coefficient of the fuselage
C_{m0f} - Fuselage pitching moment coefficient at zero angle of attack
$C_{m\alpha f}$ - Fuselage pitching moment curve slope
i_w - Wing incidence angle
C_{m0wf} - Total zero-lift pitching moment coefficient of the wing–fuselage combination
$C_{mac,w}$ - Pitching moment coefficient of the wing at its aerodynamic centre
C_{Lh} - Horizontal tail lift coefficient
C_{m0p} - Propulsion contribution to pitching moment at zero lift
h - Non-dimensional centre of gravity location (fraction of MAC)
h_0 - Non-dimensional aerodynamic centre location (fraction of MAC)
η_h - Horizontal tail efficiency factor
$C_{L\alpha h}$ - Horizontal tail finite-wing lift-curve slope
$C_{l\alpha h}$ - Horizontal tail two-dimensional (aerofoil) lift-curve slope
AR_h - Horizontal tail aspect ratio
α_h - Horizontal tail angle of attack
ε_0 - Downwash angle at zero angle of attack
C_{L0w} - Wing lift coefficient at zero angle of attack
AR_w - Wing aspect ratio
$\frac{d\varepsilon}{d\alpha}$ - Downwash gradient
$C_{L\alpha w}$ - Wing lift-curve slope
ε - Total downwash angle at the horizontal tail
α_w - Wing angle of attack
i_h - Horizontal tail incidence (setting) angle
b_h - Horizontal tail span
S_h - Horizontal tail area
$\bar{c}_h$ - Horizontal tail mean chord
$C_{h,\mathrm{root}}$ - Horizontal tail root chord
TR_h - Horizontal tail taper ratio
$C_{h,\mathrm{tip}}$ - Horizontal tail tip chord
h_t - Vertical height limit for horizontal tail placement relative to the wing MAC aerodynamic centre
l - Fuselage or tail arm length (wake placement constraint)
α_s - Wing stall angle of attack
$C_{m0,w}$ - Wing contribution to zero-lift pitching moment coefficient
$C_{m\alpha,w}$ - Wing pitching moment curve slope
$C_{L0,w}$ - Wing zero-angle-of-attack lift coefficient

$C_{m0,p}$ - Propulsion pitching moment contribution
T_{cruise} - Cruise thrust
$z_{prop,cg}$ - Vertical distance from propeller thrust line to centre of gravity
$C_{w,\text{mean}}$ - Wing mean chord (in propulsion moment equation)
$C_{m0,h}$ - Horizontal tail pitching moment contribution at zero lift
$C_{m\alpha,h}$ - Horizontal tail pitching moment curve slope
C_{m0} - Total zero-lift pitching moment coefficient
$C_{m\alpha}$ - Total pitching moment curve slope (per radian)
θ_{trim} - Trim pitch angle
NP - Neutral point location (non-dimensional, fraction of MAC)
SM - Static margin
S_v - Vertical tail area
b_w - Wing span
V_V - Vertical tail volume coefficient
l_v - Vertical tail moment arm
b_v - Vertical tail span
AR_v - Vertical tail aspect ratio
$\bar{c}_v$ - Vertical tail mean chord
$C_{v,\text{root}}$ - Vertical tail root chord
TR_v - Vertical tail taper ratio
$C_{v,\text{tip}}$ - Vertical tail tip chord
C_{D0} - Zero-lift (parasite) drag coefficient
K_C - Component build-up drag interference factor
$C_{D0,TO}$ - Zero-lift drag coefficient during take-off
$C_{D0,HLD}$ - Zero-lift drag increment due to high-lift devices
C_D - Total drag coefficient
$K = \frac{1}{\pi ARe}$ - Induced drag factor
C_L - Lift coefficient
AR - Wing aspect ratio
e - Oswald efficiency factor
R_{max} - Maximum range
R_t - Rated (total available) energy factor
n - Number of battery packs; also load factor
η_{tot} - Total propulsive efficiency
V_{batt} - Battery voltage
C_{batt} - Battery capacity
ρ_c - Air density at ceiling altitude
W - Aircraft weight
E_{max} - Maximum endurance
V_{ROC} - Best rate-of-climb airspeed
γ - Climb angle
P - Shaft power available
η - Propulsive efficiency
ROC_{max} - Maximum rate of climb
$V_{\gamma max}$ - Best-angle climb speed
P_{max} - Maximum available shaft power
$\sin \gamma_{max}$ - Sine of maximum climb angle
ROC_{steep} - Rate of climb at steepest climb angle

S_G - Ground-roll distance during take-off
m - Aircraft mass
$C_{D,TO}$ - Drag coefficient during take-off ground roll
μ - Ground friction coefficient
$C_{L,TO}$ - Lift coefficient during take-off ground roll
T - Thrust force; also oscillation period
C_{Lmax} - Maximum lift coefficient
S_R - Rotation distance during take-off
T_R - Rotation time during take-off
V_R - Rotation speed
S'_A - Slant airborne distance during take-off
T_{air} - Thrust during airborne phase
D_{air} - Drag during airborne phase
ΔV - Velocity increment during airborne phase
g - Gravitational acceleration
h_o - Obstacle clearance height
S_A - Horizontal airborne distance
S_{TO} - Total take-off distance
V^* - Corner (manoeuvre) speed
n_{max} - Maximum load factor
R_{min} - Minimum turn radius
ϕ_{tt} - Bank angle at tightest turn
$C_{l\delta A}$ - Rolling moment coefficient derivative w.r.t. aileron deflection
τ - Control-surface effectiveness parameter
C_r - Wing root chord
b - Wing span
y_i - Inner spanwise station of aileron
y_o - Outer spanwise station of aileron
λ - Wing taper ratio; also eigenvalue
L_A - Rolling moment generated by aileron lift
ΔL - Incremental lift force
y_A - Effective moment arm for aileron rolling moment
P_{ss} - Steady-state roll rate
C_{DR} - Rolling resistance drag coefficient
y_D - Reference span distance for roll drag
ϕ_1 - Bank angle at steady-state roll rate
I_{xx} - Roll-axis mass moment of inertia
P - Roll rate
L_h - Horizontal tail lift force during rotation
L_{wf} - Wing–fuselage combined lift force
x_{mg} - Main landing gear longitudinal position
x_{acwf} - Wing–fuselage aerodynamic centre position
M_{acwf} - Pitching moment at wing–fuselage AC
a_{zcg} - Vertical acceleration at CG
z_{mg} - Vertical position of main landing gear
x_{cg} - Centre of gravity location
D - Drag force
z_D - Vertical position of drag line of action

z_T - Vertical position of thrust line
I_{yy} - Pitch-axis mass moment of inertia
θ - Pitch angle; also pendulum angular displacement
x_{ach} - Horizontal tail aerodynamic centre position
$C_{m\delta E}$ - Pitching moment coefficient derivative w.r.t. elevator deflection
b_E - Elevator span
τ_e - Elevator effectiveness factor
$C_{L\delta E}$ - Aircraft lift coefficient derivative w.r.t. elevator deflection
$C_{Lh\delta E}$ - Tail lift coefficient derivative w.r.t. elevator deflection
V_T - Total airspeed under cross-wind
U_1 - Forward speed component
V_w - Cross-wind speed component
F_w - Cross-wind side force
S_S - Projected side area
C_{Dy} - Side drag coefficient
C_{n0} - Yawing moment coefficient at zero sideslip
$C_{n\beta}$ - Yawing moment derivative due to sideslip
β - Sideslip angle
σ - Heading correction angle; also real part of eigenvalue
$C_{n\delta R}$ - Yawing moment derivative w.r.t. rudder deflection
δ_R - Rudder deflection angle
d_c - Moment arm for cross-wind side force
C_{y0} - Lateral force coefficient at zero sideslip
$C_{y\beta}$ - Lateral force derivative due to sideslip
$C_{y\delta R}$ - Lateral force derivative w.r.t. rudder deflection
K_{f1} - Empirical fuselage correction factor
$C_{L\alpha V}$ - Vertical tail lift-curve slope
$\frac{d\sigma}{d\beta}$ - Sidewash gradient
η_V - Vertical tail efficiency factor
l_V - Vertical tail moment arm
S_V - Vertical tail area
K_{f2} - Empirical fuselage correction factor
τ_R - Rudder effectiveness factor
b_R - Rudder span
b_V - Vertical tail span
x - State vector
A - System matrix
B - Input matrix
u - Input vector
y - Output vector
C - Output matrix
D - Feedthrough matrix
$u(t)$ - PID controller output signal
K_p - Proportional gain
K_i - Integral gain
K_d - Derivative gain
$e(t)$ - Error signal
$q(s)$ - Pitch rate in Laplace domain

$\eta(s)$ - Elevator input in Laplace domain
k_q - Pitch-rate transfer function gain
$T_{\theta 1}$ - First numerator time constant
$T_{\theta 2}$ - Second numerator time constant
ζ_p - Phugoid damping ratio
ω_p - Phugoid undamped natural frequency
ζ_s - Short-period damping ratio
ω_s - Short-period undamped natural frequency
K - Servo motor gain constant
J - Servo motor damping coefficient
$p(s)$ - Roll rate in Laplace domain
$\xi(s)$ - Aileron input in Laplace domain
k_{ps} - Roll-rate transfer function gain
ζ_ϕ - Roll coupling damping ratio
ω_ϕ - Roll coupling frequency
T_s - Spiral mode time constant
T_r - Roll subsidence time constant
ζ_d - Dutch roll damping ratio
ω_d - Dutch roll natural frequency
w_i - Weighting factors in PID cost function
$\zeta_{sp,current}$ - Current short-period damping ratio
$\zeta_{sp,target}$ - Target short-period damping ratio
$\omega_{sp,current}$ - Current short-period undamped natural frequency
$\omega_{sp,target}$ - Target short-period undamped natural frequency
$T_{2,current}$ - Current time to double amplitude
$T_{2,target}$ - Target time to double amplitude
Δy - Vertical distance between measured points
Δx - Horizontal distance between measured points
I - Mass moment of inertia; also identity matrix
b - Pendulum configuration constant
T - Oscillation period
W - Weight of suspended system
A - Distance between bifilar filaments
l - Filament length
L - Distance between CG and oscillation axis
M - Mass of system
I_v - Virtual moment of inertia
I_s - Structural moment of inertia
I_e - Entrapped-air inertia
I_A - Added-mass moment of inertia
V - UAV body volume
ρ - Air density
M_A - Added mass coefficient
T_1 - Oscillation period using first pendulum length
T_2 - Oscillation period using second pendulum length
L_1 - First pendulum length
L_2 - Second pendulum length
q - Measured distance

x, y - Initial and final scale readings
δq - Uncertainty of measured distance
δx - Uncertainty of initial reading
δy - Uncertainty of final reading
δT - Uncertainty in oscillation period
δA - Uncertainty in filament spacing
δl - Uncertainty in filament length
λ - Eigenvalue of system matrix
ω_n - Undamped natural frequency
ζ - Damping ratio
ω_d - Damped natural frequency
ω_{sp} - Short-period undamped natural frequency
ζ_{sp} - Short-period damping ratio
M_q - Pitching moment derivative due to pitch rate
M_w - Pitching moment derivative due to vertical velocity
u_0 - Trim flight velocity
ω_{ph} - Phugoid undamped natural frequency
ζ_{ph} - Phugoid damping ratio
ω_{dr} - Dutch roll undamped natural frequency
ζ_{dr} - Dutch roll damping ratio
Y_v - Lateral force derivative due to lateral velocity
N_v - Yawing moment derivative due to lateral velocity
Y_r - Lateral force derivative due to yaw rate
N_r - Yawing moment derivative due to yaw rate
τ_r - Roll subsidence time constant
L_p - Rolling moment derivative due to roll rate
τ_s - Spiral mode time constant
L_v - Rolling moment derivative due to lateral velocity
L_r - Rolling moment derivative due to yaw rate
S_{norm} - Normalised sensitivity
X_0 - Baseline parameter value
Y_0 - Baseline modal characteristic
X_i - Modified parameter value
Y_i - Modified modal characteristic
$\%\Delta X$ - Percentage change in parameter
$\%\Delta Y$ - Percentage change in modal characteristic
I_{zz} - Yaw-axis mass moment of inertia

III. APPENDIX C : UAV Development and Fabrication

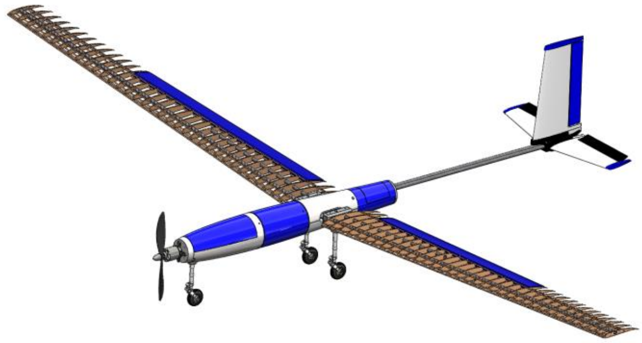


Figure III.1: 3D model of designed UAV

The fabrication process mainly includes the construction of the main wings, structural aluminum frame, fuselage, tail section, landing gear attachments, control surfaces and on-board electronic system integration.

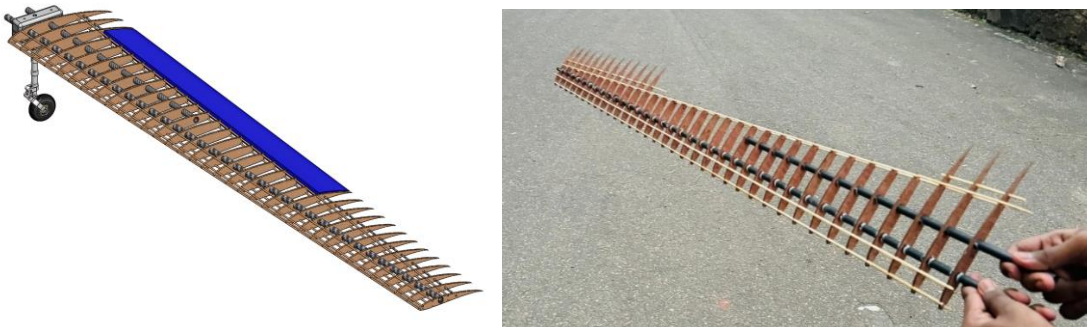


Figure III.2: Wing assembly model and construction

The main wing wings were constructed with total 2.5m wingspan. The primary load carrying members of the wing were two 8mm x 10mm hollow carbon fiber tubes, which acted as the main spars as shown in **Error! Reference source not found.**. Carbon fiber was selected as it provides high stiffness to weight ratio which is essential for reducing the wing flexing under the aerodynamic loads. The inside ribs were fabricated using 2mm thick plywood sheets. The rib profiles were laser cut to obtain accurate outer shape of the aerofoil. To ensure the correct alignment and spacing between the ribs, laser cut plywood jig was used for the assembly process of the ribs. The wing surface was covered using heat shrink film by applying the heat using an iron. The main wing geometric parameters can be listed as shown in APPENDIX D : Framework data.

The control surfaces were sized using conventional proportions available in literature rather than using the developed framework. The wing consisted of ailerons which were actuated using a single 9g servo motor. Due to the small aerofoil shape and requirement of the reduced weight, 3D printing with PLA (Polylactic Acid) material was chosen as the way of fabricating the ailerons. For the aileron hinge, 3mm diameter carbon fiber rods were used as they provide enough stiffness and resistance to bending loads created by the control surface deflections.

The wing box was fabricated using Aluminum rods joined by DC welding. This structure acted as the main structural backbone of the whole UAV. Due to the distributed weight along the fabricated wing and the aerodynamic load when at flight, the wing root experiences significant bending loads. By attaching the wing spars to the welded aluminum frame, the loads can be distributed, improving the overall structural integrity of the UAV. This also provided strong mounting points to propulsion system, fuselage, tail booms and landing gear attachments. Landing gears of the UAV were taken as tricycle type, the design was not generated using the framework. Landing gears were procured from an available store as of the limitations of manufacturing.

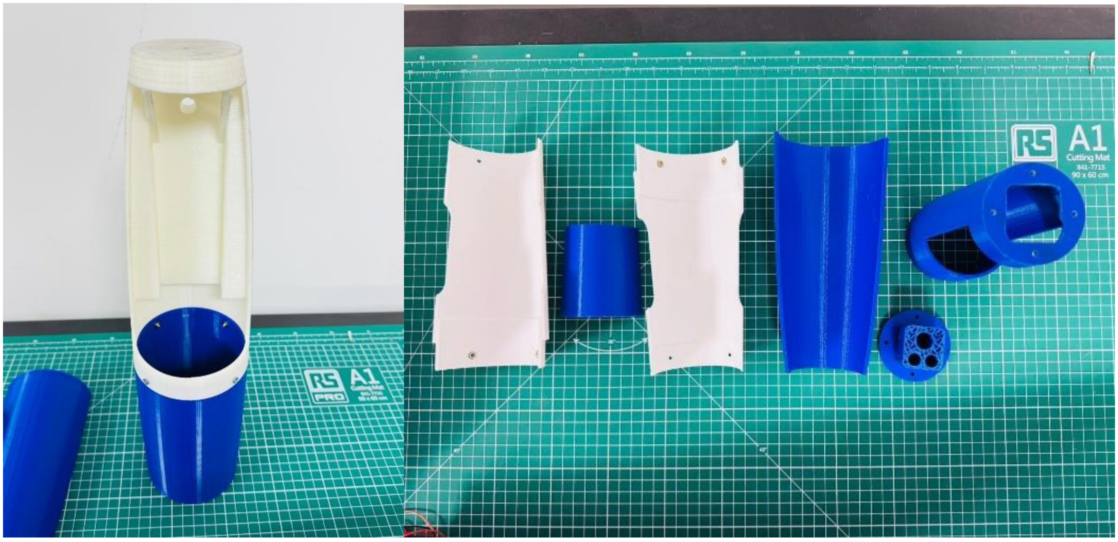


Figure III.3: 3D printed fuselage parts

Fuselage design was not obtained using the framework. Considering the manufacturing feasibility and structural integrity of the UAV, front fuselage pod and a dual tail boom design were manufactured. Fuselage pod was fabricated using 3D printing with PLA material as shown in Figure III.3. A low infill percentage was used for the prints to reduce the overall weight of the components while maintaining the required rigidity. Fuselage acts as housing to main electronic components such as the battery, ESC, radio receiver, APM 2.8 flight controller, GPS module, two servo motors and the payload mass. The layout was arranged in such a way that the battery and payload masses can be arranged to maintain the center of gravity in the desired location. The APM 2.8 flight controller was mounted close to the center of gravity location, so that the angular rates and accelerations were measured at the center of gravity.

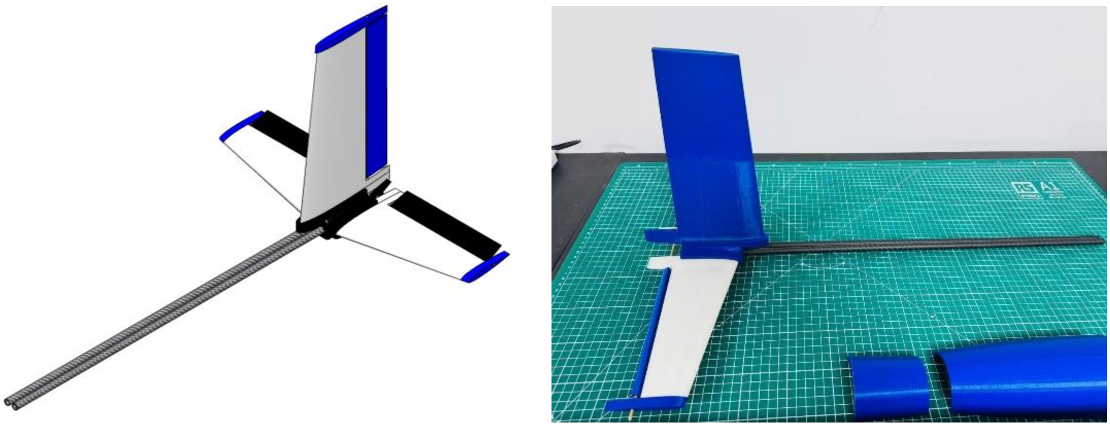


Figure III.4: Tail design model and fabrication

The UAV used a twin tail boom arrangement with two 8mm x 10mm carbon fiber rods to support the tail section as shown in **Error! Reference source not found.**. This provided higher structural rigidity against bending and torsional loads that could occur due to propeller wash induced vibrations and aerodynamic loads. Another advantage of this twin boom tail arrangement was that the push rods from two servo motors mounted at the rear fuselage section could pass through the tail booms to actuate the elevator and rudder. The main goal was to reduce the weight of the tail section as much as possible. This is crucial since any additional mass added to the tail section produced a larger moment due to the long distance from the center of gravity. If more weight is added to the tail, the center of gravity shifts backward, which can reduce the longitudinal stability. Therefore, placing the servos inside the fuselage significantly helped to maintain the center of gravity in desired position. The design parameters of whole tail section can be listed as shown in APPENDIX D : Framework data

Since the available propulsion system items were limited, available propellers, motors, and batteries were justified using the framework and used in the UAV design. The specifications of the propulsion system are listed in APPENDIX D : Framework data.

IV. APPENDIX D : Framework data

Detailed data of UAV from framework

Matching Plot

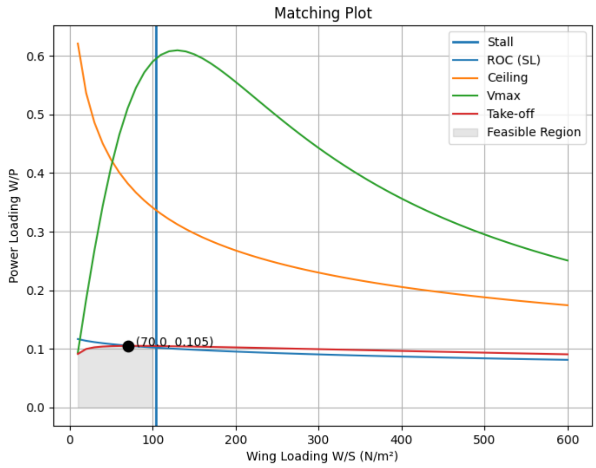


Figure IV.1: Matching plot for conceptual design of modified UAV

Table IV.1: Conceptual design phase outputs of modified UAV

Wing loading, W/S	75.34 N/m²
Power loading, W/P	0.10 N/W
Reference wing area, S	0.39 m²
Minimum power required	281.34 W
Selected aspect ratio for wing	16

Wing Design

Table IV.2: Wing Planform Geometry of modified UAV

Span	2.498 m
Root chord	0.195 m
Tip chord	0.117 m
Mean aerodynamic chord	0.159 m
Taper ratio	0.6
Dihedral angle	5.0°
LE sweep	0.0°
Wing incidence	7.5°
Aerofoil	Eppler E216

Wing Aerodynamic Characteristics:

Table IV.3 Wing aerodynamic data of modified UAV

Cruise velocity	12.71 m/s
Cruise CL	0.9346
CL,max	1.5202
CL,α	6.2304 /rad

Span efficiency, e	0.9496
L/D maximum	17.84
Cm,ac	−0.1555

Propulsion System

Table IV.4: Propulsion system details of modified UAV

Motor	SunnySky X2820
Propeller	APC 11×5.5E
Propeller diameter	0.283 m
Battery	Lipo 3S 5200 mAh
Battery voltage	11.1 V
Total propulsion mass	0.488 kg
Motor efficiency	0.917
Cruise thrust	2.02 N
Take-off thrust	13.10 N
Propeller efficiency at cruise	0.707
Propeller efficiency at climb	0.633
Propeller efficiency at take-off	0.54
Rate of climb at cruise	0.36 m/s
Rate of climb at take-off	4.06 m/s
Power at take-off	260.97 W
Total efficiency at cruise	0.59

Tail Design

Horizontal Tail:

Table IV.5: Horizontal tail geometry and details for modified UAV

Tail arm	0.800 m
Tail area	0.039 m²
Span	0.394 m
Root chord	0.118 m
Tip chord	0.077 m
Aspect ratio	4
Taper ratio	0.65
Aerofoil	NACA 0012
Tail volume coefficient	0.5
Tail incidence	−4.20°
Downwash angle	2.28°

Vertical Tail:

Table IV.6: Vertical tail geometry data for modified UAV

Tail arm	0.790 m
Tail area	0.037 m²

Span	0.256 m
Root chord	0.158 m
Tip chord	0.126 m
Aspect ratio	1.8
Taper ratio	0.8
Aerofoil	NACA 0008

Static Stability Criteria

Table IV.7: Static stability data of for modified UAV

Neutral point, NP	47% MAC
Centre of gravity	20% MAC
Static margin	27% MAC
Cm,α	−1.6798 /rad
Cm,0	0.2069
Trim angle of attack	7.06°
Cn,β	0.0620 /rad
Unshielded area of Vtail for spin recovery	51.29%

Table confirms the static stability criterion is satisfied, and the framework advanced without tail redesign iteration.

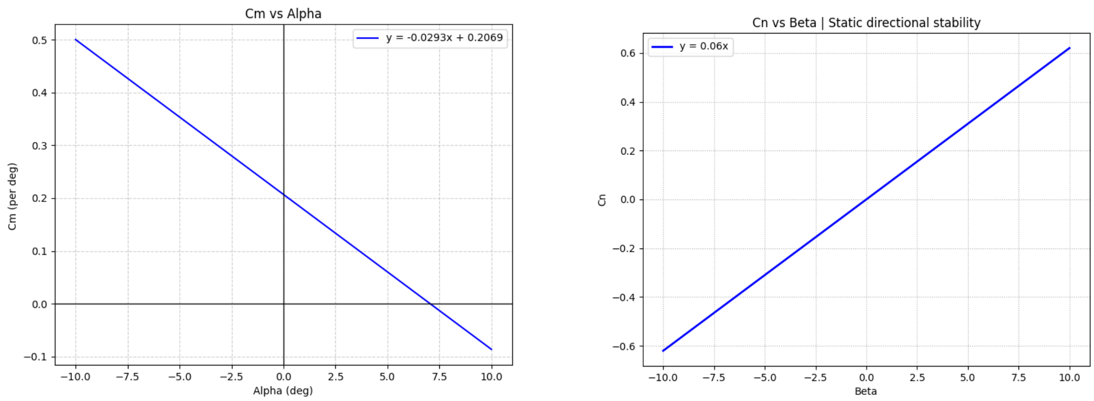


Figure IV.2: Static stability – Cm vs angle of attack and Cn vs sideslip angle

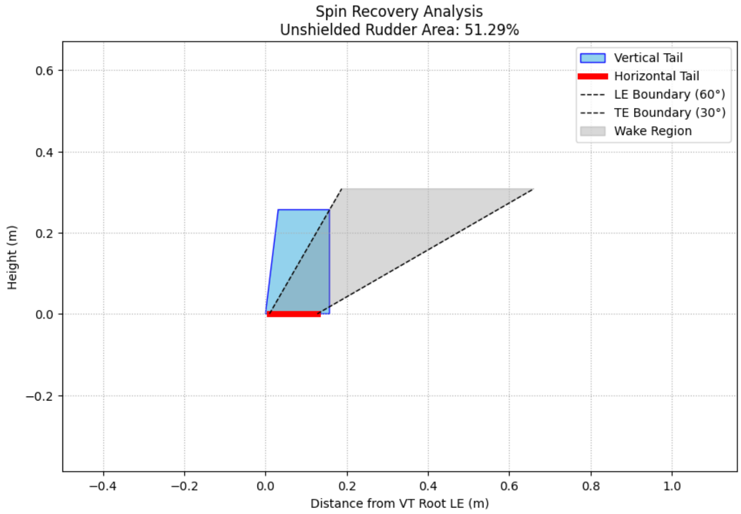


Figure IV.3: Spin recovery of vertical tail

Zero lift drag calculations

Table IV.8: Zero lift drag data of modified UAV

Zero lift drag of the aircraft at cruise	0.0622
Zero lift drag of the aircraft at take-off	0.1022
Zero lift drag of the aircraft at cruise - AVL	0.0726

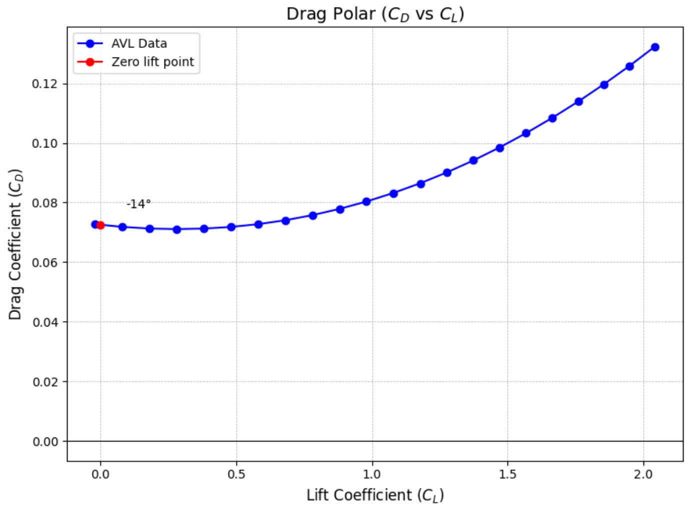


Figure IV.4: Drag polar curve for modified UAV

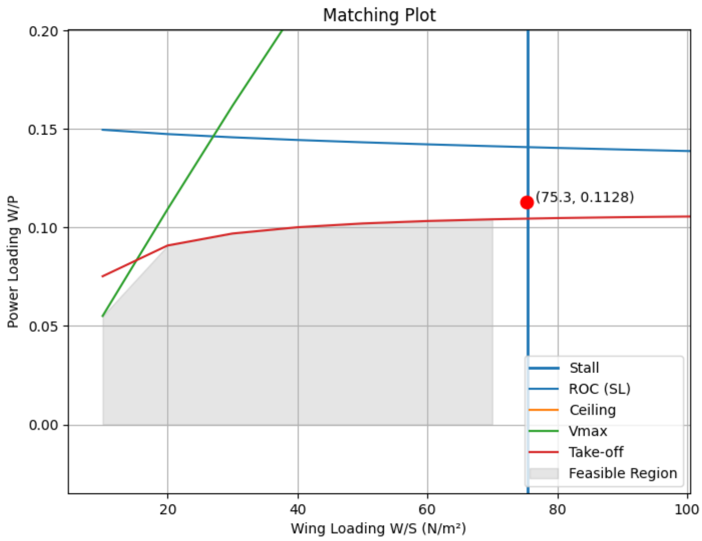


Figure IV.5: Design point assessment for updated flying data

It is noted that the selected propulsion configuration marginally underperforms the take-off power requirement derived from the matching plot. This discrepancy arises from the limited availability of motor-propeller combinations within the design mass budget and is acknowledged as a limitation with sources available to use in the UAV testing. Calculations were made for the propeller, motor and battery that are already available. The cruise and climbing requirements are satisfied, and the take-off distance computed in the performance analysis remains within acceptable bounds.

Performance Analysis

Table IV.9: Performance Analysis data for modified UAV

Performance Parameter	Required	Achieved
Cruise speed	12 m/s	12.71 m/s
Maximum speed	15 m/s	20.59 m/s
Stall speed	10 m/s	9.96 m/s
Maximum Range	-	67.81 km
Maximum Endurance	-	2.3855 h
Speed for Max ROC	-	7.11 m/s
Angle for Max ROC	-	37.85°
Maximum Rate of Climb	-	4.36 m/s
Speed for Steepest Climb	-	1.32 m/s
Maximum Climb Angle	-	90.00°
ROC at Steepest Climb	-	1.32 m/s
Total Take-off Distance	-	35.65 m
Corner Speed	-	18.64 m/s
Minimum Turn Radius	-	10.56 m

Bank angle for tightest turn	-	73.40°

Control Surfaces

Table IV.10: Control surface geometric data for modified UAV

Aileron span	0.741 m
Aileron chord	0.059 m
Elevator span	0.341 m
Elevator chord	0.032 m
Rudder span	0.234 m
Rudder chord	0.037 m

Dynamic Stability Analysis

Longitudinal State-Space Model:

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} -0.277 & 0.142 & 1.312 & -9.81 \\ -1.992 & -4.557 & 11.95 & 0 \\ -0.0342 & -0.3192 & -0.8412 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} -0.005729 \\ -0.03905 \\ -0.1317 \\ 0 \end{bmatrix} \eta$$

Lateral-Directional State-Space Model:

$$\begin{bmatrix} \dot{v} \\ \dot{p} \\ \dot{r} \\ \dot{\phi} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} -0.2516 & -1.566 & -12.37 & 9.81 & 0 \\ -0.9037 & -9.372 & 4.739 & 0 & 0 \\ 1.795 & -0.5941 & -2.902 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} v \\ p \\ r \\ \phi \\ \psi \end{bmatrix} + \begin{bmatrix} -0.03355 & -0.02665 \\ -1.923 & -0.01646 \\ 0.7213 & 0.3121 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$

Dynamic Mode Characteristics

Table IV.11: Dynamic analysis data obtained for modified UAV

Mode	**Eigenvalue**	**ωn (rad/s)**	**ζ**	**T½ or T2 (s)**
Short Period	−2.906 ± 1.159i	3.129	0.929	-
Phugoid	+0.069 ± 0.690i	0.694	−0.100	-
Dutch Roll	−1.584 ± 4.891i	5.141	0.308	-
Roll Subsidence	−9.585	-	-	Tc = 0.104 s
Spiral	0.228	-	-	Tc = −4.39 s

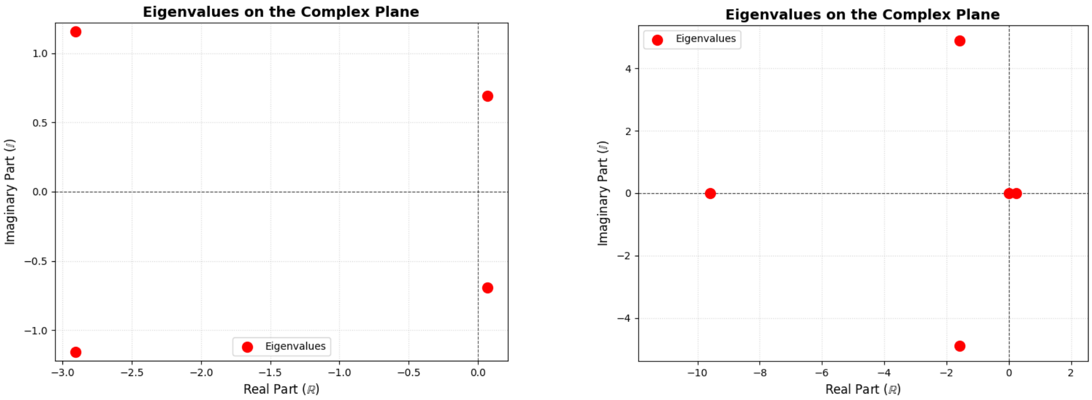


Figure IV.6: Pole location of the longitudinal and lateral-directional modes of modified UAV

Flying Quality Compliance

Table IV.12: Framework Predicted flying qualities of the Modified UAV

Mode	**Parameter**	**Requirement (Level 1)**	**Predicted**	**Level**
Short Period	ζ	0.30 – 2.00	0.929	Level 1
Phugoid	ζ	> 0.04	−0.100	Level 3
Dutch Roll	ζ	> 0.08	0.308	Level 1
Dutch Roll	ωn	> 0.5 rad/s	5.141 rad/s	Level 1
Dutch Roll	ζ*ωn	> 0.15	1.583 rad/s	Level 1
Roll Subsidence	Tc	< 1.4 s	0.104 s	Level 1
Spiral	Tc	> 28.9 s	−4.39 s	Unstable

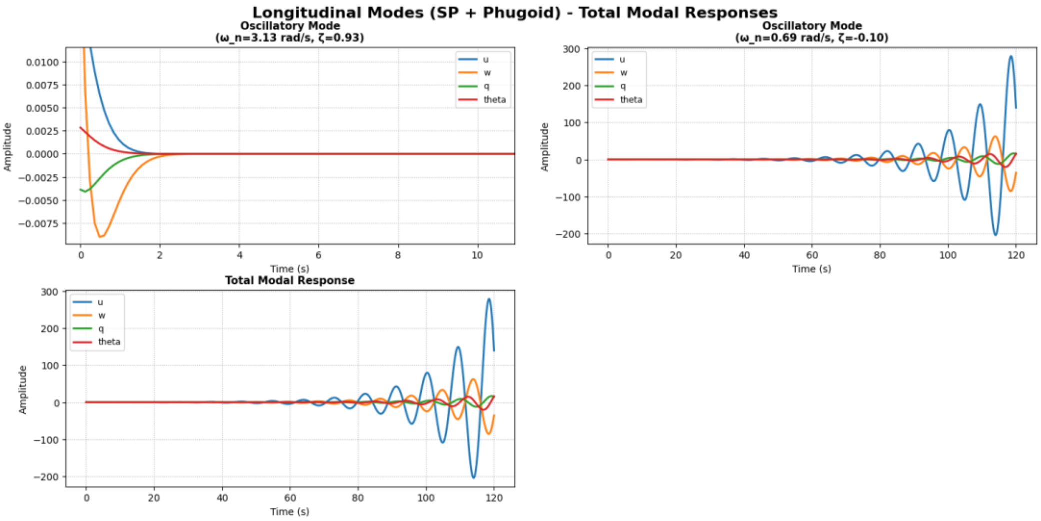


Figure IV.7: Modal response of longitudinal modes for u=0.1 initial condition.

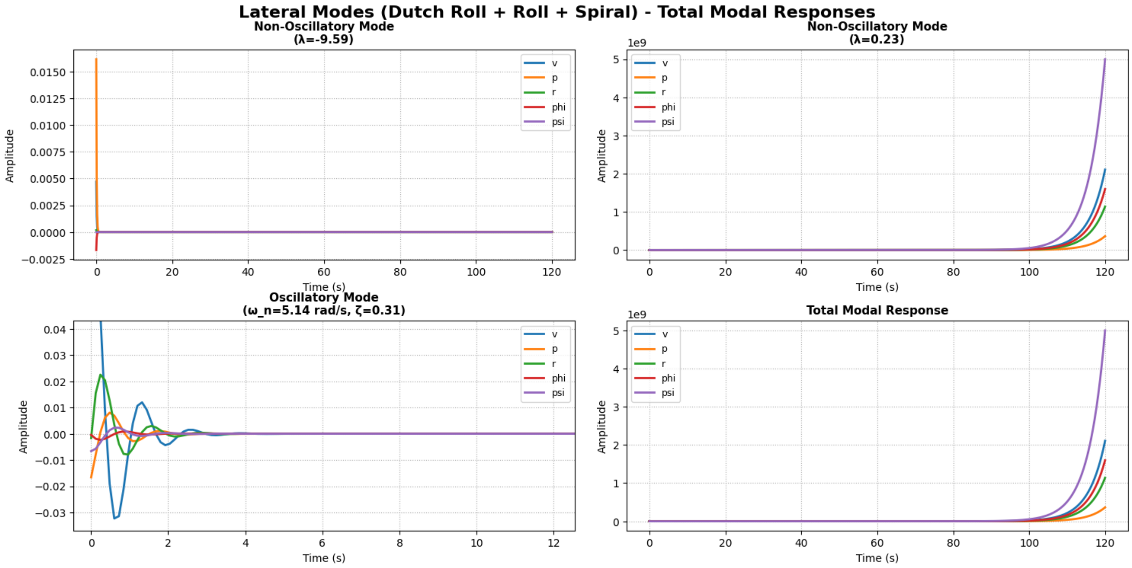


Figure IV.8: Modal response of lateral modes for v=0.1 initial condition.

Stability Augmentation Module

Table IV.13: Full comparison of pitch SAS results on developed UAV

Mode	Condition	Pole pair	Damped frequency (rad/s)	Natural frequency (rad/s)	Damping ratio ζ	Time to double T2 (s)	observation
Short period	Inherent	-2.9078 ± 1.1583j	1.1583	3.1300	0.9290	Not applicable	Stable and highly damped
Short period	Tuned with SAS	-2.8719 ± 1.1464j	1.1464	3.0922	0.9287	Not applicable	Stable; almost same short-period behaviour
Phugoid	Inherent	0.0688 ± 0.6901 j	Not available	Not available	Not available	10.08	Unstable phugoid; does not satisfy target
Phugoid	Tuned with SAS	0.0126 ± 0.7329j	Not available	Not available	Not available	55.00	Still divergent but acceptable by T2 requirement

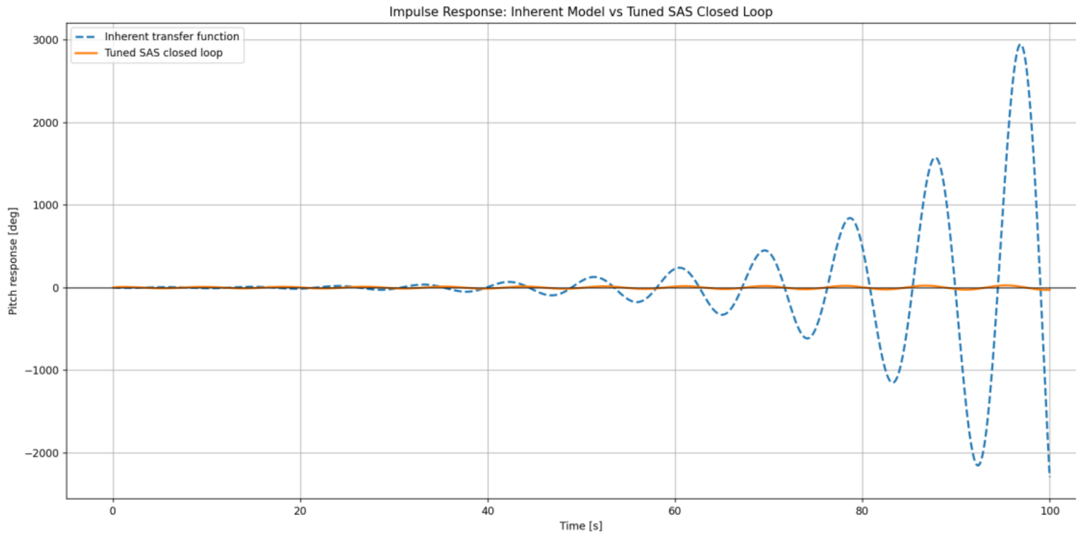


Figure IV.9: Inherent dynamic response vs SAS tuned response for developed UAV

Blu-Baby RC Aircraft | Framework Applied for Verification

Geometric Configuration and Matching Plot Verification

The Blu-Baby has a gross mass of 0.7 kg with a 0.2 kg payload capacity, operating at a cruise speed of 12.54 m/s. An aspect ratio of 4.5 was selected to match the existing airframe geometry.

Table IV.14: Conceptual design parameters of Blu-Baby RC plane

Gross mass	0.7 kg
Wing loading, W/S	42.27 N/m²
Power loading, W/P	0.0994 N/W
Reference wing area	0.162 m²
Minimum power required	69.06 W
Aspect ratio	4.5
Reynolds number (cruise)	1.26×10^5

Wing Verification

The LLT module confirmed the wing geometry using the Eppler E197 aerofoil, selected for its favourable characteristics at the low Reynolds number range of $1.26–1.57 \times 10^5$ characteristic of this aircraft class. No twist was present in the existing wing.

Table IV.15: Wing aerodynamic and gemetric data of Blu-Baby RC plane

Aerofoil	Eppler E197
Wing area, S	0.160 m²
Span, b	0.849 m
Root chord	0.207 m
Tip chord	0.170 m
Mean chord	0.189 m
Aspect ratio	4.5
Taper ratio	0.82

Dihedral	6.0°
Wing incidence	5.00°
Cruise CL	0.5457
CLmax	1.1046
CL,α	3.859 /rad
Span efficiency	0.9777
L/D maximum	12.66

Propulsion: The propulsion database for the Blu-Baby was constrained to the actual hardware installed. The APC 8×4.5MR propeller was ranked first at a cruise efficiency of $\eta = 0.713$ with the Turnigy D2826-10 motor and a single 3S 1500 mAh LiPo battery. The installed take-off power of 53.73 W does not satisfy the matching plot take-off power requirement, which is acknowledged as a limitation of the existing propulsion system for the defined mission profile; however, as the Blu-Baby is used here as a verification platform rather than a designed mission asset.

The Blu-Baby is a commercially available RC aircraft with well documented geometry and established flight characteristics. Its selection as a verification platform was motivated by the need to obtain high-quality, repeatable flight data, avoiding the trimming and controllability challenges encountered with the newly fabricated modified UAV. The framework was applied to get a closest design case for the Blu-Baby by comparing it with the experimentally measured geometric and inertial properties obtained from the measurement setup described in Chapter 3.2.

Horizontal Tail

Blu-Baby trims with the tail producing an upward force, consistent with a forward CG configuration.

Table IV.16: Horizontal tail geometry and stability data for Blu-Baby RC Plane

Aerofoil	NACA 0006
Tail area	0.047 m^2
Span	0.405 m
Root chord	0.135 m
Tip chord	0.094 m
Aspect ratio	3.5
Taper ratio	0.7
Volume coefficient	0.5
Tail arm	0.323 m
Tail incidence	+4.20°
Tail operating AoA	+1.53°

Tail CL	0.1072
CL,α,h	4.139 /rad
Cm,α (total aircraft)	−0.3137 /rad
Cm_0(total aircraft)	0.0303
Static margin	8% MAC
Neutral point	48% MAC

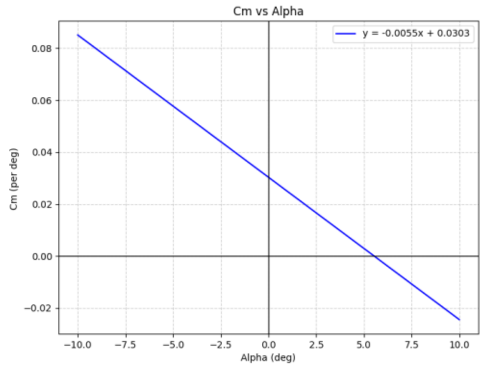


Figure IV.10: Longitudinal static stability – Cm vs angle of attack

Vertical Tail

The vertical tail framework assessment output a poor unshielded area ratio responsible for spin recovery is 9.77%, well below the 50%. This indicates that the Blu-Baby's vertical tail is largely shielded within the horizontal tail wake, which is characteristic of its compact empennage configuration. Despite this, the directional stability derivative Cn,β = +0.0549 /rad remains positive, confirming weathercock stability.

Table IV.17: Vertical tail geometry and stability data for Blu-Baby RC Plane

Aerofoil	NACA 0006
Tail area	0.011 m²
Span	0.127 m
AR / Taper	1.40 / 0.55
Tail arm	0.355 m
Cn,β	+0.0549 /rad
Unshielded area ratio	9.77%

Zero lift drag Calculations

The zero-lift drag build-up gave CD0, AC = 0.0297. Notably, the strut and landing gear contributions are zero, reflecting the clean belly-landing configuration of the Blu-Baby.

Dynamic Stability Evaluation

The measured inertial data was passed to AVL along with the geometric model. The state-space matrices assembled by the framework at the cruise trim condition are presented below.

Longitudinal State-Space Model:

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \underbrace{\begin{bmatrix} -0.0869 & 0.821 & -0.854 & -9.81 \\ -1.099 & -6.166 & 10.84 & 0 \\ 3.402 & -50.79 & -24.80 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{A_{lon}} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix} + \underbrace{\begin{bmatrix} 0.005759 \\ -0.2073 \\ -8.818 \\ 0 \end{bmatrix}}_{B_{lon}} \eta$$

Lateral-Directional State-Space Model

$$\begin{bmatrix} \dot{v} \\ \dot{p} \\ \dot{r} \\ \dot{\phi} \\ \dot{\psi} \end{bmatrix} = \underbrace{\begin{bmatrix} -0.2634 & 0.798 & -12.39 & 9.81 & 0 \\ -3.976 & -10.59 & 3.581 & 0 & 0 \\ 1.130 & -0.5398 & -0.6094 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}}_{A_{lat}} \begin{bmatrix} v \\ p \\ r \\ \phi \\ \psi \end{bmatrix} + \underbrace{\begin{bmatrix} -0.03034 & -0.03393 \\ -4.548 & -0.07608 \\ 0.0854 & 0.2555 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}}_{B_{lat}} \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$

The longitudinal and lateral-directional characteristic equations are:

Longitudinal: $\lambda^4 + 31.053\lambda^3 + 709.979\lambda^2 + 152.195\lambda + 753.358 = 0$

Lateral-Directional: $\lambda^5 + 11.463\lambda^4 + 28.510\lambda^3 + 214.777\lambda^2 - 15.927\lambda = 0$

Table IV.18: Dynamic analysis data of Blu-Baby RC Plane

Mode	Eigenvalue	ωn (rad/s)	ζ	Tc (s)
Short Period	−15.444 ± 21.572i	26.53	0.582	-
Phugoid	−0.085 ± 1.031i	1.035	0.082	-
Dutch Roll	−0.424 ± 4.486i	4.506	0.094	-
Roll Subsidence	−10.684	-	-	0.094
Spiral	0.073	-	-	−13.61

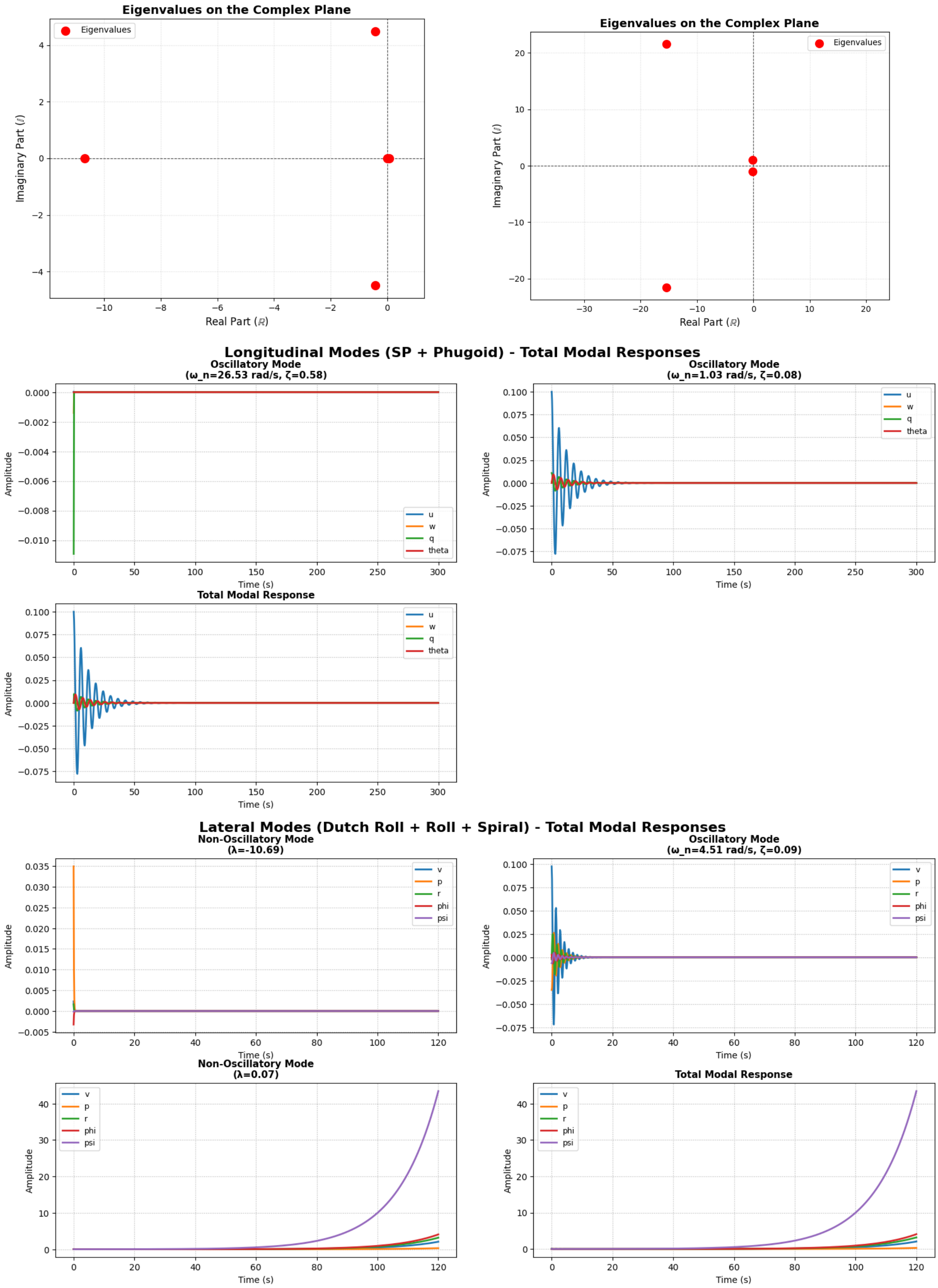


Figure IV.11: Dynamic stability evaluation

Flying Quality Assessment

Table IV.19: Flying quality assessment of Blu-Baby RC Plane

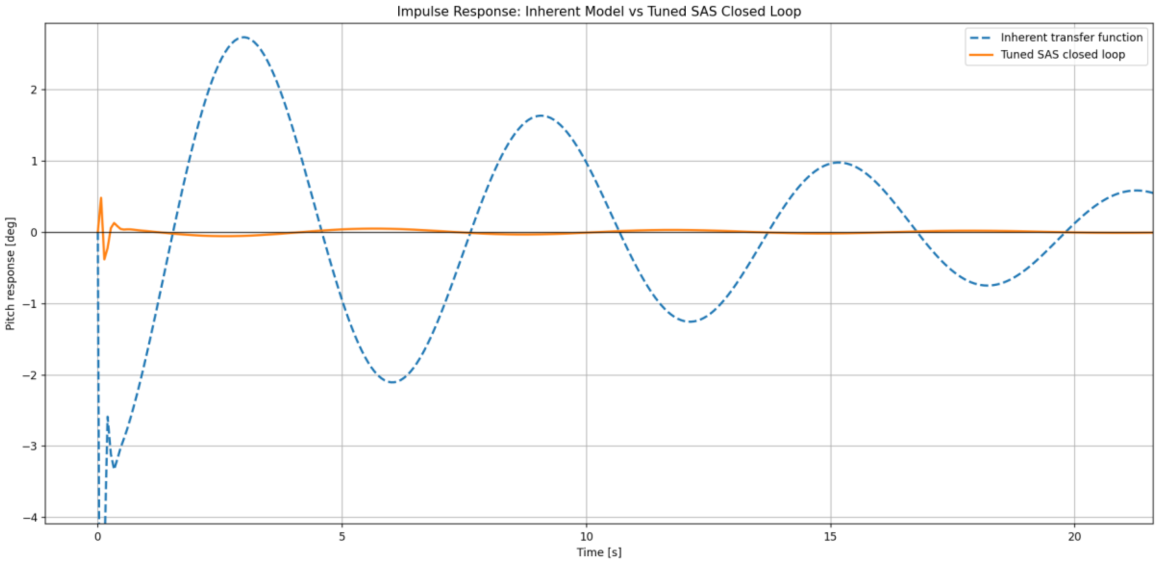
Mode	Parameter	Requirement (Level 1)	Predicted	Level
Short Period	ζ	0.30 – 2.00	0.582	Level 1
Phugoid	ζ	> 0.04	1.0347 rad/s	Level 1
Dutch Roll	ζ	> 0.08	0.0941	Level 1
Dutch Roll	ωn	> 0.5 rad/s	4.5056 rad/s	Level 1
Dutch Roll	ζ*ωn	> 0.15	0.424 rad/s	Level 1
Roll Subsidence	Tc	< 1.4 s	0.094 s	Level 1
Spiral	Tc	> 28.9 s	negative	Unstable

Stability Augmentation Module

Table IV.20: Full comparison of results after pitch SAS on Blubaby RC model

Mode	Condition	Pole pair	Damped frequency (rad/s)	Natural frequency(rad/s)	Damping ratio ζ	Time to double T2 (s)	Observation
Short period	Inherent	-15.4418± 21.5692j	21.5692	26.5270	0.5821	Not applicable	Stable but short-period frequency is too high
Short period	Tuned with SAS	-10.3210± 17.2605j	17.2605	20.1109	0.5132	Not applicable	Stable, but still above target frequency range
Phugoid	Inherent	-0.0846± 1.0312j	1.0312	1.0347	0.0818	Infinite	Stable phugoid
Phugoid	Tuned with SAS	-0.0826± 1.0298j	1.0298	1.0331	0.0799	Infinite	Stable; very small change after tuning

Table IV.21: Inherent dynamic response vs Tuned SAS response



Full Notional Example case - wing, fuselage, propulsion and LG design

Mission Requirements and Basic Configuration

Table IV.22: Conceptual design data estimation

Design gross mass	16 kg
Payload mass	4 kg
Cruise speed	19 m/s
Stall speed	14 m/s
Maximum speed	25 m/s
Service ceiling	2000 m
Rate of climb at sea level	4 m/s
Configuration	Conventional tractor, tricycle undercarriage

Full Notional Design – Wing, Propulsion, Fuselage and LG design

Weight Estimation and Matching Plot

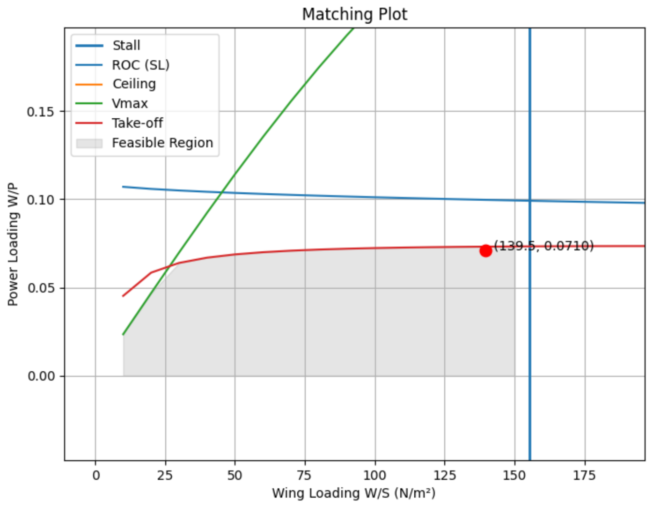


Figure IV.12: Matching plat

Wing loading, W/S	139.50 N/m²
Power loading, W/P	0.0751 N/W
Reference wing area	1.13 m²
Minimum power required	2090.66 W
Selected aspect ratio	8

Wing Design

Based on the reference area and aspect ratio from the matching plot, the wing planform was defined using the Lifting Line Theory (LLT) module.

Aerofoil	NACA 2415
Span	3.007 m
Root chord	0.442 m
Mean aerodynamic chord	0.380 m
Aspect ratio	8
Taper ratio	0.7
Linear twist	−3.00°
Dihedral angle	5.0°
LE sweep	0.0°
Wing incidence	9.00°
Cruise CL	0.8483
CLmax	1.2953
CL,α	4.074 /rad
Span efficiency, e	0.9862
L/D maximum	21.48
Cm,ac	−0.0525

Fuselage Design

The fuselage was dimensioned to accommodate the propulsion system at the nose, the avionics and payload bay in the mid-section, and the empennage tail boom.

Total fuselage length	1.804 m
Maximum diameter	0.171 m
Wetted area	0.640 m²
Wing LE position from nose	0.328 m
Horizontal tail MAC position	1.504 m
Tail moment arm, l_t	1.175 m

The fuselage pitching moment contribution was estimated as $C_{m,0,f} = -0.0001$ and $C_{m,\alpha,f} = 0.0007$ /rad, both of which are negligible relative to the wing and tail contributions.

Landing Gear Design

Landing gear height, H_LG	0.269 m
Main gear position from nose	0.546 m
Nose gear position from nose	0.031 m
Wheelbase	0.515 m
Wheel track	0.175 m
Main wheel diameter	0.073 m
Main wheel width	0.032 m
Nose wheel diameter	0.058 m
Nose wheel width	0.026 m
Main strut length	0.233 m
Nose strut length	0.121 m

Propulsion System

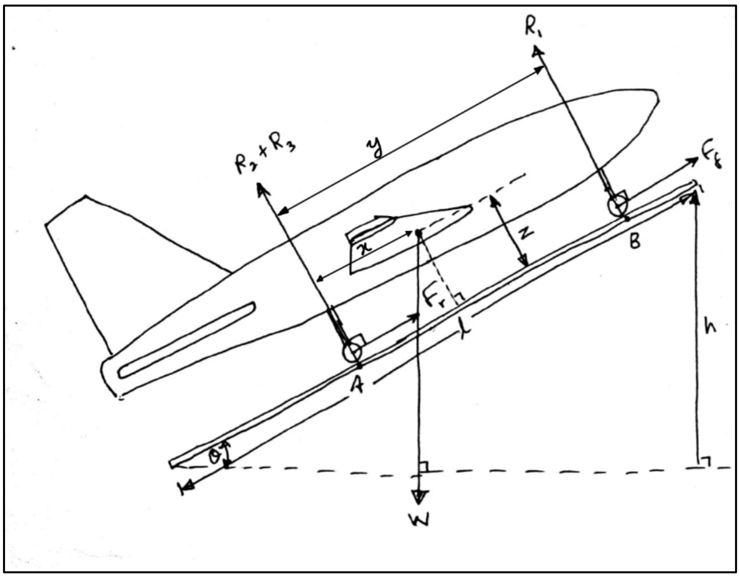
The propulsion system was selected from the component database to satisfy both the cruise thrust and take-off thrust requirements with a 10% safety factor applied.

Motor	AT7224
Propeller	APC 20×18 (F1-GT)
Propeller diameter	0.508 m
Battery	Li-ion 12S2P 12000 mAh
Battery voltage	44.4 V
Total propulsion mass	10.99 kg
Cruise thrust	14.34 N
Take-off thrust	73.85 N
Cruise propeller efficiency	0.77
Take-off propeller efficiency	0.56
Take-off power	2211.04 W
Rate of climb at take-off	6.18 m/s

V. APPENDIX E: Determination of vertical position of the centre of gravity and moments of inertia about principal axes

The vertical position of the centre of gravity of the UAV was determined using the tilted floor technique. Tilting the UAV results in variations in the normal reaction forces acting on the weighing scales on the floor. This is due to the vertical displacement of the Cg of the UAV relative to the ground level. By considering the static stability of the UAV, two simultaneous equations can be derived considering the force balance in the direction normal to the tilted floor and taking moments around one of the points A and B.

Table V.1: UAV placed on the tilted floor



The equation for the nose up configuration can be derived as follows.

$$z = \frac{x(R_1 + R_2 + R_3) - yR_1}{W \sin\theta}$$

For the nose down configuration, the equation can be derived as,

$$z = \frac{(y - x)(R_1 + R_2 + R_3) - y(R_1 + R_2)}{W \sin\theta}$$

Utilizing these equations, the following calculation was carried out.

Table V.2: Experimental data and the calculation to determine the vertical position of Cg

		h	R1	R2	R3	z	Avg	Final_avg
nose up readings	h1	0.051	0.4098	1.4026	1.4527	0.14832	0.151361	0.152434
	h2	0.082	0.377	1.4081	1.4722	0.155069		
	h3	0.095	0.3668	1.4623	1.4257	0.150694		
Nose down readings	h1	0.058	0.5098	1.33	1.42	0.151746	0.153507	
	h2	0.079	0.5323	1.286	1.4351	0.159516		
	h3	0.093	0.5397	1.2235	1.4868	0.149259		

The moments of inertia calculations were carried out for two different aircraft models. First the modified UAV model and the other being the "33-inch Blu-Baby Primary Trainer" model. Considering the first model,

Using the compound pendulum for the X body axis, the virtual moment of inertia about that axis is given by,

$$I_{Vx} = \frac{W_1 T_1^2 L_1}{4\pi^2} - \frac{W_2 T_2^2 L_2}{4\pi^2} - \left(\frac{W}{g} + V\rho + M_A\right) L^2$$

Table V.3: Experimentally obtained data for determination of Ixx

Ixx	Short		Long	
W_uav	2881.1	g	2881.1	g
W_total	4105.55	g	4105.55	g
W_frame	1224.45	g	1224.45	g
L	1462.59	mm	1635.83	mm
L_uav	1503	mm	1728	mm
L_st	1316.46	mm	1515.83	mm
T1	2.486	s	2.658	s
T2	2.22	s	2.3	s

Table V.4: Other properties requied to determine moments of inertia

weight of the frame	920.45	g
weight of the cabes	174	g
weight of the buckles	130	g
total weight of the frame	1224.45	g

volume of fusalage	4025631.034	mm^3
single wing area	185107.4702	mm^2
maxumum ordinate	20	mm
wing volume	5479181.118	mm^3
vtail area	35653.75	mm^2
vtail mean chord	142.615	mm
vtail height	250	mm

Volume V is taken from the dimension of the airplane by considering a simplified version of the fuselage. Since the fuselage consists of a revolved shape, the mean square diameter was considered while calculating the volume. Only the' fuselage and wings are considered for the considerable volume of the aircraft.

For the additional mass, only the fuselage and the vertical tail surface were considered.

$$M_{xf} = \frac{\rho \bar{c}^2 \pi (\Delta b)_1}{4}$$

$$M_{xf} = \frac{1.225 \times 0.00588568 \times \pi \times 0.25}{4} = 0.0021213506\ kg$$

Vertical tail additional mass is calculated in the same way considering mean chord length and the height as $M_{At} = 0.004287862$ kg.

Therefore, total additional mass,

$$M_A = M_{Af} + M_{At} = 0.0064111369\ kg$$

By substituting to the compound pendulum equation,

For the short pendulum,

$$I_{Vx} = \frac{4.10555 \times (2.486)^2 \times 1.46259}{4 \times \pi^2} - \frac{1.22445 \times (2.22)^2 \times 1.31646}{4 \times \pi^2} - [2.8811 + (1.225 \times 0.009504812) + 0.006411369](1.503)^2$$

$$I_{Vx} = 0.698301886\ kgm^2$$

For the long pendulum,

$$I_{Vx} = \frac{4.10555 \times (2.658)^2 \times 1.63583}{4 \times \pi^2} - \frac{1.22445 \times (2.300)^2 \times 1.51583}{4 \times \pi^2} - [2.8811 + (1.225 \times 0.009504812) + 0.006411369](1.728)^2$$

$$I_{Vx} = 0.69377871\ kgm^2$$

Therefore, an average value can be taken as,

$$I_{Vx} = 0.696040298\ kgm^2$$

How ever this is the virtual moment of inertia about the X axis of the UAV. To get the true second moment of inertia, the moment of inertia due to the additional mass must be removed from the above value. The reliability of the results can be verified through a simultaneous equation system for the two pendulum settings, without considering for the additional mass and the volume, treating Ixx and the summation of additional mass and added volume as the only two unknowns. Then,

$$I_{Vx} = 0.73909 - (V\rho + M_A)(1.503)^2$$

$$I_{Vx} = 0.74769 + (V\rho + M_A)(1.728)^2$$

$$I_{Vx} = 0.712358\ kgm^2$$

The agreement is within 2.3%.

The virtual moments of inertia about the Y axis are calculated in the same manner from the data obtained from the compound pendulum. In the case of Y axis, the additional mass in very small so that it can be neglected. The data and the results obtained are as follows.

Table V.5: Experimentally obtained data for determination of Iyy

Iyy	Short		Long	
W_uav	2881.1	g	2881.1	g
W_total	4105.55	g	4105.55	g
W_frame	1224.45	g	1224.45	g
L	1548.59	mm	1585.83	mm
L_uav	1574	mm	1720	mm
L_st	1387.46	mm	1595.83	mm
T1	2.485	s	2.693	s
T2	2.24	s	2.36	s

For the short pendulum,

$$I_{Vy} = 0.471067647\ kgm^2$$

For the long pendulum,

$$I_{Vy} = 0.4708091\ kgm^2$$

So, the average,

$$I_{Vy} = 0.470938373\ kgm^2$$

By solving two simultaneous equations,

$$I_{Vy} = 0.472401034\ kgm^2$$

Therefore, the agreement within 0.31%.

For the determination of virtual second moment of inertia about the z axis the bifilar pendulum was used with the equation,

$$I_{vz} = \frac{W_1 T_1^3 A^2}{16\pi^2 l} - \frac{W_2 T_2^3 A^2}{16\pi^2 l}$$

The experimental data obtained through the swing test are as follows.

Table V.6: Experimentally obtained data for determination of Izz

Izz	Short	
W_uav	2881.1	g
W_total	4174.25	g
W_frame	1293.15	g
A	0.81	mm

l	0.96	mm
T1	2.651612578	s
T2	2.62	s

$$I_{vz} = \frac{4.17425\times(2.652)^2\times(0.81)^2}{16\times\pi^2\times0.96} - \frac{1.29315\times(2.620)^2\times(0.81)^2}{16\times\pi^2\times0.96} = 0.869206691\ kgm^2$$

Determination of Additional moments of inertia

For the x axis, it is assumed that the contribution is only given by the wings and the horizontal tail surface. The equation is,

$$I_{xx} = \frac{k'\rho c^2\pi b^3}{48}$$

For the wings,

$$c = 0.15538m, b = 2.521m, AR = 16$$

Therefore, the k' can be taken as 1.43 from the following reference.

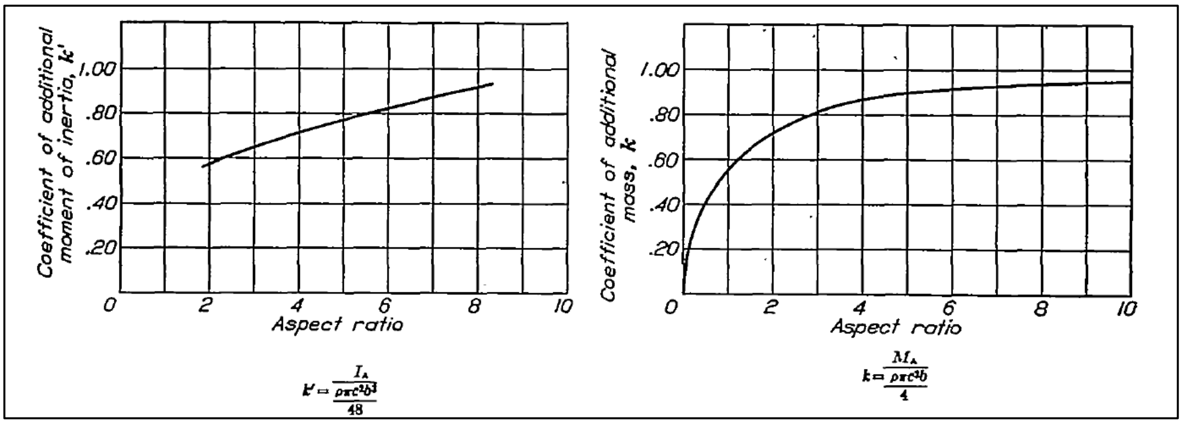


Figure V.1: coefficients of additional moments of inertia for flat plates[37]

Then,

$$I_{Aw} = \frac{1.43\times1.225\times(0.15538)^2\times\pi\times(2.521)^3}{48} = 0.044348688\ kgm^2$$

For the Horizontal tail,

$$c = 0.09548\ m, b = 0.446\ m, AR = 5, k' = 0.76$$

$$I_{At} = \frac{0.76\times1.225\times(0.09548)^2\times\pi\times(0.446)^3}{48} = 0.00004928\ kgm^2$$

Therefore, the total additional moment of inertia about x axis is,

$$I_{Ax} = 0.04439797\ kgm^2$$

The main components that contribute to the additional moments of inertia about the y axis are the fuselage and the horizontal tail surface. The fuselage is considered as a rectangle considering its projected area to the xz plane. The dimensions of this rectangle are taken by considering the square root of mean square of the different sections of the fuselage for the width and the length is taken as the fuselage length.

$$c = 0.07672\,m, b = 0.75\,m, AR = 9.78, k' = 1$$

As the Y axis is parallel to the chord at the same time is displaced from the center of the additional mass of the fuselage by a distance l,

$$I_{Af} = \frac{k'\rho c^2 \pi b^3}{48} + \frac{k\rho c^2 \pi b l^2}{4}$$

Here k is assumed to be 1 and l is measured as 0.246 m.

$$I_{Af} = \frac{1 \times 1.225 \times (0.07672)^2 \times \pi \times (0.75)^3}{48} + \frac{1.0 \times 1.225 \times (0.07672)^2 \times \pi \times 0.75 \times (0.246)^2}{4} = 0.000456091\,kgm^2$$

Since the Y axis is parallel to the span of the horizontal tail,

$$I_{Af} = \frac{k\rho c^2 \pi b l^2}{4}$$

$$c = 0.09548\,m, b = 0.446\,m, AR = 5, k = 0.9, l = 0.757m$$

$$I_{tz} = \frac{0.9 \times 1.225 \times (0.09548)^2 \times \pi \times 0.446 \times (0.757)^2}{4} = 0.002015458\,kgm^2$$

Therefore, the total additional moment of inertia about y axis is,

$$I_{Ax} = 0.002471549\,kgm^2$$

Determining the additional moment of inertia about the z axis follows the same procedure as for the y axis, only difference being that the vertical fuselage and the rail areas are considered.

Therefore, the total additional moment of inertia about z axis is calculated as,

$$I_{Az} = 0.004350343\,kgm^2$$

True Moments of Inertia

The true moment of inertia about any given axis can be found by subtracting the additional moments of inertia about that axis from the virtual moments of inertia.

Therefore, the true moments of inertia about the three principal axes can be given as follows.

$$I_{Tx} = I_{Vx} - I_{Ax} = 0.696040298 - 0.04439797 = 0.651642328\ kgm^2$$

$$I_{Ty} = I_{Vy} - I_{Ay} = 0.470938373 - 0.002471549 = 0.468466825\ kgm^2$$

$$I_{Tz} = I_{Vz} - I_{Az} = 0.869206691\ - 0.004350343 = 0.864856348\ kgm^2$$

Following the same procedure, the mass moments of inertia for the 33-Inch Blu-Baby Trainer Model were calculated.

VI. APPENDIX F: Flight Data Extraction

As with any lightly damped second order mechanical spring mass damper systems, the response is not a single clean oscillation but a superposition of multiple modes, each with its own frequency and damping characteristics. In the context of an aircraft, the response is a sum of exactly two modes, short period mode and phugoid mode. So, the total pitch response can be represented as:

$$\theta(t) \ = \ \theta_{SP}(t) \ + \ \theta_{PH}(t)$$

Here, a single vibrational response due to a mode can be represented as a damped sinusoidal function as,

$$\theta(t) \ = \ A_i \times e^{(-\sigma_i \times t)} \times \cos(\omega_i \times t + \phi_i)$$

Here,

A_i = Amplitude

σ_i = Damping constant

ω_i = Damped frequency

ϕ_i = Phase angle

The total response consists of fast decaying, high frequency short period mode, and slower, low frequency phugoid mode. This can be written as,

$$\theta(t) \ = \ A_{SP} \times e^{(-\sigma_{SP} \times t)} \times \cos(\omega_{SP} \times t + \phi_{SP}) \ + \ A_{PH} \times e^{(-\sigma_{PH} \times t)} \times \cos(\omega_{PH} \times t + \phi_{PH})$$

Prony's Method

Prony's method is a technique used to represent a sampled time domain signal as a sum of damped sinusoids, each of which corresponds to a single mode of a dynamic system. Estimation of the modal frequencies and damping ratios is done using the Prony's method. It represents a signal as a summation of damped sinusoidal components as shown below.

$$x(t) \ = \ \sum_{k=1}^{n} A_k \times e^{(\sigma_k \times t)} \times \cos(\omega_k t \ + \ \phi_k)$$

Where each term represents one mode with,

A_k = Amplitude

σ_k = Decay constant

ω_k = Damped frequency

ϕ_k = Phase angle

Considering only the short period response, n = 1 with single term representing the short period mode. As the flight data is sampled at discrete, evenly spaced time intervals Δt, discrete version of Prony's method is used.

$$x(m) = \sum_{k=0}^{n} C_k Z_k^m$$

Where $x(m)$ is the mth sample and C_k is a complex amplitude containing both A_k and ϕ_k. $Z_i = e^{(\sigma_k + j\omega_k)\Delta t}$ is a complex pole that includes both damping and frequency of mode k. The main step in Prony's method is to determine the unknown discrete poles. So, instead of solving directly for the poles, Prony's method first assumes a linear prediction equation. For a second order Prony's model, pitch response is assumed to satisfy,

$$x[m] + a_1 x[m-1] + a_2 x[m-2] = 0$$

This shows that a pitch sample value can be represented by previous two pitch values. The coefficients are then obtained from the measured pitch data. All of these equations can be arranged as a matrix. Since the number of pitch data samples is usually much larger than the number of unknown coefficients, this system is over determined. So, least square method is being used to find the coefficients. Once the coefficients are found, the characteristic polynomial of the system is formed as,

$$z^2 + a_1 \mathrm{z} + a_2 = 0$$

The roots of this equation are the discrete time poles of the response, and they are complex conjugate pairs. The single pair represents the short period mode. In this case of short period mode, there are two poles. So, 2nd order Prony approximation is used to represent a single sample. The expanded version can be written as,

$$y(k) = C_1 Z_1^k + C_2 Z_2^k$$

Fast Fourier Transformation Method:

Fast Fourier Transform method was also considered for the identification of the modal frequencies of the pitch vs time response. FFT converts the measured time domain response into the frequency domain and shows the dominant frequency components present in the signal. Therefore, FFT is useful for identifying the frequencies in longitudinal modes. However, FFT can only provide frequency distribution of the wave, and it does not provide the damping ratio of each mode. This is a huge limitation for using this FFT method, since damping ratio is also an important parameter for aircraft dynamic stability analysis.

FFT Analysis of Pitch Response

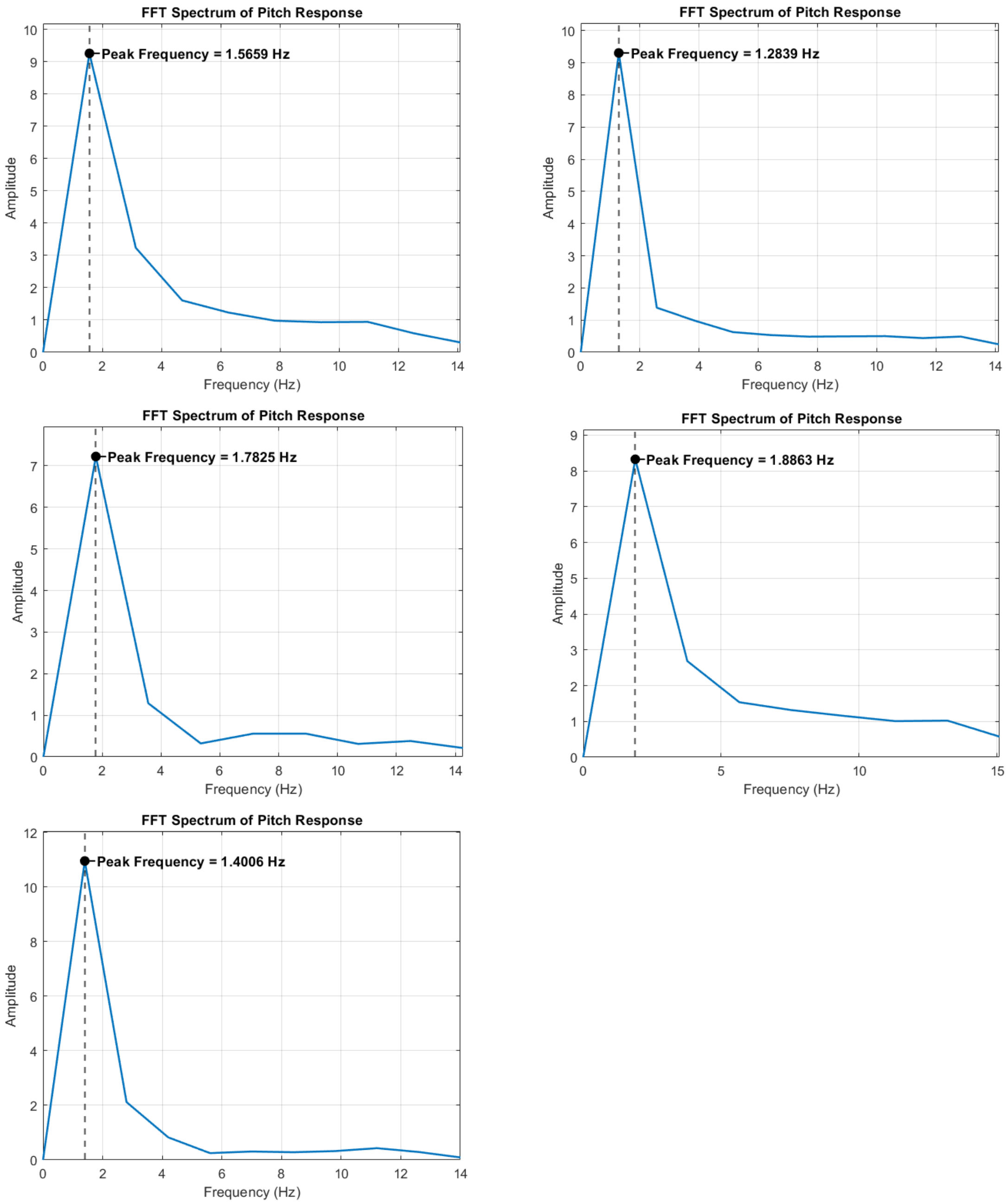


Figure VI.1: FFT analysis results of pitch response

FFT Analysis of Pitch Rate Response

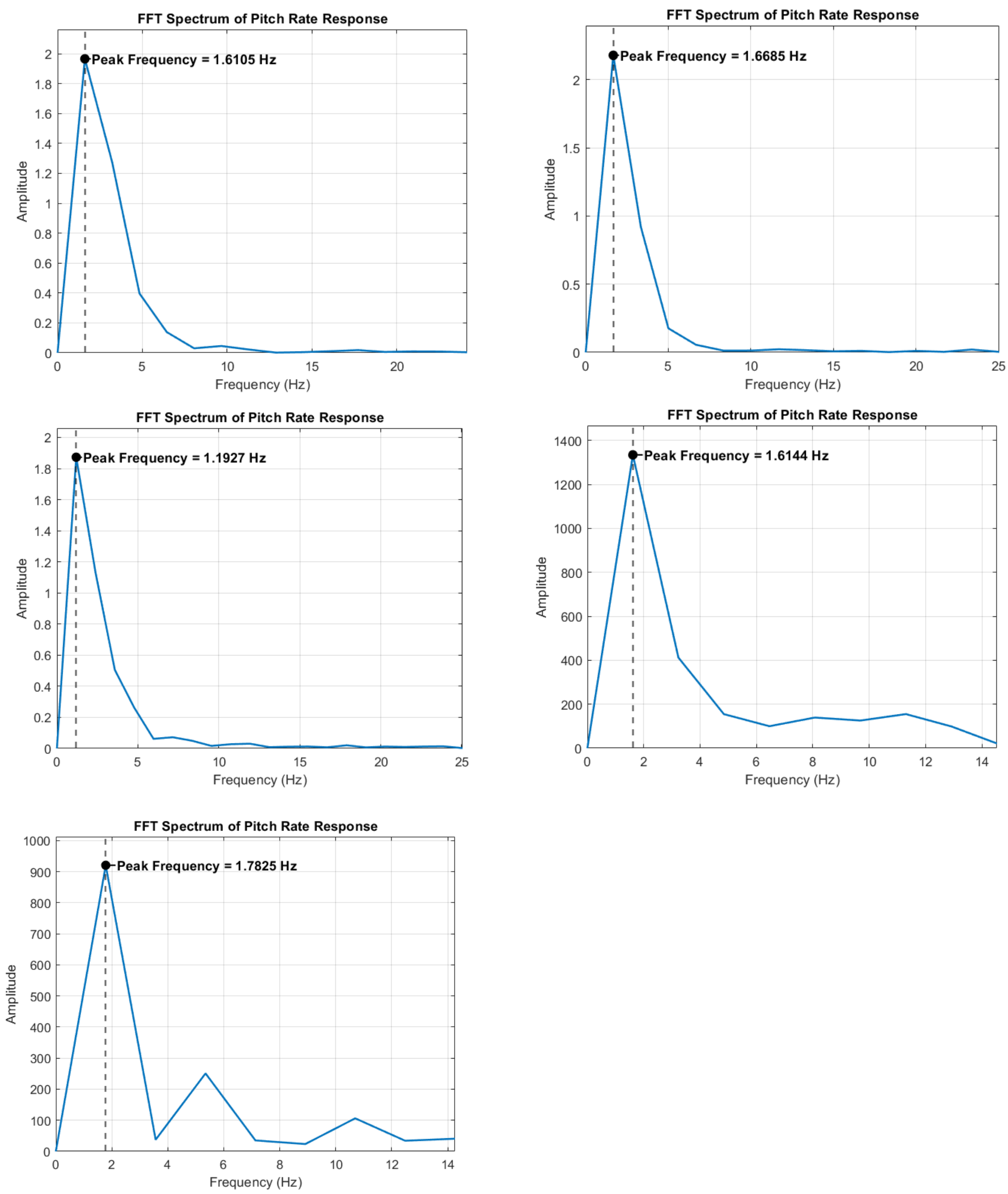


Figure VI.2: FFT analysis results of pitch response

Prony Analysis of Pitch Response

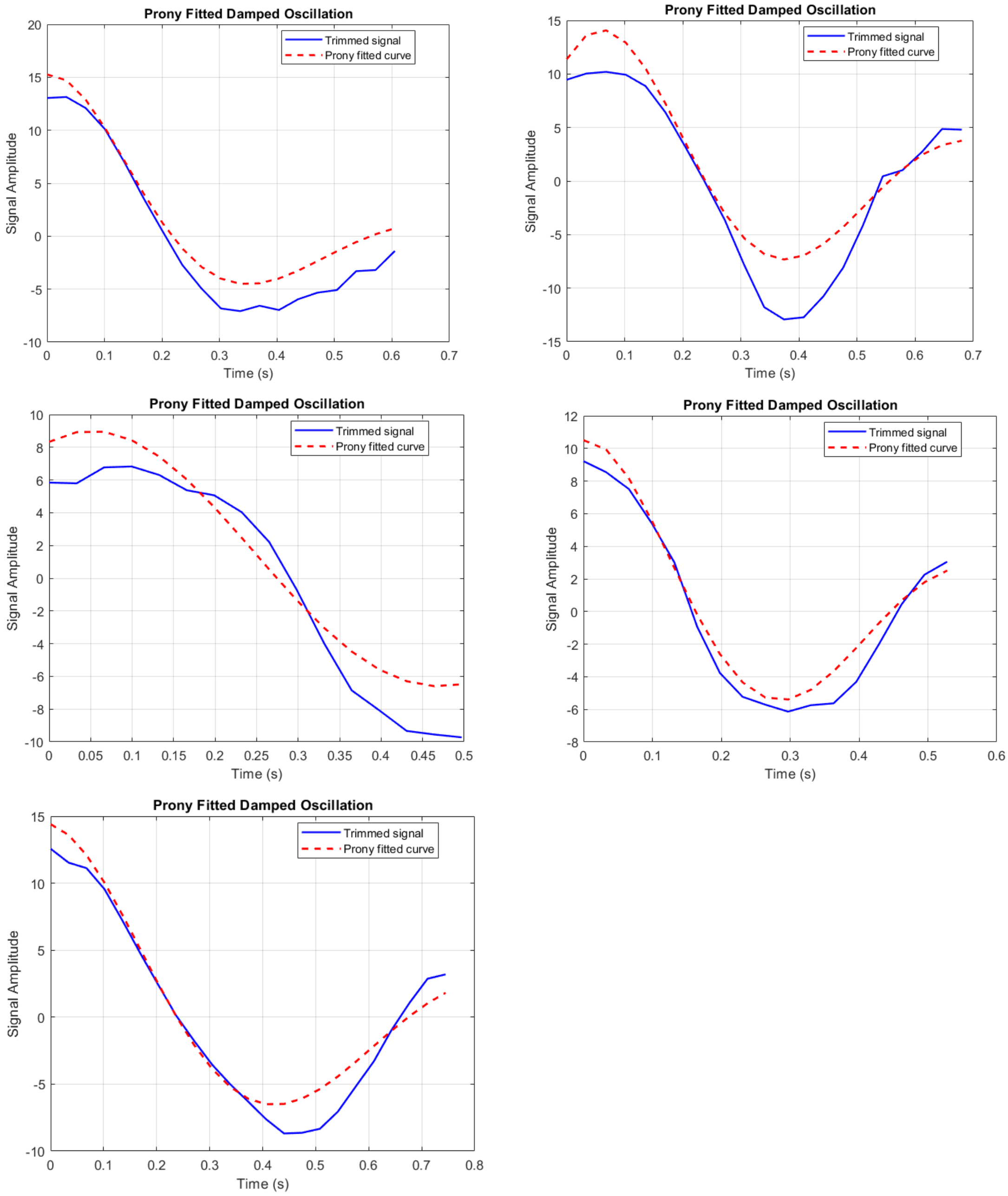


Figure VI.3: Prony approximation results of pitch

Prony Analysis of Pitch Rate Response

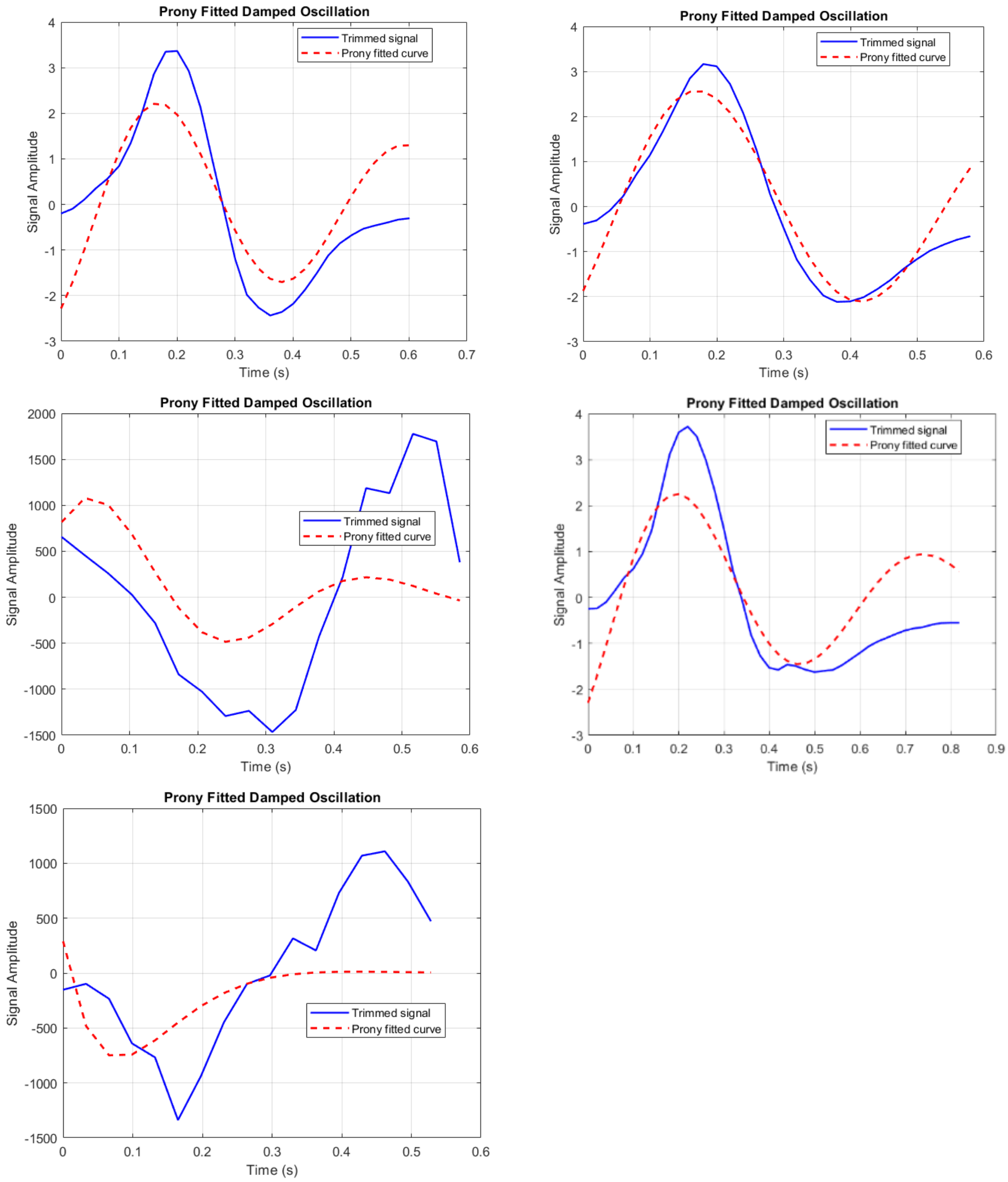


Figure VI.4: Prony approximation results of pitch rate

VII. APPENDIX G: Sensitivity Analysis Results

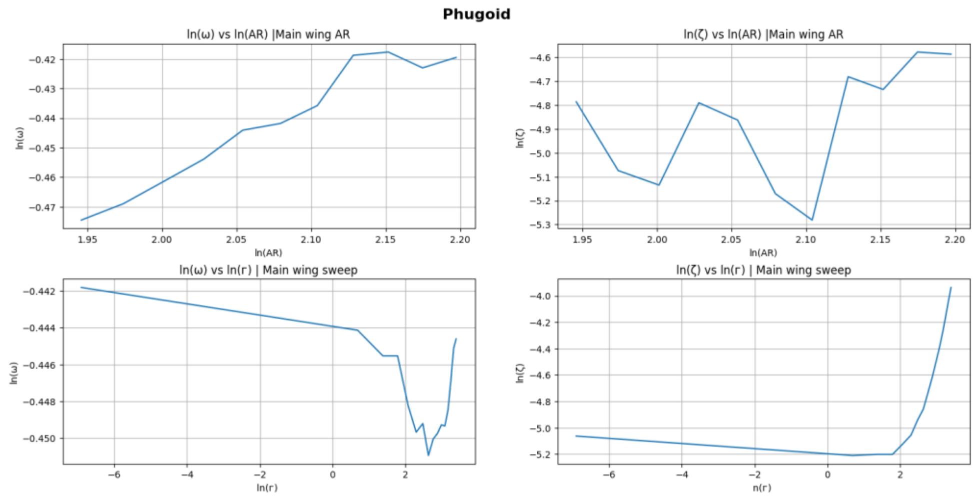


Figure VII.1: Phugoid sensitivity analysis results (1)

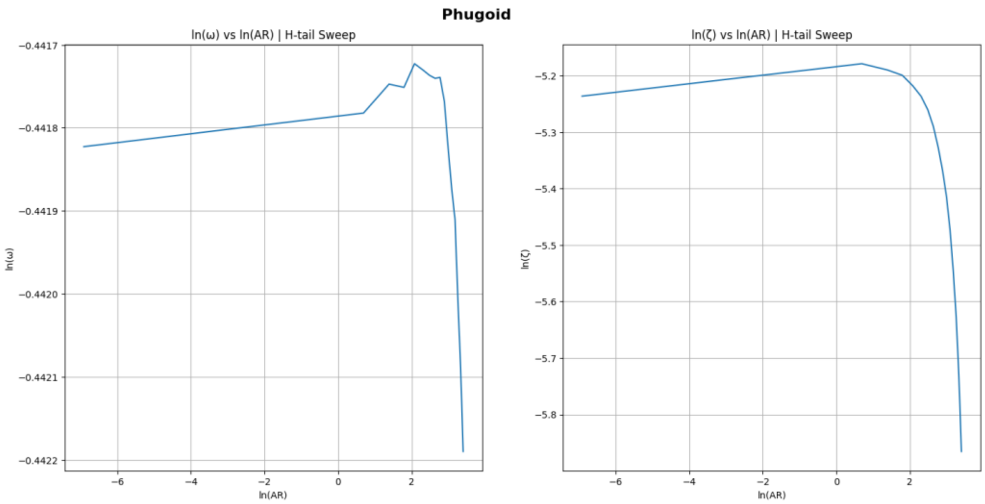


Figure VII.2: Phugoid sensitivity analysis results (2)

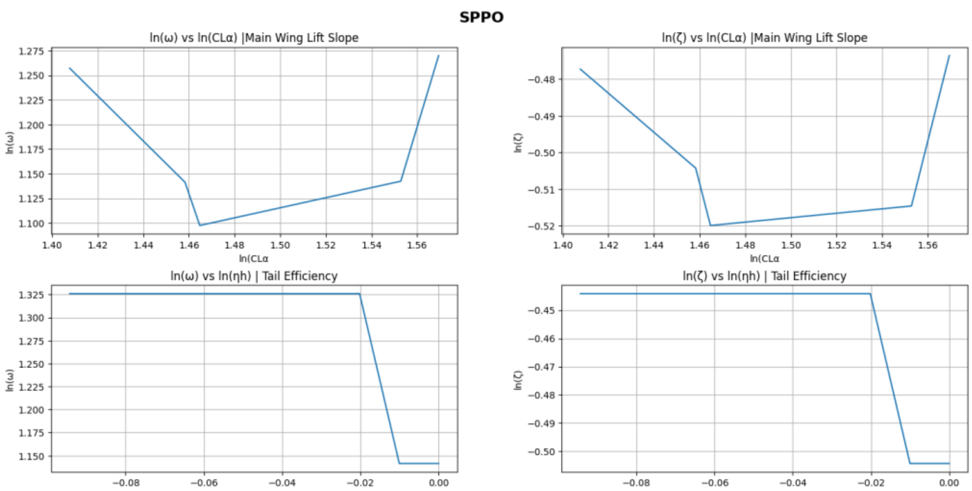


Figure VII.3:Short period sensitivity analysis results (1)

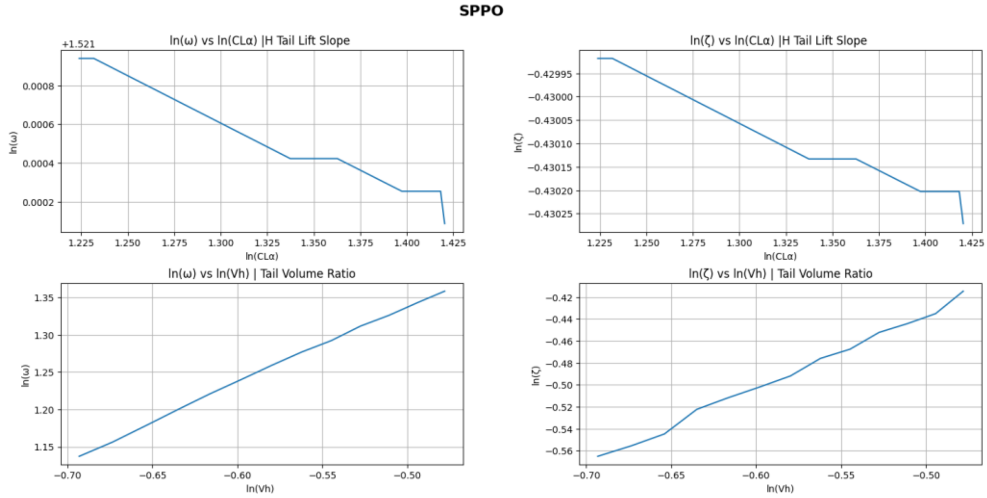


Figure VII.4:Short period sensitivity analysis results (2)

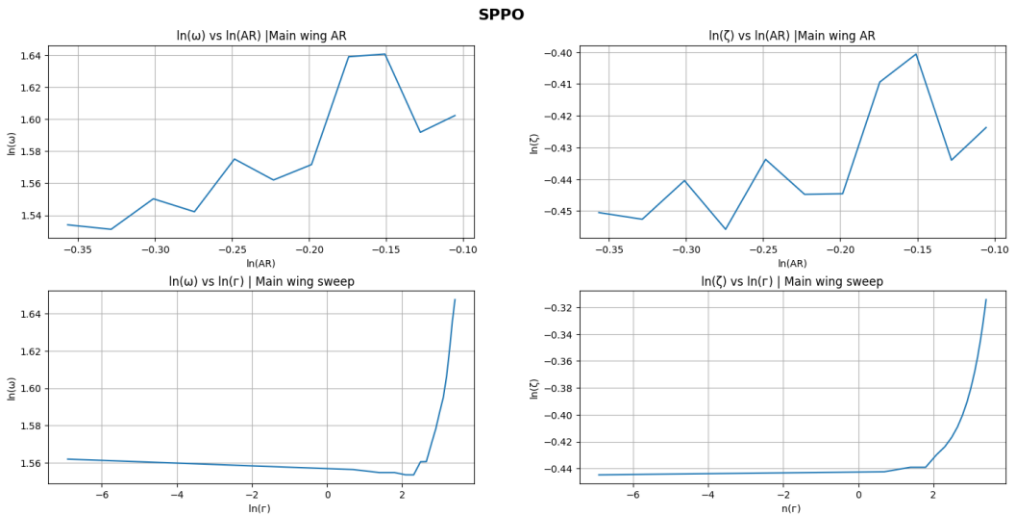


Figure VII.5:Short period sensitivity analysis results (3)

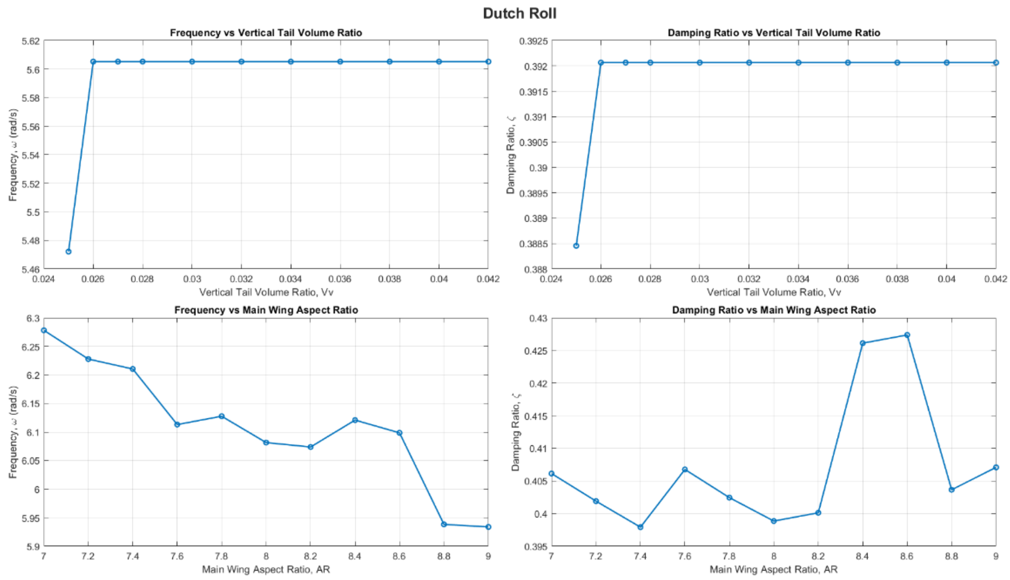


Figure VII.6:Dutch roll sensitivity analysis results

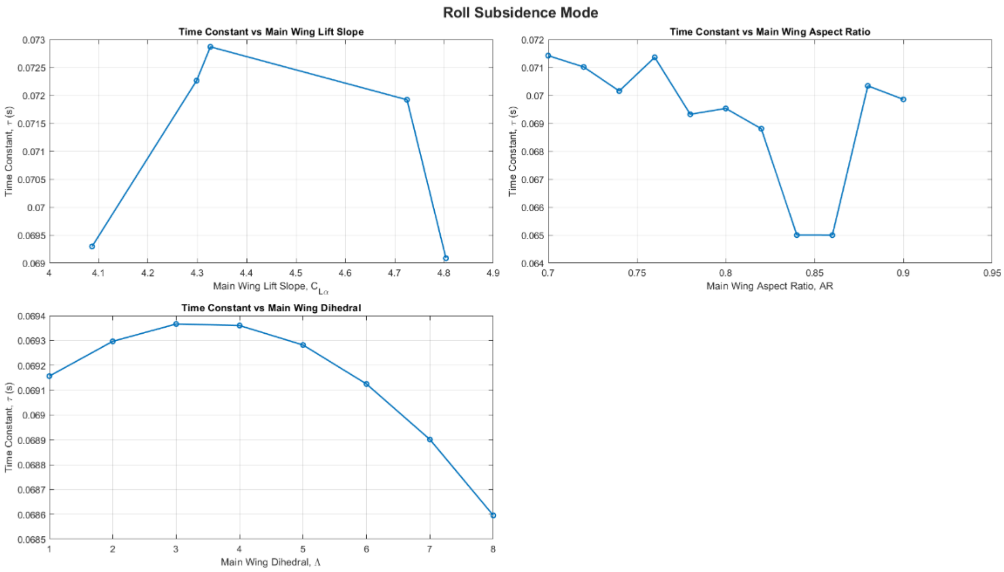


Figure VII.7:Roll subsidence sensitivity analysis results

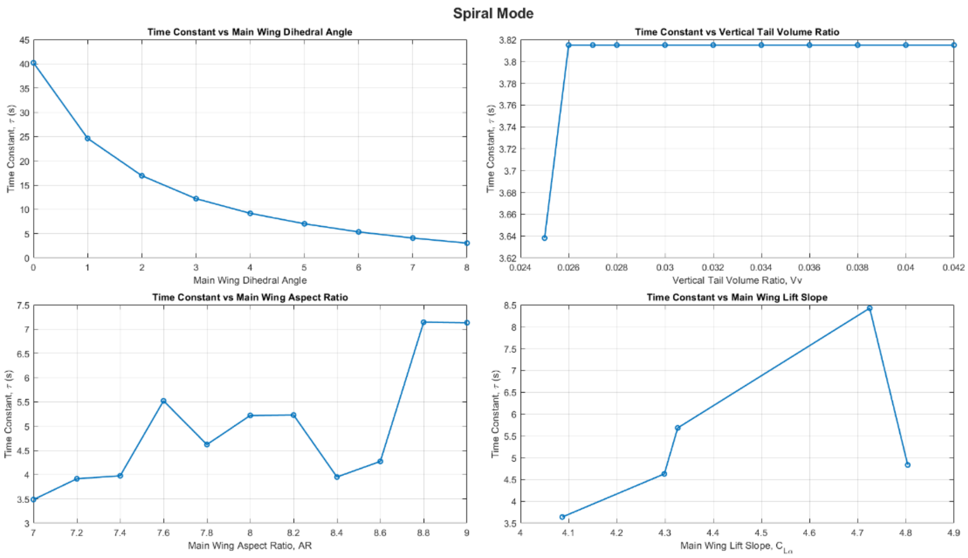


Figure VII.8:Spiral sensitivity analysis results